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WATERMARKING BASED ON UNIFIED PATTERN

RECOGNITION FRAMEWORK

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WATERMARKING BASED ON UNIFIED PATTERN RECOGNITION FRAMEWORK

A Dissertation Presented to the Graduate Faculty of

School of Engineering

Southern Methodist University

in

Partial Fulfillment of the Requirements

for the degree of

Doctor of Philosophy

with a

Major in Electrical Engineering

by

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December 15, 2007

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Watermarking based on Unified Pattern Recognition Framework

Advisor: Professor Mandyam D. Srinath

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Dissertation completed: February 20, 2015

Digital watermarking has emerged as a potentially effective tool for multimedia copyright protection, authentication and tamper proofing. Various approaches to watermarking have been proposed which exploit image properties and image processing techniques in different domains. The majority of the algorithms developed to date are concentrated in spatial, frequency, and wavelet domains. In recent years, image watermarking based on image moments has drawn more interest. This is due to the fact that image moments have been shown to be robust with respect to image variations such as in scale or position. Recently, H. S. Kim and H. K. Lee [24] proposed a new image watermarking technique using Zernike moments (ZM). ZM are rotation invariant and they can be made to be both scale and translation invariant by properly normalizing the image. In addition, it also has very good resistance to noise in the image. The fundamental problem with this the existing moments based watermarking techniques is that the watermark synchronization between source and receiver is highly dependent

upon the same image centroid location is detected at both the source and receiver. As a result, they are vulnerable against geometric attacks involves image cropping or padding.

In the research, a new blind image watermarking scheme using a novel pattern recognition framework is presented. In both watermark embedding and extraction, we first utilize an affine invariant point detector to determine an invariant point as the center and use the scale associated with this point to determine the region to embed/detect the watermark. When embedding the watermark, a self-similar sequence serves as a quantization function to modulate the amplitude of the image moments with a pseudo random sequence. The watermark detection consists of a weighted correlation of the original pseudo random sequence with a binary sequence extracted by demodulating the amplitude function by searching through the same self-similar quantization function. The performances of two types of image moments (ZM and Radial Tchebichef Moments (RTM)) in this framework are studied. Experiments conducted using Stirmark4 were demonstrate that this proposed framework overcomes the limitation of existing moments-based approaches in dealing with geometric attacks involves image cropping/padding operations. The framework is then extended to video watermarking realm, where two video watermarking implementations that utilize RTM and Hu's moment invariants (MI) respectively are discussed.

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Chapter 1

INTRODUCTION

1.1 Background of Digital Watermarking

Watermarks originally referred to the identification marks produced during the paper making process. The first watermarks appeared in Italy during the 13th century. They were used as a means to identify the papermaker or the trade guild that manufactured the paper [52], one example of such mark is shown in Figure 1. Today Watermarks are among the most popular and well-known security features of paper money.



Figure 1. Paschal Lamb Watermark from the 15th Century

Since mid 1990s, there has been an extraordinary growth of both interest and demand for techniques for copyright protection of different types of data, especially multimedia information. This interest is not surprising in view of the simplicity of digital copying and dissemination: digital copies can be made identical to the original and later reused or even manipulated. Cryptography is an effective solution to the distribution problem, but

most cryptographic protocols are concerned with secure communications rather than with copyright infringements. For instance, the High-Bandwidth Digital Content Protection (HDCP) protocol in High-Definition Multimedia Interface (HDMI) for digital video decoding succeeds in avoiding unauthorized access to programs that are being broadcast in scrambled form but fails in precluding further storage and illegal reproductions [54].

In this regard, digital watermarking as an alternative content protection mechanism has some advantages. Digital watermarking is a form of information hiding, which seeks to embed information inside a cover message with little to no degradation of the cover-object. This concept owes its origin to a conference paper by Tanaka, Nakamura, and Matsui in 1990 [53]. In real life applications, digital watermarks generally co-exist with various other data security mechanisms. There are several reasons that watermarks have advantages over other data security mechanisms: first of all, it allows us to hide encrypted messages in mediums less likely to attract attention. Secondly, other security approaches such as cryptographic techniques, generally rely on that of a safety deposit box metaphor – if the box is open, all of the information security is lost. In watermarking or information hiding techniques, the key is embedded into the information itself, and therefore provides added security.

In general a watermarking application consists of watermark embedding at the source and watermarking extraction at the receiver. To increase the security of the watermark message, the watermark can optionally be encrypted before sending to the receiver. In order to detect the embedded watermark, the receiver will incorporate a watermark detector which recovers the embedded watermark. The whole process can be conceptually described in Figure 2.

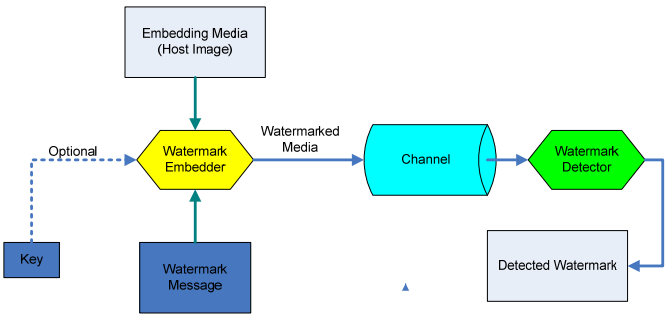


Figure 2. Generic Watermark Systems

In general, an ideal watermark should satisfy the following requirements: [15]

- Perceptibility;
- Robustness;
- Capacity;
- Speed;

A watermarking system is of no use to anyone if it distorts the cover image to the point of being useless, or even highly distracting. Ideally the watermarked imaged should look indistinguishable from the original even on the highest quality equipment.

The ideal watermark also should be highly robust, resistant to distortion introduced during either normal use (unintentional attack), or a deliberate attempt to disable or remove the watermark present (intentional, or malicious attack).

Other requirements of an ideal watermarking system that are slightly less important are capacity and speed. A watermarking system must allow for a useful amount of information to be embedded into the image. Furthermore, the complexity of the watermark algorithm is highly dependent on the application environment: If it is a real time processing environment, then the watermark algorithm may not be overly computationally intensive. On the other hand, when off-line processing is permissible,

computation load and processing time turns to become a secondary consideration. Since our priority is in the work reported here the robustness of the watermarking framework, we regard the concern of processing time and computational load and watermark capacity as secondary issues.

The existing watermarking techniques in the traditional transform domain such as DCT, DFT or DWT based watermarking algorithms all exhibit deficiency in their robustness against attacks, especially geometric attacks – since these transformations all require the synchronization of the spatial coordinates between the source and receiver. Another issue with transform domain based watermarking is that in general the performance would greatly degraded when the source has undergone lossy transcoding (i.e., a video stream converted from AVI to H.264 to MPEG2), the issue will be made worse when such transcoding is coupled with geometric transformations such as rotation and scaling.

One way of making the watermark immune to geometric attacks is to use some property of the image which is invariant to geometric transformations as the medium in which to embed the watermark. Image moments are attractive for this purpose as one of the unique properties of many of the image moments such as Zernike Moments (ZM) is that the amplitudes of the moments are invariant to scale, rotation, and translation of the image. Image moments have been widely used in image processing, remote sensing, and pattern recognition because of this invariance property. This is also the motivation behind the exploration of watermark embedding in the image moments domain.

Many different moments-based watermarking algorithms have been proposed over the last few years. The methods in general share one commonality in that at the source, they use as the watermark region, a circular region of the image with the image centroid as the

origin. The amplitudes of the moments are modulated with watermark information. Then, at the receiver, the amplitude of the image moments over the chosen region is analyzed to detect the existence of watermarks.

One fundamental problem in these techniques is that image cropping or padding can desynchronize the image centroid of the source and receiver. The resulting desynchronization of the image centroids at the source and receiver leads to errors in reconstructing the image moments and hence in retrieving the watermark.

The watermarking framework developed in this research offers a solution to overcome the above drawbacks against image cropping/padding related attacks. Instead of relying on achieving the synchronization between source and receiver by using image centroid as origin, we introduce another affine interest point component into the framework. Since it has been shown in computer vision applications that such interest point is rotation, scaling and translation invariant, therefore using such local feature to achieve the synchronization between source and receiver would be more robust than image centroid. Since the affine interest points do not depend on the global image structure, therefore, the synchronization can be maintained under the geometric attacks. Additionally, this work integrates two different image description techniques – image moments type global image descriptors and affine interest point type local image descriptors. Such integration offers more flexibility in composing different types of watermarking schemes.

In addition, we will attempt to extend this framework to the problem of watermarking digital video sequences. In the video watermarking environment, the temporal aspect of the medium can be used to improve the watermark capacity.

1.2 Image Moments and Affine Interest Point Detection

As noted earlier, image moments have been widely used in image processing, remote sensing, and pattern recognition. The image moments are global image descriptors that are invariant to scale, rotation, and translation of the image. Invariant pattern recognition is performed by classification in the multidimensional moment invariant feature space.

Image moments have been widely used in image processing, remote sensing, and pattern recognition. The moment invariants are global image descriptors that are invariant to scale, rotation, and translation of the image. Invariant pattern recognition is performed by classification in the multidimensional moment invariant feature space.

The moment invariants were introduced to be used in pattern recognition by Hu' as nonlinear combinations of the geometric moments. It has been shown that Hu's moment invariants are equivalent to some special orders of the more general circular Fourier Mellin moments (FMMs) [45]

In 1934, Zernike proposed a set of complex polynomials that form a complete orthogonal basis set defined on the unit disc [46]. In comparison with other image moments, including Legendre moments or Orthogonal Fourier-Mellin moments (OFMM) in terms of noise sensitivity, information redundancy, and capability of image description, the Zernike moments were found to have the optimum overall performance [56]. In the pattern recognition field, Complex Zernike moments are constructed using a set of complex Zernike polynomials. Besides its orthogonal characteristics, another attractive property of the Zernike Moments is that the absolute value of a Zernike moment is rotation invariant as reflected in the mapping of the image to the unit disc.

Recently, the Radial Tchebichef moments (RTM) were proposed, which offers an alternative to the Zernike moments for scale- and rotation-invariant pattern recognition. One unique advantage of the RTM is that unlike many continuous moment function such as Zernike moments or many other moments function, RTM is a discrete moment function. Furthermore, for small images the RTMs are shown to be superior to the Zernike moments with regard to image reconstruction error.

Applications of the moment methods above all require the origin of the coordinate system to be at the image centroid (x, y), which is determined by the zero- and first-order geometric moments m_{00}, m_{10}, and m_{01} as:

$$x_c = \frac{m_{10}}{m_{00}} \quad y_c = \frac{m_{01}}{m_{00}} \tag{1.1}$$

As such, the image centroid is critical to the accuracy of the image moments. The main factor that can impact the image centroid location is image cropping, since it will change values of the geometric moments m_{00}, m_{10}. This is a not major concern for pattern recognition applications, since in general the entire pattern is available to the classifier.

One deficiency of global features such as moments is that even though they are simple representations of the information in an image, they have difficulty in coping with image detail or varying image attributes. Therefore, as in pattern recognition application, it seems natural to design the watermarking scheme to incorporate both global and local image descriptors - that is, first computing the most reliable or more informative global features, and then applying further computations only to local features of interest.

The idea behind local approaches is to find a more compact representation of the image content, by localizing the main areas carrying the most useful information, while

discarding poor or noisy data. They are called local image descriptors, since they are based for instance, on the detection of the edges or the corners present in the image. Many local features have been proposed and appropriate descriptions for them have been discussed [47]: a good descriptor should be associated with meaningful, detectable parts of the image, invariant or at least tolerant to small variations of illumination and pose, and it should tackle scale variations.

When a human looks at a scene, he tends to select some points that are richer of information than others. In computer vision these are called *points of interest* [47]. Usually these are the points of the image for which the signal changes two-dimensionally. Thus, corners that the signals change in two dimensions are more significant than edges that the signals vary along a single direction.

Point of interest detectors can be used to selectively process images by concentrating effort at visually significant locations in the image; they can identify salient features and compare the similarities between them.

Loy and Zelinsky [49] present a fast gradient-based interest operator that uses a radial symmetry transform on a set of pre-determined radii to detect points of high radial symmetry since it has been demonstrated that human eyes tend to concentrate more on the symmetric areas of an image [47].

However, the main drawback of the above method is that to it requires external input to determine the radius R to detect features of different size, since the transform investigate a pixel neighborhood of dimension strictly related to the radius of the transform. An obvious enhancement would be an algorithm can automatically determine the scales

associated with each of the interest points, which leads us to the scale-space representation, since it laid the foundation for the automatic scale selection.

Scale-space is a simplified representation of the image which is seen at different resolution levels while its affine-scale structures are deleted. The resulted description is a well defined process to guarantee linearity and spatial shift invariance. Moreover scale space representations will not create new structures in the transformation from fine to coarse levels.

A common approach to build scale-space representation [50] can be described by the following steps:

- The original signal is convolved with a Gaussian kernel;
- The convolution is repeated until the signal is meaningful to obtain a parametric family of signals;
- The family of signals is scanned in search of classical features as corners, edges or ridges, which now have significance relative to the level of their scale.

A very important result from the scale-space approach is the *Principle for scale selection* proposed by Lindeberg [34], which is the theoretical foundation that allows automatically choosing the resolution level at which performs the search of features.

Lindeberg composed a two-stage scale-space interest point detector that integrates the automatic scale selection and Scale space maxima. This detector has the ability to correctly select the characteristic scale for each interest point, however, one deficiency is that it works in the linear scale space; therefore it can not comprehend general affine transforms. Mikolajczyk and Schmid [22] further expanded this algorithm into affine scale space through the introduction of a normalization transform proposed by Baumberg

[51]. Their work resulted in a robust affine interest point detector that plays an important role in the new watermarking framework to be developed in this research.

1.3 Outline of the Dissertation

The dissertation is organized as following: In the next chapter, we will provide some background on watermarking and analyze some conventional watermarking schemes in the traditional spatial, or transform domain; In Chapter 3 we will provide an overview of some global image description techniques such as Hu's moment invariants, Zernike Moments and Radial Tchebichef moments. Chapter 4 will present several watermarking techniques that fall into the "invariant domain", where the properties of image moment invariants are exploited for watermark embedding, such as the existing Zernike moments based watermarking techniques. In Chapter 5 we will introduce the local image descriptors such as affine interest point detectors. Chapter 6 will present the new unified watermarking framework based on the integration of image moments and affine interest point detection. We will also extend this framework to the video watermarking domain. Chapter 7 will present some experiment results. We will then conclude the dissertation with a summary of the proposed algorithm in Chapter 8.

Chapter 2

DIGITAL WATERMARKING OVERVIEW

2.1 Information Hiding

Digital watermarking, like cryptography is a form of information hiding. The explosive growth of various multimedia material distributions in digital form has led to large scale piracy of film, music, image and software. Digital watermarking has hence become an essential tool to combat the wide spread piracy of multimedia products

To introduce digital watermarking, we will start from steganography. Steganography is derived from the Greek root which means “covered writing” or “to hide in plain sight”. This definition can be used to distinguish between cryptography and steganography: cryptography is about protecting the content of messages (their meaning), steganography is about concealing their very existence.

As defined by Cachin [8], steganography is the art and science of communicating in such a way that the presence of a message cannot be detected. Steganography can be used to hide a message intended for later retrieval by a specific individual or group. The aim is to prevent the message being detected by any other party.

The history of steganography is recorded in many early human civilizations. Herodotus tells an ancient Greek story of Histiaeus revolting against Persia. The revolt was hinged on a series of secret communications. Histiaeus achieved the secret communications by tattooing a message on the shaved head of his slave, and waiting for new hair to grow.

The recipient recovered the image by shave the slave's head once he reached his destination.

The other major area of steganography is copyright marking, where the message to be inserted is used to assert copyright over a document. Copyright marking can be further divided into watermarking and fingerprinting. Fingerprinting is equivalent to pattern matching – statistically matching the object to the known content stored in the database. Watermarking on the other hand, involves embedding a message inside the content of each object (i.e., image, video). Figure 3 shows how steganography can be broken down into different areas.

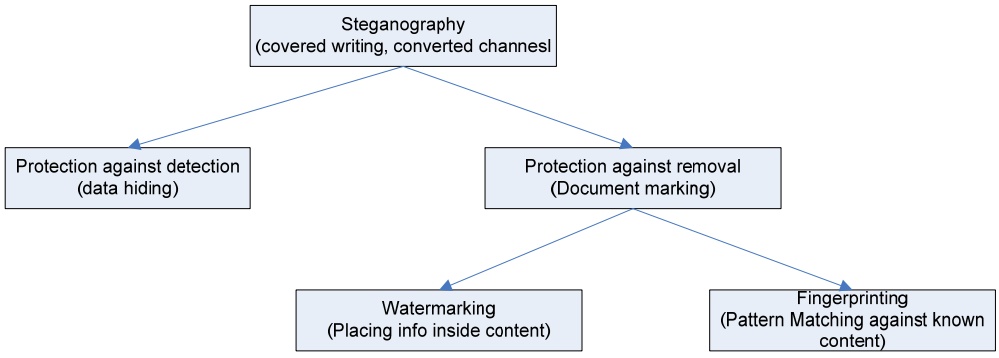


Figure 3. Steganography Classifications

Compared to the general definition of steganography, watermarking has some uniqueness. Since steganography is to have a covert communication between two parties whose existence is unknown to a possible attacker, a successful attack is to detect the existence of this communication. Watermarking, on the other hand, has the additional requirement of "survival" against possible attacks. In addition, some applications may

require visible watermarks, such as company logo or copyright sign overlaid on digital images.

2.2 Watermark Applications

Having clarified the relationships among steganography, cryptography and watermarking, we will now more specifically discuss digital watermarking. Digital watermarking is a method of providing a persistent digital identity for audio, video, print and images. Digital watermarks are data embedded directly within content which are imperceptible to humans but readable by computers. Watermarks can be recognized by enabled software or hardware to provide copyright information, or to authenticate, track, monitor, enhance security and enable access to additional data, information and ecommerce opportunities across devices and networks.

The reason that digital watermark embedding is possible is that there is always a redundancy between the source material and what a human can hear/see. For example, digital image watermarking technology makes use of the fact that the human eye has only a limited ability to observe differences between two images. Minor modifications in the color of an image are subconsciously corrected by the eye, so that the observer does not notice any difference.

Currently digital watermarks are broadly deployed with billions of watermarked objects and hundreds of millions of watermark detectors in the market. The following is an overview of the main applications of digital watermarking:

A. Ownership Declaration

One of the greatest advantages of the Internet is the ability to easily share information, including text, images, audio and video. On the other hand, Internet also poses challenge

to the authors and photographers want to maintain their rights under copyright law. For digital photographs, one of the best ways to identify or protect copyrighted or original material is watermarking.

Visible watermarks are the most obvious and recognizable form of this method of protecting photos published on the Web. A visible watermark covers some portion of the image with an identifying or authenticating symbol or text translucently overlaid on the primary image, thus making it hard to reuse the image without the origin being known. In this regard, digital watermark acts in the same way as the original watermarks in the paper workshops in 13th century Europe.



Figure 4. Images with Visible Watermark for Ownership Declaration

While some software packages or algorithms are able to remove digital watermarks in images, there is almost invariably some level of distortion, since the original image information "under" the watermark is lost or changed. Therefore it is important to overlay

the watermark in a way which makes it difficult to remove to assure the goal of indicating property rights is to be achieved.

B. Copyright Protection

Watermarking can be used to protect redistribution of copyrighted material over an unprotected network like Internet or peer-to-peer (P2P) networks. Content aware networks could incorporate watermarking technologies to report or filter out copyrighted material from such networks. For example, if the system recognizes that the material is sent by a licensed user but that this particular file is not authorized, it may substitute an authorized version for redistribution by the user.

C. Content Archiving

Watermarking can be used to insert a digital object identifier or serial number to help archive digital content such as images, audio or video. It can also be used for classifying and organizing digital content. Normally digital contents are identified by their file names; however, this is a very fragile technique as file names can be easily changed. Hence embedding the object identifier within the object itself reduces the possibility of tampering and hence can be effectively used in archiving systems.

D. Broadcast Monitoring

Watermarking can also be used for broadcast monitoring which refers to verifying whether the content that was supposed to be broadcast (on TV or Radio) has really been broadcast, for example when an advertiser wants to know whether the advertisement was actually broadcast at the right time and for right duration.

E. Tamper Detection

With the powerful image processing software like Adobe PhotoShop, it is possible to combine portions of several different images together while leaving barely detectable traces. This makes digital forgeries a reality. Therefore, tamper detection is very important for some highly sensitive digital imagery t like satellite or medical images.

Digital content can be detected for tampering by embedding fragile watermarks. If the fragile watermark is destroyed or degraded, it indicates the presence of tampering and hence the digital content cannot be trusted. Some commonly used fragile watermarks include embedding the checksum obtained by summing the numbers determined by the K most significant bits of all pixels and taking a remainder operation with a large integer N [55]. The technique then randomly chooses the locations of the pixels that are to contain one bit of the checksum. The pixel locations of the checksum, together with the checksum itself, form the watermark. Another method is to embed a PN sequence. For an $N \times N$ image, a sequence of length N is randomly shifted and added to the image rows. The phase of the sequence carries the watermark information. A simple cross-correlation is used to test for the presence of the watermark.

F. Digital Fingerprinting

Digital fingerprinting is one level above tamper detection. While content can be detected for tampering by embedding fragile watermarks, digital fingerprinting requires the determination of the origin of tampered material. If the fragile watermark is destroyed or degraded, it indicates the presence of tampering and hence the digital content cannot be trusted; for digital fingerprinting, however, it is necessary to extract the

embedded watermarks. Therefore, for the purpose of digital fingerprinting, watermarks need to be robust, and should be able to be extracted even after malicious attacks.

From the description of the watermark applications, it is clear that many different varieties of watermarks are required for different application environment. We now briefly review the different classifications of watermarks.

2.3 Watermark Classification

There are different criterion can be used for classifying watermarking techniques. First from the perceptivity aspect, watermarks can be separated into *visible watermarks* which obviously mean the watermark can be seen by human without any preprocessing and *invisible watermarks*, in which case a certain detection technique is needed to extract the embedded watermark.



Figure 5. (a) "Good" (b) "Bad" Visible Watermarks

Visible watermarks in general would cause alteration to the original object, but such alteration is regarded as acceptable if the intended information from the original object can still be delivered. The trick here is to embed a watermark that is visible without becoming annoying. However the threshold between the two is subjective.

In the case of the invisible watermarks, the associated watermarking algorithms are required to assure that the quality of the original objects is not degraded after embedding the watermark; In general, the majority of technical journal papers in this field are devoted to the discussion about improving various invisible watermarking techniques.

Watermarks can also be classified based on their robustness into *fragile watermarks, semi-fragile watermarks* or *robust watermarks*. A fragile watermark is one that will not be able to survive any tampering. In this case, the assumption is that there always will be a watermark embedded in the original objects; therefore if one can't be detected as the receiver, then the sources are either not authentic or tampered with before reaching the destination. The semi-fragile watermarks supposedly can survive certain types of processing of the original objects, while they will be destroyed by other types of processing. For example, [40] describes a semi-fragile watermarking technique that will survive JPEG compression to a certain quality factor, but will be destroyed by other tampering such as geometric transformations. Robust watermarks obviously are supposedly to survive all types of malicious attacks. Because of the uncertainty of the types of attacks in the field, designing a robust watermark is one of the most challenging aspects of watermarking techniques.

The data needed for extracting a watermark can be used as another criterion for classification. Based on this, watermarking techniques can be separated into: *informed watermarking, semi-blind watermarking* and *blind watermarking*. In the domain of informed watermarking, the original object as well as every aspect of the watermark (i.e., encryption key, location, strength of the watermark etc.) is available at the detector. For semi-blind watermarking, in general the original objects are not available, but the

receiver would have some reference information about the watermark to assist in the detection process. Finally, in a blind watermarking scheme, the receiver does not have any information about the original objects or the watermark. However, in general, the receiver will have to aware of the implementation of the watermark embedding process. Without this agreement, watermark detection would not be possible. Therefore the separation between semi-blind and blind watermarking is not always clear cut.

The type of watermarks embedded can also be different – an *image watermark* overlays an actual image on top of the original image; a *noise watermark* modulates the image with a binary Pseudo-Noise sequence and takes the advantage of the autocorrelation property of such sequences for the detection of the watermark.

Finally watermarks can be classified by the techniques used to embed the watermarks. We will discuss in detail the different watermarking techniques in the next section.

2.4 Watermarking Techniques

Watermarking techniques can be classified as belonging to the spatial domain, frequency (transform) domain or wavelet domain.

2.4.1 Spatial Domain Techniques

The most straightforward method of watermark embedding would be to embed the watermark into the least significant bits (LSB) of the cover object [24]. If the channel capacity is high, a smaller object may be embedded multiple times. Even if most of these watermarks were lost due to attacks, a single surviving watermark would be considered a success.

Despite its simplicity, one drawback of LSB substitution is its sensitivity toward noise. Any addition of noise or lossy compression is likely to defeat the watermark. A simple

artificial attack would be to set the LSB bits of each pixel to one, fully defeating the watermark with negligible impact on the cover object.

Another technique for watermark embedding is to exploit the correlation properties of additive pseudo-random noise patterns [13].A pseudo-random noise (PN) pattern *W(x, y)* is added to the luminance value of the pixels of the cover image *I(x, y)*, according to the equation shown below:

$$I_w(x, y) = I(x, y) + k * W(x, y) \tag{2.2}$$

where k denotes a gain factor, and I_w the resulting watermarked image. Increasing k increases the robustness of the watermark at the expense of the quality of the watermarked image. The watermark inserting is done in a line scanning fashion, from left to right, top to bottom.

To retrieve the watermark, the same pseudo-random noise generator algorithm is seeded with the same key, and the correlation between the noise pattern and possibly watermarked image computed. If the correlation exceeds a certain threshold T, the watermark is detected, and a single bit is set. This method can easily be extended to a multiple-bit watermark by dividing the image up into blocks, and performing the above procedure independently on each block.

To improve the payload of the watermark scheme, spectrum spreading techniques similar to the code division multiple-access (CDMA) in wireless communications can be used, as shown in Figure 6 and Figure 7 below. Rather than determining the values of the watermark from "blocks" in the spatial domain, spread-spectrum techniques are employed to scatter each of the bits randomly throughout the cover image, increasing

capacity. The watermark is first formatted as a long string rather then a 2D image. For each value of the watermark, a PN sequence is generated using an independent seed. These seeds could either be stored, or themselves generated through PN methods. The summation of all of these PN sequences represents the watermark, which is then scaled and added to the cover image [17].

$\mathbf{RP_0}$	-1	1	1	-1	-1	1	-1	-1	1	1	-1	$\mathbf{b_0}$:	0	→	$\mathbf{+RP_0}$	-1	1	1	-1	-1	1	-1	-1	1	1	-1
$\mathbf{RP_1}$	1	1	-1	-1	1	-1	-1	1	1	-1	1	$\mathbf{b_1}$:	0	→	$\mathbf{+RP_1}$	1	1	-1	-1	1	-1	-1	1	1	-1	1
$\mathbf{RP_2}$	1	-1	-1	1	-1	-1	1	1	-1	1	-1	$\mathbf{b_2}$:	1	→	$\mathbf{-RP_2}$	-1	1	1	-1	1	1	-1	-1	1	-1	1
$\mathbf{RP_3}$	-1	-1	1	-1	-1	1	1	-1	1	-1	-1	$\mathbf{b_3}$:	1	→	$\mathbf{-RP_3}$	1	1	-1	1	1	-1	-1	1	-1	1	1
$\mathbf{RP_4}$	-1	1	-1	-1	1	1	-1	1	-1	-1	1	$\mathbf{b_4}$:	0	→	$\mathbf{+RP_4}$	-1	1	-1	-1	1	1	-1	1	-1	-1	1
$\mathbf{RP_5}$	1	-1	-1	1	1	-1	1	-1	-1	1	1	$\mathbf{b_5}$:	1	→	$\mathbf{-RP_5}$	-1	1	1	-1	-1	1	-1	1	1	-1	-1
$\mathbf{RP_6}$	-1	-1	1	1	-1	1	-1	-1	1	1	1	$\mathbf{b_6}$:	0	→	$\mathbf{+RP_6}$	-1	-1	1	1	-1	1	-1	-1	1	1	1
															MARK	-3	5	1	-3	1	3	-7	1	3	-1	3
															L_{org}	98	98	97	98	97	96	97	96	95	94	94
															$\mathbf{L_{mark}}$	**95**	**103**	**98**	**95**	**98**	**99**	**90**	**97**	**98**	**93**	**97**

Figure 6. An Example of CDMA Watermark Generation

$\mathbf{E[(RP_0-E(RP_0))\bullet(L_{MARK}-E(LMARK))]}$	=	15.6	→	$\mathbf{b_0}$:	0
$\mathbf{E[(RP_1-E(RP_1))\bullet(L_{MARK}-E(LMARK))]}$	=	16.4	→	$\mathbf{b_1}$:	0
$\mathbf{E[(RP_2-E(RP_2))\bullet(L_{MARK}-E(LMARK))]}$	=	-26.4	→	$\mathbf{b_2}$:	1
$\mathbf{E[(RP_3-E(RP_3))\bullet(L_{MARK}-E(LMARK))]}$	=	-3.1	→	$\mathbf{b_3}$:	1
$\mathbf{E[(RP_4-E(RP_4))\bullet(L_{MARK}-E(LMARK))]}$	=	21.6	→	$\mathbf{b_4}$:	0
$\mathbf{E[(RP_5-E(RP_5))\bullet(L_{MARK}-E(LMARK))]}$	=	-23.6	→	$\mathbf{b_5}$:	1
$\mathbf{E[(RP_6-E(RP_6))\bullet(L_{MARK}-E(LMARK))]}$	=	0.4	→	$\mathbf{b_6}$:	0

Figure 7. Example of the CDMA Watermark Detection

To detect the watermark, each seed is used to generate its PN sequence, which is then correlated with the entire image. If the correlation is high, that bit in the watermark is set to “1”, otherwise a “0”. The process is then repeated for all the values of the watermark.

2.4.2 Frequency Domain Techniques

An advantage of the spatial techniques discussed above is that they can be easily applied to any image, regardless of subsequent processing (whether they survive this processing, however, is a different matter entirely). A possible disadvantage of spatial techniques is that they do not allow for the exploitation of this subsequent processing in order to increase the robustness of the watermark.

In addition, adaptive watermarking techniques are more difficult in the spatial domain. Both the robustness and quality of the watermark could be improved if the properties of the cover image could similarly be exploited. For instance, it is preferable to hide the watermark in noisy regions and edges of images, rather than in smoother regions. The benefit is two-fold: degradation in smoother regions of an image is more noticeable to the human visual system (HVS), and becomes a prime target for lossy compression schemes.

Taking these aspects into consideration, working in a frequency domain becomes very attractive. The classic domain for image processing is the Discrete Cosine Transform (DCT). The DCT allows an image to be broken up into different frequency bands, making it easier to embed the watermark into the middle frequency bands of an image. The middle frequency bands are chosen such that they avoid the visually most important parts of the image (low frequencies) without over-exposing themselves to removal through compression and noise attacks (high frequencies) [13].

One such technique utilizes the comparison of middle-band DCT coefficients to encode a single bit into a DCT block. To begin, we define the middle-band frequencies (FM) of an 8x8 DCT block as shown below in Fig. 2.8, where FL, FM and FH are used to denote the low, middle and high frequency components of the block. FM is chosen as the embedding region so as to provide additional resistance to lossy compression techniques, while avoiding significant modification of the cover image [2].

Next two locations $B_i(u_1,v_1)$ and $B_i(u_2,v_2)$ are chosen from the FM region for comparison. The DCT block will encode a "1" if $B_i(u_1,v_1) > B_i(u_2,v_2)$; otherwise it will encode a "0". The coefficients are then swapped if the relative size of each coefficient does not agree with the bit that is to be encoded [2].

The swapping of such coefficients should not alter the watermarked image significantly, as it is generally believed that DCT coefficients of middle frequencies have similar magnitudes. The robustness of the watermark can be improved by introducing a watermark "strength" constant k, such that $B_i(u_1,v_1) - B_i(u_2,v_2) > k$. Coefficients that do not meet this criteria are modified though the use of random noise so as to then satisfy this relation. Increasing k thus reduces the chance of detection errors at the expense of additional image degradation [2].

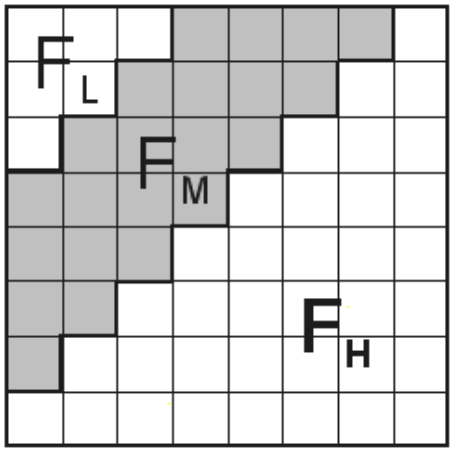


Figure 8. Definitions of DCT Regions

Another possible technique is to embed a PN sequence *W* into the middle frequencies of the DCT block. We can modulate a given DCT block using the equation shown below:

$$I_{W_{x,y}}(u,v) = \begin{cases} I_{x,y}(u,v) + k * W_{x,y}(u,v), & u,v \in F_M \\ I_{x,y}(u,v), & u,v \notin F_M \end{cases} \tag{2.3}$$

For each 8x8 block of the image, the DCT for the block is first calculated. In that block, the middle frequency components FM are added to the PN sequence *W*, multiplied by a gain factor *k*. Coefficients in the low and middle frequencies are copied over to the transformed image unaffected. Each block is then inverse-transformed to give us our final watermarked image I_W [13].

The watermarking procedure can be made somewhat more adaptive by slightly altering the embedding process as shown below:

$$I_{W_{x,y}}(u,v) = \begin{cases} I_{x,y}(u,v) * (1 + k * W_{x,y}(u,v)), & u,v \in F_M \\ I_{x,y}(u,v), & u,v \notin F_M \end{cases} \quad (2.4)$$

This modification scales the strength of the watermarking based on the size of the coefficients used. Larger *k*'s can be used for coefficients of higher magnitude [14].

For detection, the image is broken up into the same 8x8 blocks, and a DCT performed. The same PN sequence is then compared to the middle frequency values of the transformed block. If the correlation between the sequences exceeds some threshold *T,* a "1" is detected for that block; otherwise a "0" is detected. Again *k* denotes the strength of the watermarking, where increasing *k* increases the robustness of the watermark at the expense of quality [13].

2.4.3 Wavelet Watermarking Techniques

Another possible domain for watermark embedding is the wavelet domain. The DWT (Discrete Wavelet Transform) separates an image into a lower resolution approximation image (*LL*) as well as horizontal (*HL*), vertical (*LH*) and diagonal (*HH*) detail components. The process can then be repeated to compute multiple-scale wavelet decompositions, as in the 2-scale wavelet transform shown below in Figure 9.

One of the many advantages of the wavelet transform is that that it accurately models aspects of the HVS as compared to the FFT or DCT. This allows us to use higher energy watermarks in regions that the HVS is known to be less sensitive to, such as the high-resolution detail bands {*LH, HL, HH*). Embedding watermarks in these regions allow us to increase the robustness of our watermark, at little to no additional impact on image quality [13].

One of the most straightforward techniques is to use an embedding technique similar to that used in the DCT, that is, to embed a pseudo-random sequence W in the detail bands according to the equation shown below:

$$I_{Wu,v} = \begin{cases} W_i + a \mid W_i \mid x_i, & u,v \in HL, LH \\ W_i & u,v \in LL, HH \end{cases} \tag{2.5}$$

where W_i denotes the coefficient of the transformed image, x_i the bit of the watermark to be embedded, and a is a scaling factor. To detect the watermark we generate the same pseudo-random sequence used in generation of the watermark and determine its correlation with the two transformed detail bands. If the correlation exceeds some threshold T, the watermark is detected.

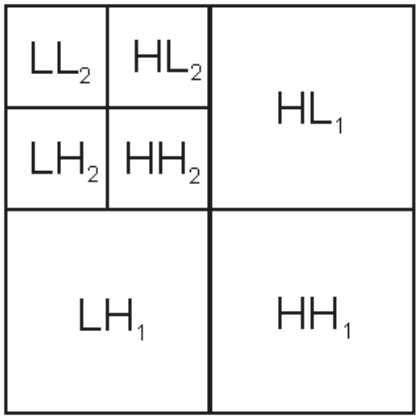


Figure 9. 2-Scale 2-Dimensional Discrete Wavelet Transform

This can be easily extended to multiple bit messages by embedding multiple watermarks into the image. As in the spatial version, a separate seed is used for each PN sequence, which is then added to the detail coefficients as per Eq. (2.4). During detection, if the correlation exceeds the threshold for a particular sequence, a "1" is recovered; otherwise

a zero. The detection process then iterates through the entire PN sequence until all the bits of the watermark have been detected.

Even though the above three domains are the most commonly used for watermark embedding, however, there is a growing interest in exploring other domains for more robust watermarking techniques. There is a close relation between pattern recognition and watermarking applications. Therefore, it is natural that researchers are interested in exploring some techniques that are commonly used in pattern recognition field. One such technique is the image description using global image features such as Zernike Moments. We will introduce the image moments in Chapter 3, followed by a discussion of the watermark embedding in the image moments domain in Chapter 4.

Chapter 3

IMAGE MOMENTS BASED GLOBAL DESCRIPTORS

3.1 Pattern Recognition

Pattern recognition and machine learning is a field concerned with machine recognition of meaningful regularities in noisy or complex environments. [52] In the pattern recognition application, the first step will require an accurate measurement of the external environment, for example a raw image that accurately reflects the scene to be classified. The next step is to extract a feature vector to represent the external environment. During this step, the goal is to construct a feature vector with reduced dimensionality that represents the external environment that is intended to classify.

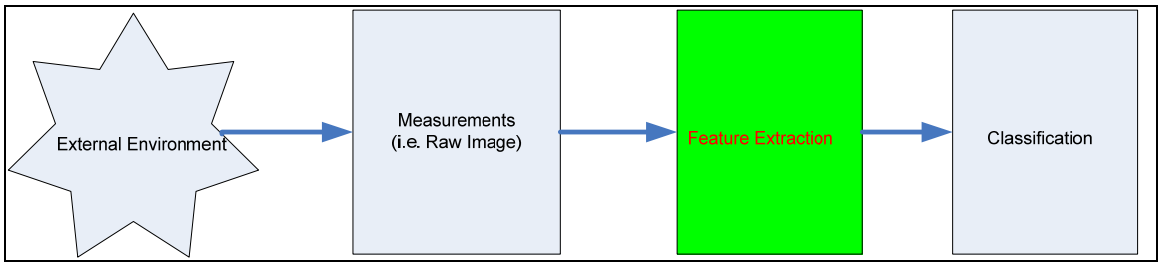


Figure 10. General Process of Pattern Recognition

For visual pattern recognition applications that involve the classification of image scenes, one key area of study is the methodology to extract a feature vector that is robust

and invariant to image geometric transformation such as rotation, translation, scaling as well as image processing operations such as filtering or compression.

Two types of commonly used feature vectors for visual pattern recognition are *Global Image Descriptors* and *Local Image Descriptors*. For image classification, an example of the former are moment invariants associated with the image while an example of the latter would be an interest point such as a corner and the associated support region. In the rest of this chapter, we will describe the fundamentals of the global image descriptors – image moments.

3.2 Global Image Descriptor – Image Moments

3.2.1 Fundamentals of Image Moments

Image moments and moment invariants play a very important role in pattern recognition and machine vision. The general two-dimensional $(p+q)^{th}$ order moments of a grey-level image $f(x, y)$ are defined as:

$$m_{pq} = \iint_{\Gamma} x^p y^q f(x,y) dxdy \tag{3.6}$$

where $p,q = 0,1,2\ldots$, and Γ is the support of image $f(x, y)$.

In the case of a digital image, the double integral in the equation above will be replaced by a summation. The most common way to do that is to employ the rectangular (i.e., zero order) method of numerical integration:

$$m_{pq} = \sum_{\Gamma} x^p y^q f(x,y) \tag{3.7}$$

where the summation extends over all the elements in Γ.

The operations of scale change, translation and rotation of an image are easily defined in term of moments:

- Scaling: The moments of *f(x, y)* after a scale change by a factor λ are given by:

$$m'_{pq} = \sum_{\Gamma} x^p y^q f(\frac{x}{\lambda}, \frac{y}{\lambda}) = \lambda^{2+p+q} m_{pq} \tag{3.8}$$

- Translation: The moments of *f(x, y)* translated by an amount (*a, b*) are given by

$$m'_{pq} = \sum_{\Gamma} (x+a)^p (y+b)^q f(x,y) = \sum_{r=0}^{p} \sum_{s=0}^{q} \binom{p}{r} \binom{q}{s} a^{p-r} b^{q-s} m_{rs} \tag{3.9}$$

- Rotation: The moments of *f(x, y*) after a rotation by an angle θ about the origin are given by

$$m'_{pq} = \sum_{\Gamma} (x\cos\theta + y\sin\theta)^p (y\cos\theta - x\sin\theta)^q f(x,y) \tag{3.10}$$

The moments of the rotated image may be expressed in terms of the original moment:

$$m'_{pq} = \sum_{r=0}^{p} \sum_{s=0}^{q} \binom{p}{r} \binom{q}{s} (\cos\theta)^{p-r+s} (\sin\theta)^{q+r-s} m_{m+q-r-s,r+s} \tag{3.11}$$

The rotation of an image about an arbitrary point may be achieved in three steps. First, translate the moments so that the point of rotation is at the origin using Eq. (3.4), then rotate the moments using Eq. (3.5), and finally, translate the moments such that the point of rotation is returned to its original position.

- Reflection: A reflection about the *y*-axis results in a change of sign of all moments which depend on an odd power of *y*, i.e.,

$$m'_{pq} = \sum_{\Gamma} x^p y^q f(x,-y) = (-1)^q m_{pq} \tag{3.12}$$

A reflection about an arbitrary axis may be achieved in three steps. First the moments are rotated so that the reflection axis is aligned with the *y*-axis. Secondly, the moments are reflected about the *y*-axis by using the transform specified above. Finally, the reflected moments must be rotated to restore the reflection axis to its original orientation.

The image central moments are defined as:

$$\mu_{pq} = \sum_{\Gamma} (x - x_c)^p (y - y_c)^q f(x,y) \tag{3.13}$$

where $\bar{x}$ and $\bar{y}$ are the co-ordinates of the image centroid *s* defined and determined as follows :

$$x_c = \frac{m_{10}}{m_{00}}, \qquad y_c = \frac{m_{01}}{m_{00}} \tag{3.14}$$

One important property of the central moments is that they are invariant to translation: if μ_{pq}' is the central moments of *f(x-a,y-b)*, then $\mu_{pq}' = \mu_{pq}$.

The image centroid is a critical component in most moments based analysis. It is therefore useful to consider the effect of some basic operations on image centroids too. Let (*xc', yc'*) represent the centroid of image obtained by performing on of these operations.

- Scaling: Under scaling$(x_c', y_c') = (\alpha x_c, \alpha y_c)$.
- Translation: If the image is translated by (a,b), then $(x_c', y_c') = (x_c+a, y_c+b)$.
- Rotation: For an image rotated by the angle θ about a point $(\bar{x}, \bar{y})$, then the new centroids is:

$$\begin{aligned} x_c' &= \cos\theta(x_c - \bar{x}) - \sin\theta(y_c - \bar{y}) + \bar{x} \\ y_c' &= \sin\theta(x_c - \bar{x}) + \cos\theta(y_c - \bar{y}) + \bar{y} \end{aligned}, \tag{3.15}$$

Using the translation properties of the image centroids, the translation invariant property of the central moments can easily be shown:

$$\begin{aligned} \mu_{pq}' &= \sum_{\Gamma} (x' - \bar{x}')^p (y' - \bar{y}')^q f(x, y) \\ &= \sum_{\Gamma} ((x - a) - (\bar{x} - a))^p ((y - b) - (\bar{y} - b))^q f(x, y) \\ &= \mu_{pq} \end{aligned} \tag{3.16}$$

Several geometric moments have been proposed in the literature for use as pattern features because of the rotation and scale invariance properties. In the following sections, we will introduce a few such image moments. Our primary interest is to determine the applicability of these moments in digital watermarking applications. We start with a description of the moment invariants, which were the first image moments function that was proposed for pattern recognition applications; we will then introduce Zernike moments (ZM) which have been used in watermarking applications [24],[49] and also present other moments which may also be used in such applications.

3.2.2 Moments Invariants

Recognition/classification of objects and patterns independent of their position, size, orientation and other variations in geometry and colors has been the goal of many researchers in the pattern recognition field. Finding efficient invariant object descriptors is the key to solving this problem. Several groups of features have been used for this purpose, such as simple visual features (edges, contours, textures, etc.), Fourier and Hadamard coefficients, and moment invariants among others.

In 1962, M. K. Hu [20] first exploited the technique of constructing feature vector using image moment invariants for visual pattern recognition. He derived the theory behind the moment invariants and provided some examples of using moment invariants for pattern recognition. In his work, Hu has developed the basic theorem that is the foundation of moment based pattern recognition scheme. Moments invariants capture global information about the image and do not require closed boundaries as required by boundary-based methods such as Fourier descriptors.

The moment invariants use the normalized central moments $\eta_{p,q}$:

$$\eta_{p,q} = \frac{\mu_{p,q}}{(\mu_{0,0})^{\gamma}}, \gamma = (p+q+2)/2 \tag{3.17}$$

Based on the definition of normalized central moments, Hu [20] introduced a set of *moment invariants* that are invariant to orthogonal transformations.

$$\begin{aligned}
\phi_1 &= \eta_{2,0} + \eta_{0,2} \\
\phi_2 &= (\eta_{2,0} - \eta_{0,2})^2 + 4\eta_{1,1}^2 \\
\phi_3 &= (\eta_{3,0} - 3\eta_{1,2})^2 + (\eta_{0,3} - 3\eta_{2,1})^2 \\
\phi_4 &= (\eta_{3,0} + \eta_{1,2})^2 + (\eta_{0,3} + \eta_{2,1})^2 \\
\phi_5 &= (\eta_{3,0} - 3\eta_{1,2})(\eta_{3,0} + \eta_{1,2})[(\eta_{3,0} + \eta_{1,2})^2 - 3(\eta_{2,1} + \eta_{0,3})^2] \\
&+ (\eta_{0,3} - 3\eta_{2,1})(\eta_{0,3} + \eta_{2,1})[(\eta_{0,3} + \eta_{2,1})^2 - 3(\eta_{1,2} + \eta_{3,0})^2] \\
\phi_6 &= (\eta_{2,0} - \eta_{0,2})(\eta_{3,0} + \eta_{1,2})[(\eta_{3,0} + \eta_{1,2})^2 - (\eta_{0,3} + \eta_{1,2})^2] \\
&+ 4\eta_{1,1}(\eta_{3,0} + \eta_{1,2})(\eta_{0,3} + \eta_{2,1}) \\
\phi_7 &= (3\eta_{2,1} - \eta_{0,3})(\eta_{3,0} + \eta_{1,2})[(\eta_{3,0} + \eta_{1,2})^2 - 3(\eta_{2,1} + \eta_{0,3})^2] \\
&+ (\eta_{3,0} - 3\eta_{2,1})(\eta_{2,1} + \eta_{0,3})[(\eta_{0,3} + \eta_{2,1})^2 - 3(\eta_{1,2} + \eta_{3,0})^2]
\end{aligned} \tag{3.18}$$

In addition, four other functions that are invariant under general affine transformation were also presented by Hu and revised by Flusser [14].

3.2.3 Zernike Moments

Zernike moments are a set of continuous image moments that consist of a set of complex polynomials that form a complete orthogonal set over the interior of the unit circle: $x^2 + y^2 \le 1$. Let the set of these polynomials be denoted by $\{V_{nm}(x, y)\}$. The form of these polynomials is:

$$V_{nm}(x,y) = V_{nm}(\rho,\theta) = R_{nm}(\rho)\exp(jm\theta) \tag{3.19}$$

where

n: Positive integer or zero

m: Positive and negative integers subject to constraints *n*-|*m*| even, $|m| \le n$

ρ Length of vector from origin to pixel *(x, y)*

θ: Angle between vector ρ and *x*-axis in counterclockwise direction.

$R_{nm}(\rho)$: Radial polynomial defined as:

$$R_{nm}(\rho) = \sum_{s=0}^{n-|m|/2} (-1)^s \frac{(n-s)!}{s!\left(\frac{n+|m|}{2}-s\right)!\left(\frac{n-|m|}{2}-s\right)!}\rho^{n-2s} \tag{3.20}$$

One can observe that $R_{n,-m}(\rho) = R_{nm}(\rho)$.

These polynomials are orthogonal and satisfy:

$$\iint_{x^2+y^2\le 1} [V_{nm}(x,y)]^* V_{pq}(x,y)\,dxdy = \frac{\pi}{n+1}\delta_{np}\delta_{mq} \tag{3.21}$$

with $\delta_{ab} = \begin{cases} 1 & a = b \\ 0 & otherwise \end{cases}$

Zernike moments are the projection of the image function onto these orthogonal basis functions. The Zernike moment of order n with repetition m for a continuous image function $f(x, y)$ that vanishes outside the unit circle is

$$A_{nm} = \frac{n+1}{\pi} \iint_{x^2+y^2 \le 1} f(x,y) V_{nm}^*(\rho,\theta) dxdy. \tag{3.22}$$

For a digital image, the integrals are replaced by summations:

$$A_{nm} = \frac{n+1}{\pi} \sum_x \sum_y f(x,y) V_{nm}^*(\rho,\theta), \qquad x^2 + y^2 \le 1 \tag{3.23}$$

Given the Zernike moments A_{nm} of $f(x, y)$ up to a given order n_{max} is known, then based on the orthogonality of the Zernike polynomial, the reconstructed image function $\hat{f}(x,y)$ can be obtained as

$$\hat{f}(x,y) = \frac{n+1}{\pi} \sum_{n=0}^{n_{\max}} \sum_m A_{nm} V_{nm}^*(\rho,\theta) \tag{3.24}$$

As $n_{\max} \to \infty$, $\hat{f}(x,y)$ will approach f(x, y).

If the image is rotated by an angle α to get the rotated image f^r, it follows that the relationship between the Zernike moments of the original image and the rotated image is given by [23]:

$$\begin{aligned} A_{nm}^r &= A_{nm} \exp(-jm\theta) \\ |A_{nm}^r| &= |A_{nm}| \end{aligned} \tag{3.25}$$

so that the Zernike moments merely acquire a phase shift in rotation and their amplitudes are rotation invariant. Image normalization prior to computing the Zernike moment can

further achieve rotation, translation and scale (RTS) invariance. First, the image centroid is obtained using the following equations:

$$X_{cent} = \frac{\sum_x \sum_y x * f(x,y)}{\sum_x \sum_y f(x,y)} \qquad y_{cent} = \frac{\sum_x \sum_y y * f(x,y)}{\sum_x \sum_y f(x,y)} \tag{3.26}$$

The image origin is then moved to the centroid coordinates and scaled to have the majority pixels within the unit circle to get the normalized image $f_{norm}(x, y)$:

$$f_{norm}(x,y) = f(\frac{x}{a} + x_{cent}, \frac{y}{a} + y_{cent}) \tag{3.27}$$

where $m_{00} = \sum_x \sum_y f(x,y)$ is the zero order geometric moment, β is a predefined value and $a = \sqrt{\beta / m_{00}}$. Only the pixels within the unit circle will be used in Zernike moments computation.

3.2.4 Radial Tchebichef Moments [37]

The Tchebichef moments of order $p+q$ of an image $I(\mathbf{x})$ of size N are defined using the scaled orthogonal Tchebichef polynomials $t_{p,N}()$, as follows:

$$T_{pq} = \frac{1}{n\rho(p,m)} \sum_{r=0}^{m-1} \sum_{k=0}^{n-1} t_{p,m}(r) e^{-j\frac{2q\pi k}{n}} f(r,\theta_k) \tag{3.28}$$

where $m=N/2$, $\theta_k = \frac{2\pi k}{n}, k = 0,1,2 \ldots n-1$, and n denotes the maximum number of pixels along the circumference of the circle in Fig.1; The polynomial functions are given by:

$$t_{p,m}(r)=\frac{p!}{m^p}\sum_{k=0}^{p}(-1)^{P-K}\binom{m-1-k}{p-k}\binom{p+k}{p}\binom{r}{k} \tag{3.29}$$

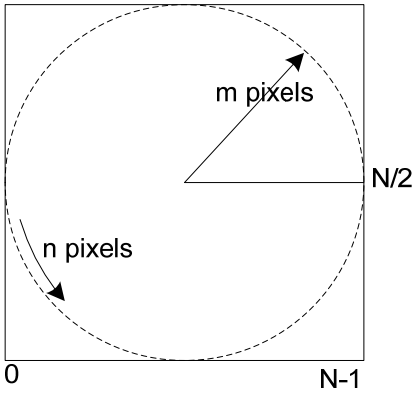


Figure 11. Domain of Definition of RTM Functions

The polynomials $t_{p,N}(r)$, satisfy the recurrence formula:

$$\begin{aligned}&(p+1)t_{p+1,m}(r)-(2p+1)(2r-m+1)t_{p,m}(r)\\&+p(1-p^2/N^2)t_{p-1,m}(r)=0\\&\qquad p=1,2\ldots m-2;\qquad r=0,1,\ldots m-1\end{aligned} \tag{3.30}$$

The squared norm $\rho(p,m)$ is equivalent to:

$$\rho(p,m)=\sum_{x=0}^{m-1}\left\{t_{p,m}(x)\right\}^2=\frac{m\left(1-\frac{1}{m^2}\right)\left(1-\frac{2^2}{m^2}\right)\ldots\left(1-\frac{p^2}{m^2}\right)}{2p+1} \tag{3.31}$$

where $p = 0, 1 \ldots m-1$

The squared norm can be further simplified by the following recurrence relation:

$$\rho(p,m) = \frac{(m^2 - p^2)}{m^2}\frac{(2p-1)}{(2p+1)}\rho(p-1,m) \tag{3.32}$$

One important property of the Tchebichef polynomial is its orthogonality in discrete domain:

$$\sum_{r=0}^{m-1} t_{p,m}(r)t_{q,m}(r) = \rho(p,m)\delta_{pq} \tag{3.33}$$

Like Zernike Moments, Radial Tchebichef Moments possess the rotational invariant property: if T'_{pq} is the RTM of rotating the original image by a degree of α, then T'_{pq} is related to the original moments T_{pq} via the following equation:

$$|T'_{pq}| = |T_{pq}\, exp(-q\alpha)| \tag{3.34}$$

The inverse moment transform can be used to reconstruct the image, which is given by the following equation:

$$\hat{f}(\mathbf{x}) \approx \sum_{p=0}^{p_{max}} \sum_{q=1}^{q_{max}} T_{pq} t_{p,m}(r) e^{jq\theta} \tag{3.35}$$

where p_{max}, q_{max} denote the maximum order and repletion of the moments used.

3.2.5 Orthogonal Fourier-Mellin Moments [48]

Orthogonal Fourier-Mellin Moments (OFMMs) was originally introduced by Sheng and Shen for invariant pattern recognition. The Fourier-Mellin moments (FMMs) are defined in the polar coordinate system (r,θ) as

$$M_{pq} = \int_0^{2\pi}\int_0^{\infty} r^p f(r,\theta) exp(-iq\theta) r dr d\theta \tag{3.36}$$

where $f(r, \theta)$ is the image function and is the q = 0, ±1, ±2, **...** is the circular harmonic order and the order of the Mellin radial transform is a positive integer $p \geq 0$. In the polar

system (r,θ), $r=\sqrt{x^2+y^2}$ is the distance from the origin to the pixel *(x, y)*, and $\theta=\arctan(y/x)$ is the angle between *r* and the *x* axis in the counter-clockwise direction.

The OFFMMs are based on a set of polynomials $Q_n(r)$ which are orthogonal over the range of $0 \le r \le 1$:

$$\int_0^1 Q_j(r)Q_k(r)rdr = a_j\delta_{jk} \tag{3.37}$$

where δ_{jk} is the Kronecker delta function, $a_p = \frac{1}{2(p+1)}$ is a normalization constant and r=1 is the maximum size of the objects that can be encountered in the application.

Here $Q_p(r)$ is a polynomial in *r* of degree *p* defined as [48]:

$$Q_p(r) = \sum_{k=0}^{p} \alpha_{pk} r^{k-1/2} \tag{3.38}$$

with

$$\alpha_{pk} = (-1)^{p+k} \frac{(p+k+1)!}{(p-k)!\,k!\,(k+1)!} \tag{3.39}$$

It can be verified that the set $Q_p(r)$ defined above is orthogonal over the range $0 \le r \le 1$. The explicit expressions of $Q_p(r)$ are:

$$\begin{aligned}
&Q_0(r) = r^{-1/2} \\
&Q_1(r) = -(2-3r)r^{-1/2} \\
&Q_2(r) = (-3-12+10r^2)r^{-1/2} \\
&Q_3(r) = -(4-30r+60r^2-35r^3)r^{-1/2} \\
&Q_4(r) = (-5-60r+210r^2-280r^3+126r^4)r^{-1/2} \\
&\ldots
\end{aligned} \tag{3.40}$$

The *(p, q)* order OFMM basis function $U_{pq}(x,y)$ and the OFMM moments Φ_{pq} can be defined in the polar coordinate system (r,θ) as

$$U_{pq}(x,y) = Q_p(r)e^{-iq\theta}, \qquad r = \sqrt{x^2 + y^2} \le 1 \tag{3.41}$$

$$\Phi_{pq} = \frac{1}{2\pi a_p}\int_0^{2\pi}\int_0^1 f(r,\theta)U_{pq}(r,\theta)rdrd\theta \tag{3.42}$$

where the normalization factor $a_p=1/[2(n+1)]$.

It follows from the definition of the polynomial set $Q_p(r)$ that the basis functions $U_{pq}(x,y)$ of the OFMMs are orthogonal over the interior of the unit circle. The discrete version of OFMMs can be expressed in rectangular coordinates *(x, y)* as

$$\Phi_{pq} = \frac{p+1}{\pi}\sum_x \sum_y f(x,y)Q_p(r)e^{-iq\theta}\Delta x\Delta y \tag{3.43}$$

where $x^2 + y^2 \le 1, \quad x = r\cos\theta, \quad y = r\sin\theta$

The OFMMs are integrable when the degree p of $Q_p(r)$ is $p \ge 0$. By substituting equation (3.31) and equation (3.37) into equation (3.38), the OFMMs, when $0 \le r \le 1$, can be expressed as linear combinations of FMMs as

$$\Phi_{pq} = \frac{p+1}{\pi}\sum_{k=0}^{p}\alpha_{pk}M_{kq} \tag{3.44}$$

If the image is rotated through angle ϕ, then the relationship between M'_{kq} and M_{kq} is

$$M'_{pq} = M_{pq}e^{-iq\phi} \tag{3.45}$$

so that

$$\Phi'_{pq} = \Phi_{pq} e^{-iq\phi}$$
$$\left|\Phi'_{pq}\right| = \left|\Phi_{pq}\right| \tag{3.46}$$

Therefore, the OFMM $|\Phi_{pq}|$, has the property that the magnitude is rotation-invariant – the same property that the ZM and RTM possess.

It has also been shown that both the polynomial set $R_p(r)$, used in ZM transform and the polynomial set $Q_p(r)$, used in OFMM belong to the family of Jacobi polynomial, and there exists an relationship between the two given by [48]:

$$rQ_p(r^2) = R_{2p+1}(r) \tag{3.47}$$

Like RTM and ZM, the summation over a finite number of OFMM can give a reconstructed image.

$$\hat{f}(r,\theta) = \sum_{p=0}^{P} \sum_{q=-Q}^{Q} \Phi_{pq} Q_p(r) exp(jq\theta) \tag{3.48}$$

Because of the orthogonality of the set $Q_p(r)exp(jq\theta)$, each OFMM makes an independent contribution to the reconstructed image. The sum given by equation (3.43) converges to the original image when the original image is nonzero in a limited region, is continuous or has a finite number of finite jumps so that $f(r,\theta)$ and $|f(r,\theta)|^2$ are integrable and the polynomials $Q_p(r)$ are bounded over (0,1) as p becomes infinite.

The structure of the OFMM has provided a recipe to construct other image moment functions that can be used for image recognition applications. Ping, Wu and Sheng presented Chebyshev–Fourier Moments (CHFM) in [42]; Ren, Ping, Bo, Wu and Sheng presented Radial Harmonic Fourier Moments (RHFM) in [43].

In CHFM, while maintaining the same structure as the OFMM, the radial polynomial $Q_p(r)$ is replaced by the radial polynomial based on shifted Chebyshev polynomial of the second kind. The shifted Chebyshev polynomial of the second kind is

$$U_n^* = U_n(2r-1) \tag{3.49}$$

where the function $U_n(2r-1)$ is the Chebyshev polynomial of the second defined as

$$U_n(\cos\theta) = \frac{\sin(n+1)\theta}{\sin\theta} \tag{3.50}$$

With some variable substitution, we can express $U_n^*(r)$ as

$$U_n^*(r) = U_n(\cos\theta) = \sum_{k=0}^{(n+2)/2}(-1)^k \frac{(n-k)!}{k!(n-2k)!}[2(2r-1)]^{n-2k} \tag{3.51}$$

It can be shown that $U_n^*(r)$ satisfy the following relationship:

$$\int_0^1 U_n^*(r)U_k^*(r)w(r)dr = \frac{\pi}{8}\delta_{nk}$$
$$w(r) = (r-r^2)^{1/2} \tag{3.52}$$

Therefore the following polynomial $R_n(r)$

$$R_n(r) = \sqrt{\frac{8}{\pi}}\left(\frac{1-r}{r}\right)^{\frac{1}{4}} \sum_{k=0}^{(n+2)/2}(-1)^k \frac{(n-k)!}{k!(n-2k)!}[2(2r-1)]^{n-2k} \tag{3.53}$$

satisfies the orthogonal relation

$$\int_0^1 R_n(r)R_k(r)rdr = \delta_{nk} \tag{3.54}$$

Therefore in CHFM, the basis function is $C_{pq}(r,\theta) = R_p(r)\exp(-jq\theta)$, and CHFM has similar form as OFMM

$$\Psi_{pq} = \int_0^{2\pi}\int_0^1 C_{pq}(r,\theta) f(r,\theta) r dr = \int_0^{2\pi}\int_0^1 R_p(r) \exp(jq\theta) f(r,\theta) r dr \tag{3.55}$$

The image can be reconstructed from Ψ_{pq}

$$\hat{f}(r,\theta) = \sum_{p=0}^{P}\sum_{q=-Q}^{Q} \Psi_{pq} R_p(r) \exp(jq\theta) \tag{3.56}$$

With Radial Harmonic Fourier Moments (RHFM), again the difference is the radial polynomial used in the basis function. Here the following harmonic polynomial is used:

$$T_p(r)\begin{cases} \dfrac{1}{\sqrt{r}} & if\ \ p = 0 \\ \sqrt{\dfrac{2}{r}}\sin[(\ p+1)\pi r\ ] & if\ \ p\ is\ odd \\ \sqrt{\dfrac{2}{r}}\cos(\ p\pi r\ ) & if\ \ p\ is\ even \end{cases} \tag{3.57}$$

Then the basis function is $P_{pq}(r,\theta) = T_p(r)\exp(-jq\theta)$ and RHFM is defined as:

$$\Pi_{pq} = \int_0^{2\pi}\int_0^1 P_{pq}(r,\theta) f(r,\theta) r dr = \int_0^{2\pi}\int_0^1 T_p(r) \exp(jq\theta) f(r,\theta) r dr \tag{3.58}$$

While each of these variations provides certain unique property, because of their similarity, they will be grouped together with OFMM in future discussions.

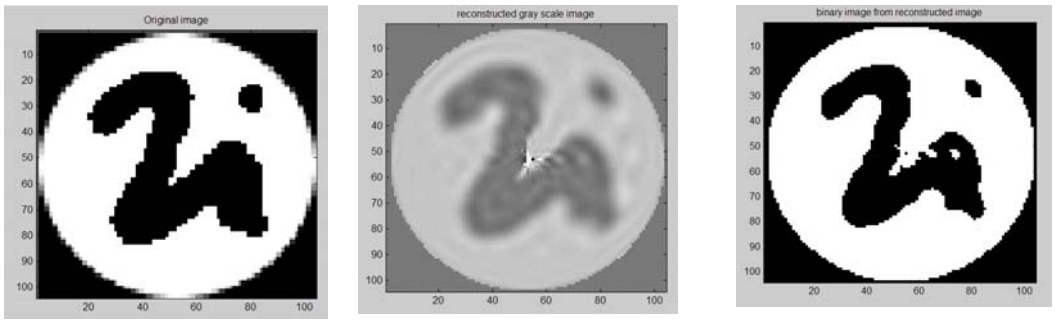
3.3 Image Reconstruction Performance

All three types of image moments discussed above: ZM, RTM, and OFMM are orthogonal moments defined inside the unit circle. If we reconstruct the image using a subset of independent moments, the reconstruction error is reduced with the use of additional higher order moments.

One issue in the use of OFMM and its variants that determines their effectiveness in reconstruction is that the polynomial has a factor of $r^{-1/2}$, therefore, near the origin, the polynomial approaches infinity. If an image has edges near the origin, the high order

moments are non-zero there. This will create signification visual distortion in the form of a spot of saturated peak white appearing around the origin due to the nature of the underlying polynomial. The distortion would be most obvious when the brightness near the origin is at the mid level intensity instead of peak white.

For binary images in many pattern recognition applications such as optical character recognition (OCR), this is not a major issue, since the process of converting the reconstructed image to binary image will actually help to mitigate the distortion.



(a) Original Image (b) Reconstructed Grey Scale (c) Binary Image from Reconstructed

Figure 12. Image Reconstruction using OFMM

In our application, the watermark is defined only over a small region, since the moments will be used to reconstruct small image patches; it is useful to analyze the performance of image reconstruction for each of the three different moments. To conduct such evaluation of ZM and RTM, images of different size are reconstructed with ZM and RTM with different order of moments. The resulting reconstruction error is compared in the table below.

As shown in the tables below, both types of moments show improved reconstruction results with increased moment order; Both techniques show better reconstruction results for lager images; RTM shows lower reconstruction using the same order of moments. Most importantly RTM shows better reconstruction results for smaller image compared to ZM.

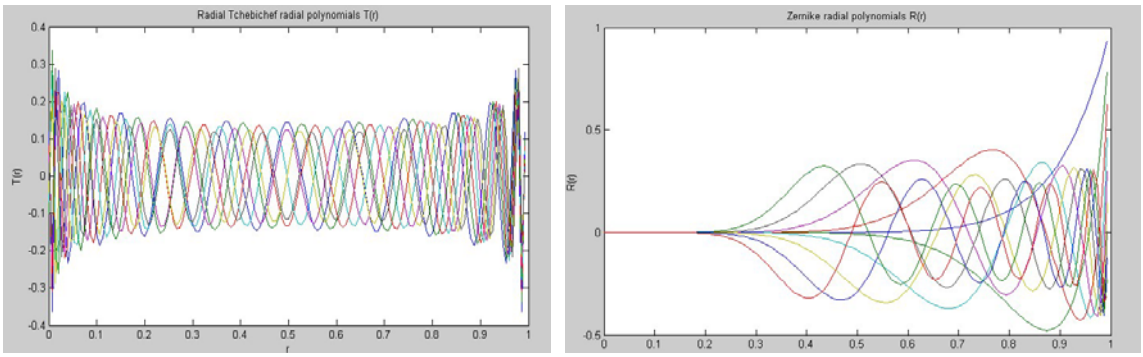
Table 1. ZM and RTM Reconstruction Errors

ZM Reconstruction Error					
Max Moments / Image Size	10	15	20	25	30
40x40	21.00%	12.50%	9.60%	8.30%	6.90%
60x60	17.60%	9.10%	6.20%	4.90%	3.70%
100x100	19.00%	10.20%	7.30%	6.00%	4.70%

RTM Reconstruction Error					
Max Moments / Image Size	10	15	20	25	30
40x40	10.20%	7.00%	5.50%	4.60%	4.00%
60x60	7.20%	4.60%	3.40%	2.90%	2.80%
100x100	7.80%	4.80%	3.40%	2.60%	2.10%

With regard to the explanation for the better performance of RTM for smaller images, we follow the same reasoning in [48] that Sheng used to compare OFMM and ZM performance. Sheng has noted that the ability of an image moments reconstruction to describe high spatial-frequency components of the image can be evaluated by the distribution of the zeros in the polynomials used in that type of moments, since the distributions of the zeros corresponds approximately to the sampling frequency and sampling positions in the image. Therefore we will compare the number and position of

the zeros of the Zernike polynomials and Radial Tchebichef polynomials. As shown in Figure 13(a) Tchebichef polynomial zeros are uniformly distributed in $0<r<1$; Zernike polynomial zeros, as shown in Figure 13(b), are concentrated in the region of large r, the first zero is close to r = .05. This explains the reason that RTM performs better in reconstructing smaller images.



(a) (b)

Figure 13. (a) Tchebichef Polynomial $T_{p,10}(r)$ of Radial Tchebichef Moments; (b) Zernike Polynomial $R_{p,10}(r)$ of Zernike Moments

In this research the watermark embedding media will be grey scale images. Due to the deficiency in handling grey scale images pointed out above, in our investigations, we exclude OFMM from the list of potential candidates for use with our watermarking algorithm. In Chapter 4, we present the image watermarking works utilizing ZM, RTM and Moment Invariants.

Chapter 4

INVARIANT DOMAIN IMAGE WATERMARKING

As noted in Chapter 3, there are some intrinsic limitations of embedding watermarks in traditional transform domains such as DCT/DFT or wavelet domain. We have also noted that orthogonal moments have been used in pattern recognition applications because of their invariance properties. These properties also make the use of such moments attractive in watermarking applications because of their robustness in the face of attacks. We will discuss the implementation of a moment-based watermarking scheme in the next chapter. However, before discussing invariant domain image watermarking, we provide an overview of the possible attacks against digital watermarking and some deficiencies of the techniques describe d in Chapter 3.

4.1 Watermark Attacks

In order to understand the shortcomings of the common attack strategies that can be found in the field. In general, a watermarking scheme will face the following different kind of attacks:

- Attacker knows nothing about the watermark or media content – *uses general weakness of watermarking schemes*
- Attacker has more than one watermarked work – *different host data watermarked with the same watermark* or *same host data watermarked with different watermarks*

- Attacker knows the algorithm – *exploits specific weakness of the algorithm*

Since often the attacker can get a general idea of what types of algorithm are used to embed the watermark, the last scenario is the most common scenario. Therefore, the success or robustness of an algorithm can be achieved not only by the secrecy of the algorithm itself, but by incorporating the ability to withstand the attacks within the internal mechanism of the algorithm, for example, by using longer P-N sequences to avoid descrambling the watermark. . In addition, it should be understood that for different types of watermarks, the goals of the attackers are different. For example, for robust watermark, the attackers want to make the detector unable to detect the watermark while maintaining the perceptual quality of the object under attack. For fragile watermarks, since the watermark is supposed to be destroyed if the object is tampered, the goal of the attackers is to make the watermark still valid after tampering. Since the algorithm developed in this research is oriented toward robust watermarking techniques, we will concentrate on attacks that target robust watermarking schemes.

Some of the main attack families we are concerned with are targeting robust blind watermarking, especially under the condition that the attackers have general information about the watermarking algorithm (i.e., embedding in *n* highest amplitudes of the DCT coefficients etc.). However, we can assume that the attackers do not have access to the actual detector. Generally, in this situation, the attack families can be classified into the following categories [39]:

Geometric attacks – through geometric transformation to disrupt or even destroy the synchronization between the source and receiver. As a result, even though the watermark still exists in the material, the receiver has no way of detecting its existence. This type of

attack in general would not severely degrade the quality of the original objects, and it is fairly straightforward and simple to implement. For this reason, geometric attacks are the first and foremost concern for a given watermarking algorithm. This is especially true for any watermarking techniques that embed watermarks in the transform domain since an image transform is defined on a local image coordinate system. As such, disturbance to the underlying coordinate system will drastically alter the transform coefficients. *Affine transforms* is one type of geometric attack. Among them, image rotation is the most common attack operation, since rotation will not in general degrade the image quality. Scaling is another affine transform operation that can be used to attack watermarking. Obviously, use of an arbitrary affine transform would be the ultimate way to alter the image; however such operations may end up in introducing too much distortion in the image.

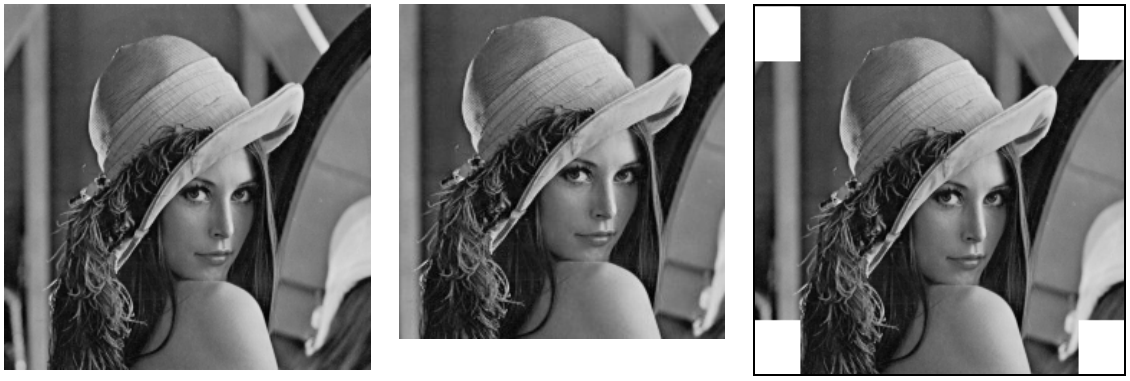


(a) (b) (c)

Figure 14. Examples of Different Affine Transform Operations
(a) Original image; (b) Scaling (c) Rotation.

Another geometric operation that is commonly used to attack watermarking is image cropping/padding. When it is done properly (i.e. not cropping out essential information in the image), then there will be no significant degradation to the quality of the image. Cropping/padding could be done in more than one form. While, in general, it is done parallel to the sides of the image, other types of cropping/padding are also being used. In Figure 15 below, the cropping operation cropped all sides of the image in (b), while in (c), only a small section on each corner is cropped.

In any video scene or an image, the central portion of the image is the portion that is used to tell the story; therefore, it is conceivable to design other types of cropping masks to alter the images without degradation.



(a) (b) (c)

Figure 15. Examples of Different Cropping Operations (a) Original Image; (b) Cropping on All Sides (c) Cropping out Four Corners.

The operations described above can be used in different combinations, which are summarized in Table 2 below. The combinations that involve cropping are specifically

highlighted because of their impact on the image moments based watermarking techniques to be discussed later.

Table 2. Different Combinations of Geometric Operations for Attacks

(CP: Cropping/Padding, R: Rotation, S: Scaling)

CP	R	S
CP →S	R→S	S→R
CP →R	R→ CP	S→ CP
CP →S→R	R→S→ CP	S→R→ CP
CP →R→S	R→ CP →S	S→ CP →R

Image processing de-synchronization is another type of operation that can be used in attacking watermarks. Since the invisible watermarks are able to be embedded without degrading the images, there must be redundancy in the original images. Therefore, image processing that identifies and alters such redundancy can be used to disrupt the embedded watermarks.

One operation is to recognize that the low bits of any image data turn to be quantization noise, therefore dithering or randomization of LSB bits of the image data would not significantly impact the visual effect of an image.

As shown in the figure 16 below, replacing the lower 2 bits of the image with artificially created random noise does not cause much degradation to the image. Another similar type of processing is to add noise to the image, again with the hope that the addition of noise will interrupt the watermarking patterns in the LSB bits of the image.

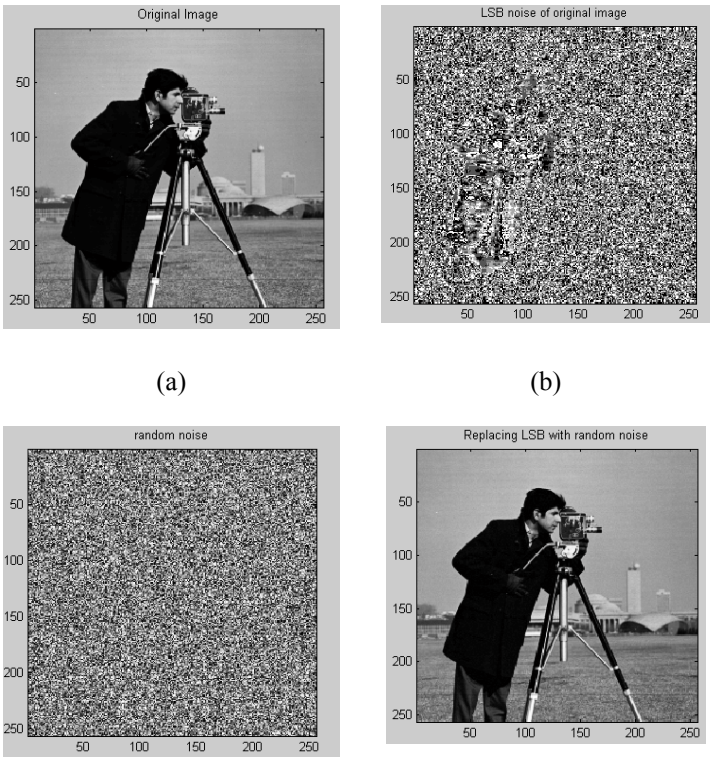
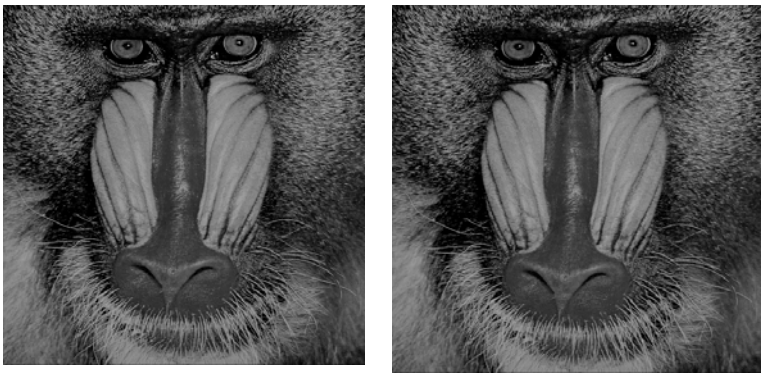


Figure 16. Image LSB Randomization (a) Original Image; (b) Image LSB Data (c) Random Noise (d) New Image with Random Noise as LSB.

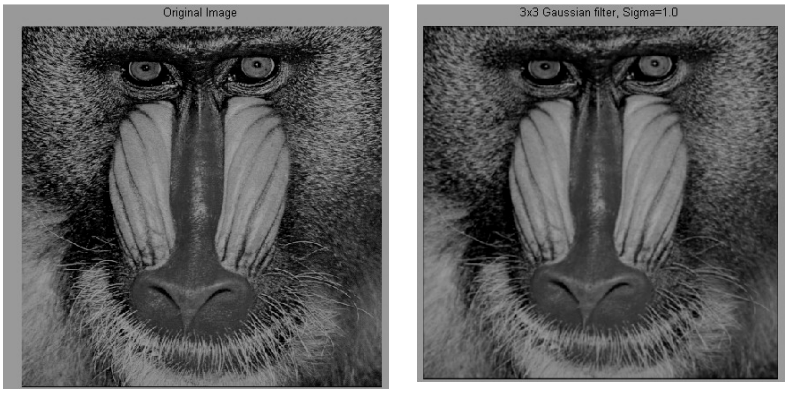
Based on the same foundation of image data redundancy, lossy compression can be another technique that can disrupt the watermarks. Such an operation can be malicious attack, or it could be simply to meet bandwidth limitation requirements. One example is the 512x512 Baboon image which takes 768KB when stored in BMP format and when stored in JPEG format with 90% quality factor, it takes only 131KB, with no obvious different in image visualization. However, with the reduction of the redundancy, watermarks detection can be disrupted.



(a) (b)

Figure 17. (a) Original Image in Window BMP Format (b) JPEG Compression with 90% Quality Factor

Another potential processing is lowpass filtering or blurring/denoising by reducing the high frequency content of the image. Again, such operation removes redundancy in high frequency component, but has limited impact on image quality.



(a) (b)

Figure 18. Lowpass filtering (a) Original (b) Gaussian Lowpass filter (σ=1.0)

Transcoding is another operation on images and video. Technically this is also a type of image processing, however, since it could be a combination of image processing (filtering, compression) and geometric transformation (scaling), we would list it as a separate category. Image and Video transcoding is very common. While JPEG is the most popular image format – almost all digital cameras use JPEG formats to store image, TIFF, PNG, BMP and JPEG2000 are all widely used image formats. Since these formats all involve certain image transforms such as the DCT or wavelet transform, during the process of the transcoding, the transform coefficients tend to be altered by the different formats.

All three categories of attacks listed above are trying to achieve one or both of the following goals: (i) Generating de-synchronization between source and receiver or (ii) Removing the watermarks.

4.2 Common Deficiency

For the watermarking techniques described in Chapter 3, geometric attacks turn to be a serious threat. One common assumption in embedding watermark in spatial, and DCT and wavelet transform domains is their dependency on the underlying pixel coordinates. This dependency reduces their effectiveness when the image undergone affine transform.

In the spread spectrum spatial technique, the watermark is a PN sequence embedded in the line scanning mode in spatial domain. Image rotation may skew the order the embedded sequence and result in the failure of the cross correlation with the original PN sequence.

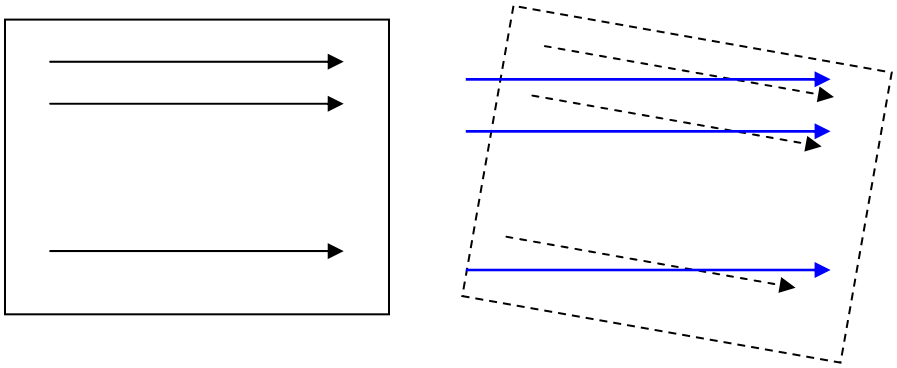


Figure 19. Rotation Disrupts Line Scanning of PN Sequence

Scaling could also pose similar problems for spread spectrum spatial domain watermarking techniques, since the alignment of the embedded PN-sequence will be interrupted. This effect will be aggravated if the scaling is combined with a cropping operation. For example, if a 4 bit sequence [1 -1 1 -1] is embedded into the image using one pixel per bit, when the image is scaled to 200% the original size, conceivably the sequence will now become a 8 bit sequence [1 1 -1 -1 1 1 -1 -1], therefore, the cross-correlation between the original sequence and the sequence in the scaled image will not yield the expected signature of a typical autocorrelation function. On the other hand, the same problem exists if multiple pixels are used to embed one bit in the sequence. For example, if 2 pixels are used to embed one bit, then a down-scaling the image to 50% will resulted in half of the embedded sequence missing.

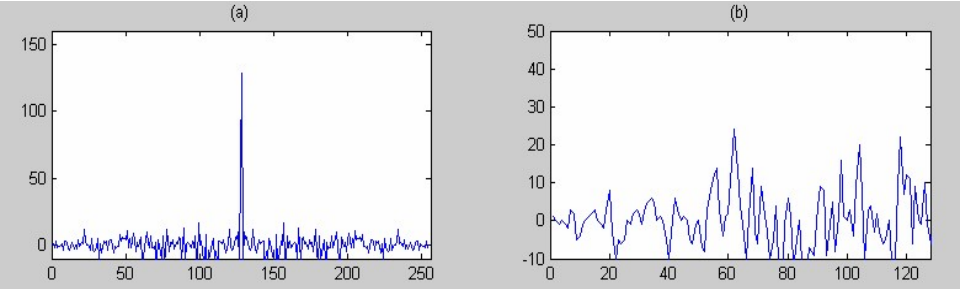


Figure 20. Scaling Disrupts Watermark Synchronization (a) Autocorrelation (b) Cross-Correlation with Sequence Sub-sampled by Half.

Another effective attack against the spatial watermarking techniques is image processing operations such as the randomization of image LSB or image lowpass filtering or smoothing. Such operations essentially can remove the watermarks embedded in the image LSB.

While image processing attacks in general are not very effective against transform domain watermarking methods, geometric transform does pose serious challenges for these techniques. One key common technique used in DFT/DCT based watermarking algorithms is that the DFT/DCT is applied in blocks of 8 × 8 pixels. This will allow the algorithm to take advantage of some of the common processing infrastructure that already requires DFT/DCT transformation, such as JPEG compression etc. However, such a technique intrinsically lends itself to image scaling attacks or image cropping attacks. With scaling, a 8x8 block in the new image occupies different size on the image; with cropping the alignment of the block grids is lost.

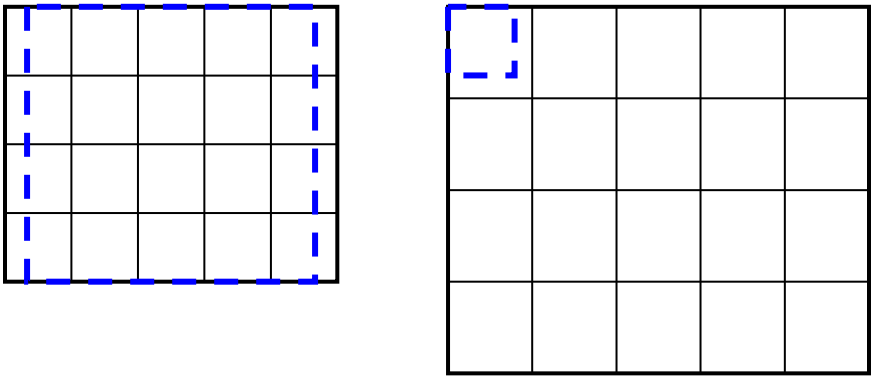


Figure 21. Cropping and Scaling Disrupts the Synchronizations of DFT/DCT block Boundary (a) Cropping (b) Scaling.

One way to avoid such attacks is to forego the processing speed advantage associated with the block based DFT/DCT processing and instead, perform DFT/DCT over the whole image. In this case, scaling or (non-excessive) cropping will have less impact on the resulted DCT coefficients, resulting in turn to less disruption to the embedded watermarks.

However, on the other hand, rotation remains an intrinsic threat to DFT, DCT and wavelet domain watermarking techniques. Such limitation is related to the definition of both transforms.

It is well known that DFT based transformation has the property of rotation invariants:

$$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \theta + \theta_0) \tag{4.59}$$

where f is the spatial domain image and F is its DFT coefficients. This indicates that a rotation in spatial domain will result in same phase shift in the DFT coefficients without amplitudes change. However, one catch in this property is that the rotation has to be done around the same origin in the original image to maintain such rotation invariant property.

Therefore, if an image has first undergone cropping then followed by rotation, and then the above property will not hold.

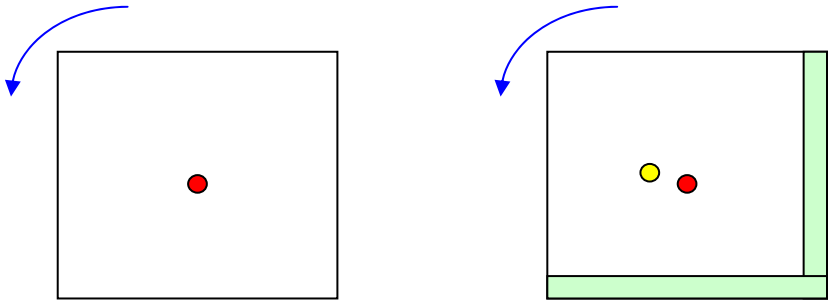


Figure 22. DFT Rotation Invariant Disrupted after Cropping (a) Rotation against Same Origin (red) (b) Rotation against New Origin (Yellow)

In addition to the above shortcomings for image watermarks in spatial or transform domain, the issues with transcoding type attacks pose serious problems for video watermarking. There are a variety of video coding standard each with different compression/processing schemes. Therefore any watermarking algorithm that is tied to a certain coding standard is very likely to be removed either maliciously or unintentionally through the transcoding process.

It is in this background, that more attention has been drawn to image moments based watermarking techniques.

4.3 Image Moment Based Watermarking

Interest in the image moments is because of their invariance to certain types of distortion. These techniques are mostly inspired by the successful use of these moments

in the area of pattern recognition. In the following sections, several existing invariant domain watermarking techniques will be briefly reviewed.

4.3.1 Moment Preserving based Copyright Protection

Moment Preserving (MP) algorithm is a non-parametric algorithm to convert gray scale images into binary images [29].

Given a gray-scale image *f* which contains *N* pixels, let the pixel value at coordinate *(x, y)* be *f(x, y)*. Let z_0 and z_1 be the two gray levels corresponding to 0 and 1 in the binary image respectively: $z0 < z1$. The first three moments are given by:

$$\begin{aligned} 1^{st}\ moment: &\quad m_1 = \frac{1}{n}\sum_x\sum_y f(x,y) = z_0 P + z_1(1-P) \\ 2^{nd}\ moment: &\quad m_2 = \frac{1}{n}\sum_x\sum_y f^2(x,y) = {z_0}^2 P + {z_1}^2(1-P) \\ 3^{rd}\ moment: &\quad m_3 = \frac{1}{n}\sum_x\sum_y f^3(x,y) = {z_0}^3 P + {z_1}^3(1-P) \end{aligned} \tag{4.60}$$

These three equations can be solved for the three unknown parameters z_0, z_1 and *P*. z_0 and z_1 are used to preserve the moments of the input data – since if z_0 and z_1 satisfies Eq. (4.2), then the resulting binary image will have the same first, second and third moments as the original image; *P* is used to find the threshold *T* such that: $\sum_{i=0}^{T} p(i) = P$, where

$$p(i) = \frac{Frequencies \quad of \quad gray \quad level \quad i}{n} \tag{4.61}$$

A copyright ownership-proving algorithm using moment preserving was reported by Chang, Hwang and Lin [29]. The MP technique is used to retrieve the image characteristics from the original image. The retrieved feature is then employed to

generate a secret key. An author who creates the original image can hold the secret key to compute his/her watermark. In other words, the secret key is used to prove the copyright ownership of that image. The secret key contains two parts: K_s and K_m. K_s is the seed of a pseudo-random number generator, and K_m is the key generated using the following algorithm based on MP technique. Given an N x N gray scale image $O=\{O_{i,j}| \ 1 \le i,j \le N\}$ and an M x M watermark $W = \{W_{i,j}| \ 1 \le i,j \le M\}$, $W_{i,j} \in \{0,1\}\}$. Define a block B with the same dimension M, as the watermark and divide the block into sub-blacks $B_{i,j}$ each of size n x n. Then $B = \{B_{i,j}| \ 1 \le i,j \le M\}$, where

$$B_{i,j} = \begin{bmatrix} O_{k,l} & O_{k,l+1} & \cdots & O_{k,l+n-1} \\ O_{k+1,l} & O_{k+1,l+1} & \cdots & O_{k+1,l+n-1} \\ \cdots & \cdots & \cdots & \cdots \\ O_{k+n-1,l} & \cdots & \cdots & O_{k+n-1,l+n-1} \end{bmatrix} \tag{4.62}$$

The starting (k, l) for each block $B_{i,j}$ is determined by the random number generator and the pre-defined seed K_s.

Next, the MP technique is performed on B. The intensity of two levels $h_{i,j}^{1}$ and $h_{i,j}^{2}$ are obtained using MP technique outlined above. Let $d_{i,j} = | h_{i,j}^{1} - h_{i,j}^{2}|$ and Avg be the average of $d_{i,j}$, for i,j=1,2…M. A temporary key $T = \{T_{i,j}| \ 1 \le i,j \le M\}$ is formed as

$$T_{i,j} = \begin{cases} 1, & d_{i,j} \ge Avg \\ 0, & otherwise \end{cases} \tag{4.63}$$

Finally, the MP dependent secrete key K_m is generated by the following exclusive OR (XOR) operation:

$$K_m = T \oplus W. \tag{4.64}$$

If we are required to authenticate an image O', we perform another simple XOR operation to extract the watermark

$$W' = T \oplus K_m. \tag{4.65}$$

where T is computed from image O'. Given the same K_s and K_m, then if the new watermark W' matches the original watermark W, O' is authenticated; conversely, if W' does not match W, the image is not authenticated.

In the strict sense, this application is more of a steganography application than watermarking one, since the image content was not altered to embed any information. Instead a secrete key that is dependent on the image content was generated based on the predefined bit sequence referred to by the author as watermark.

4.3.2 Image Watermarking by Moment Invariants

In [4] M. Alghoniemy and A. Tewfik presented a watermarking technique based on moment invariants. The motivation for their algorithm is to improve the robustness of image watermarking in the presence of image geometric manipulation/attack utilizing some results/success from the pattern recognition field.

The idea behind the moment invariants based watermark design is to modify the image in a way such that a predefined function of the moment invariants lies within a predetermined value ν. Let Φ be the vector whose elements are the moment invariants ϕ_1 *to* ϕ_7 of an image.

The image is declared watermarked if:

$$\nu - \varepsilon \le f(\Phi) \le \nu + \varepsilon \tag{4.66}$$

where $f(\Phi)$ is a predefined function of Φ and ε is a small tolerance determined experimentally. This can be achieved by finding perturbation ΔI added to the original image I such that $\widetilde{I} = I + \Delta I$ satisfies the above inequality. The choice of ΔI suggested is $\Delta I = \beta \log(I)$, where β is a weighting factor. The choice of β can be determined by exhaustive search around $f(\Phi)$ of the original image. This choice of ΔI has the advantage of transparency – since only a nonlinear transformation of the original image is added to itself, the image transparency is guaranteed up to a contrast variation.

To authenticate a received image I_r, its moment invariants function $f(\Phi | I_r)$ is computed. Let the error E be defined as $E = \left| f(\Phi \mid I_r) - \nu \right|$, then the received image I_r is declared authentic if $E \le \varepsilon$.

One requirement of this algorithm is that the predetermined MI value ν will need to be provided externally, furthermore, ν will have different values for different image. Therefore, this technique belongs to the non-blind watermarking category.

4.3.3 Zernike Moment Based Image Watermarking

Geometric moments represent a projection of the image function $f(x, y)$ onto the space of monomials $x^p y^q$. Unfortunately, this basis set is not orthogonal. Consequently, there is a great deal of redundancy in the contents of the moments m_{pq}. Teague has suggested the use of orthogonal moments based on orthogonal polynomials such as the Zernike or Legendre polynomials to overcome the problems associated with the

geometric moments. Zernike moments have been used in a wide variety of applications such as character recognition [23].

Farzam and Shirani [49] have proposed embedding watermarks in the Zernike moments domain to utilize the invariant geometric characteristics of the Zernike moments. The input image is divided into sub-images which are selected as concentric rings in order to preserve the rotation-invariant property of the Zernike moments. The Zernike transforms of each of the sub-images is computed. Based on the watermark signal, the Zernike moments of the sub-image are modified. The modified sub-images are inverse Zernike transformed to obtain the watermarked image. The watermark signal is embedded by modulating the amplitude of the Zernike moments of the input image as follow:

$$A_j^{'} = A_j + B(j)r \qquad \forall j \in W \tag{4.67}$$

where *B(j)* is the *j*th bit of the watermark signal, A_j is a vector consisting of the magnitudes of all the Zernike moments (up to N_{max}) belonging to the jth sub-image, $A_j^{'}$ is the vector of modified magnitude of the moment of the same sub-image. *r* is a vector of random numbers and *W* is the watermark set. In order to maintain the quality of the reconstructed sub-image, a value of N_{max} =30 has been suggested in [49].

The detection process consists of calculating the normalized correlation of the magnitude of the Zernike moments of the received image with a random vector and comparing the result with a threshold (*T*):

$$\frac{|A_j^{'}|.r}{\|A_j^{'}\|^2} \underset{0}{\overset{1}{\gtrless}} T \tag{4.68}$$

The original image is not required at the receiver for the detection.

Recently, Kim and Lee [24] have provided another image watermarking algorithm in Zernike domain. Similar to Farzam and Shirani's method in [49], the magnitudes of Zernike moments is chosen for watermark embedding since they are RST invariant:

$$A'_{n,m} = A_{n,m} + \alpha \tag{4.69}$$

where $A_{n,m}$ is the moment chosen in which to embed watermark, α is the embedded watermark strength. In [49], inserting the watermark requires determining high-order Zernike moments (N_{max}=30), which is very computationally demanding. With multiple sub-images, such a computational load can be prohibitively high. To avoid this problem, Kim and Lee first consider the relation between one single Zernike moment and its equivalent spatial pattern. For an image consisting of one non-zero Zernike moment pair ($A_{k,,-l}$, $A_{k,l}$):

$$\{A'_{n,m}\} = \begin{cases} \alpha & n = k, m = -l, l \\ 0 & otherwise \end{cases} \tag{4.70}$$

the corresponding spatial pattern $\hat{g}(x, y)$ is:

$$\begin{aligned} \hat{g}(x,y) &= \sum_{n=0}^{N_{\max}} \sum_{m} A'_{n,m} V_{n,m}(\rho,\theta) \\ &= A_{0,-N_{\max}} V_{0,-N_{\max}} + A_{0,-N_{\max}+1} V_{0,-N_{\max}+1} + \ldots \\ &\quad + A_{k,-l} V_{k,-l} + A_{k,l} V_{k,l} + \ldots \\ &\quad + A_{N_{\max},N_{\max}} V_{N_{\max},N_{\max}} \\ &= \alpha (V_{k,l} + V_{k,-l}) \end{aligned} \tag{4.71}$$

Since $|V_{k,l}|$, $|V_{k,-l}| \leq 1$, it can be easily derived that $|\hat{g}(x, y)| \leq 2\alpha$. Therefore, the watermark can be inserted into the spatial pattern for Zernike moment (k,l) with intensity α; and by adjusting α, the maximum pixel distortion in the image can be controlled.

Because of the orthogonality of the Zernike polynomials, modification of one Zernike moment $A_{n,m}$ will not impact another moment $A_{n',m'}$, therefore once the watermark vector is determined, the summation of the spatial patterns of all these moments is the final watermark pattern in the spatial domain.

An iterative feature modification and verification procedure method is used to determine the proper watermarking intensity level α. Two noise levels *r1* and *r2* are used in the iterative process. *r1* is chosen as

$$r1 = \max\{s(\vec{p}_i, \vec{q}_i) \mid i = 0,...,M\} \tag{4.72}$$

where $\vec{p}_i$ is the feature vector of the original image; $\vec{q}_i$ is the feature vector of the image after RTS distortions. That is, the similarity *s* is measured between the original image and the image after it has undergone different types of transforms/attacks (rotation, scale, random geometric attack and JPEG compression etc.). The maximum value of *s* is chosen as *r1*. Similarly *r2* is defined as

$$r2 = \min\{s(\vec{p}_i, \vec{p}_j) \mid i = 0,...,M\} \tag{4.73}$$

where $\vec{p}_i$ and $\vec{p}_j$ are the feature vectors of two different unmarked test images. Thus the smallest value of the similarity *s* is chosen for *r2*. During watermark insertion, the watermark intensity α is adjusted iteratively to satisfy a pre-defined PSNR level and also to satisfy the inequality

$$r1 \le s(\vec{p}, \vec{p}^{\,*}) \le r2 \tag{4.74}$$

where $\vec{p}$ is the feature vector of the original image; $\vec{p}^{\,*}$ is the feature vector at the receiver. See Figure 23.

At the detector, the Zernike feature vector is estimated from the test image and the similarity between this and the watermark vector is computed as

$$s(\vec{P},\vec{w}) = (\frac{1}{N}\sum_{i=1}^{N}[p_i - w_i]^2)^{\frac{1}{2}} \tag{4.75}$$

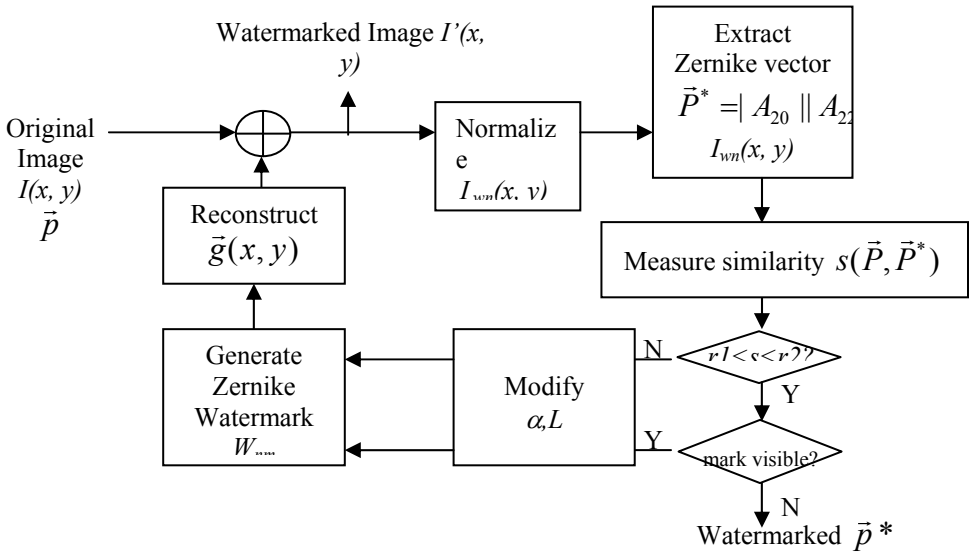


Figure 23. Watermark Insertion in Zernike Domain [24]

If this value is smaller than the threshold, the watermark is detected. Again, the original image is not required at the receiver for detection. The watermark detection process is described in Figure 24.

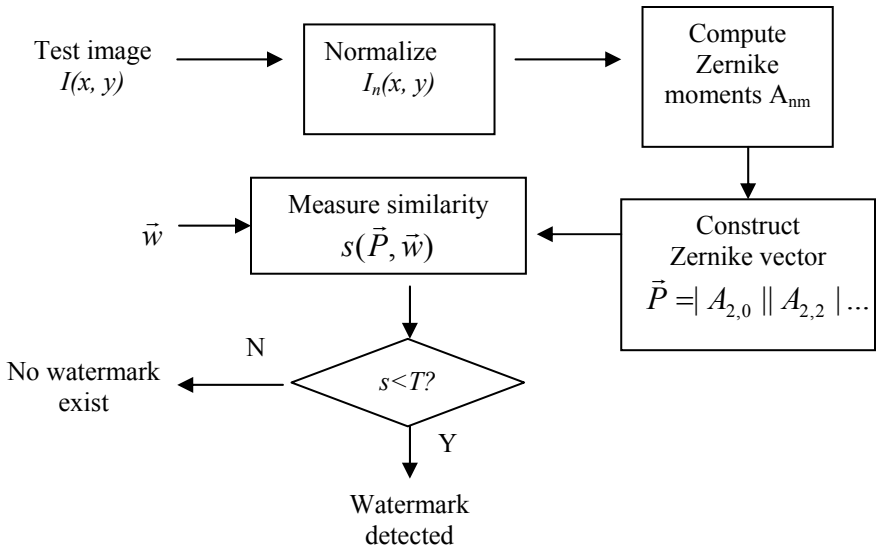


Figure 24. Watermark Detection in Zernike Domain [24]

Image moments based algorithms open a new route in the field of image watermarking. The unique rotation, scaling and translation invariants property make them excellent medium to embed watermarks.

Having described some existing image moments-based watermarking techniques, in the next chapter we will start with pointing out their deficiencies, especially when facing the geometric attacks outlined at the start of this chapter.

Chapter 5

AFFINE INVARIANT INTEREST POINT DETECTION

5.1 Deficiency in existing moments based watermarking techniques

Existing image moments-based watermarking techniques take advantage of their rotation-, scale- and translation-invariant nature. Therefore, compared to other transform-domain based techniques, intrinsically these techniques are immune to rotation type attacks. In addition, these existing moments-based watermarking techniques also demonstrated good resistance toward image processing type attacks such as filtering or lossy compression.

As noted in Chapter 4, image moments have been used in watermarking applications due to their success in the field of pattern recognition. Because of this, some of the common practices associated with the use of image moments in pattern recognition have also been carried over to the study of image watermarking, for example, the use of the image centroid as the origin of the region over the image moments are computed. This assumes that the image centroid will be invariant between the source (i.e., original image) and the receiver (i.e., modified image). While in a pattern classification application, the image centroid will be invariant to affine transforms since the moments are computed over the entire image as shown in Figure 25, it is not necessarily true for watermarking applications. This difference arises from the nature of the two applications.



Figure 25. Whole Objects with Different Orientations in Pattern Recognition Application (a) Original (b) Rotate and Scaled

Any distortion of the image in pattern recognition applications is because of the relationship between the sensor and the objects to be classified, for example different object orientations, different distance from sensor to the objects etc. In general, there are no intentional or malicious attempts to distort the object. On the other hand, the distortions concerned in the watermarking applications are in general of malicious nature. Because of this, the key for a robust watermarking technique is to mask any apparent weakness to potential attackers.

Since the image centroid is the key to the synchronization of the computation of the image moments at the source and at the receiver, it obviously will become the target of attack: if synchronization is destroyed, then watermark detection will fail. Since the location of the image centroid is dependent on the 0th-order moment m_{00} and first-order moments m_{10} and m_{01} as shown in Eq. (3.9), if the attacks can alter these moments, then they would be successful in destroying the detection of the embedded watermarks.

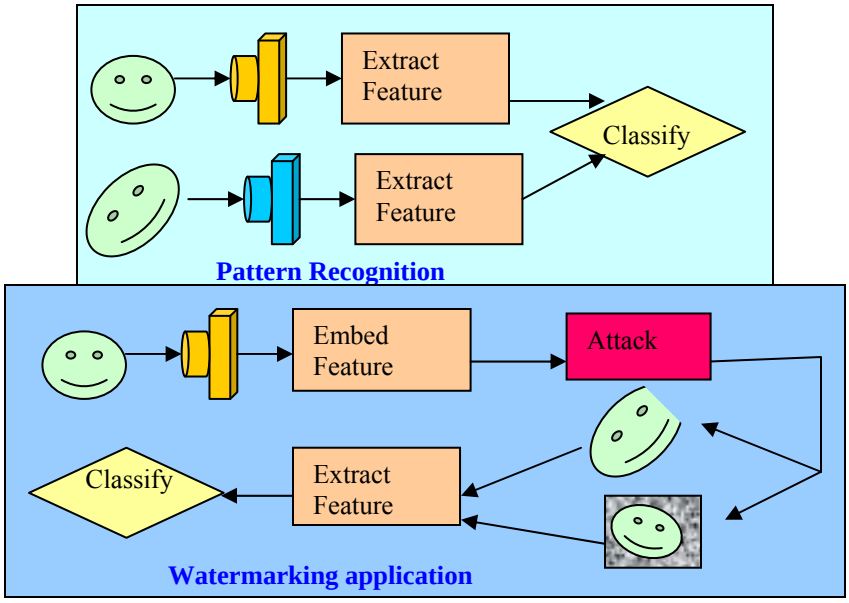


Figure 26. Distortion Sources Different in Pattern Recognition and Watermarking

Various geometric attacks can achieve the intended goal of altering geometric moments. Two very straightforward approaches are image cropping and image padding. Now let us look at some examples of such attempted attacks.

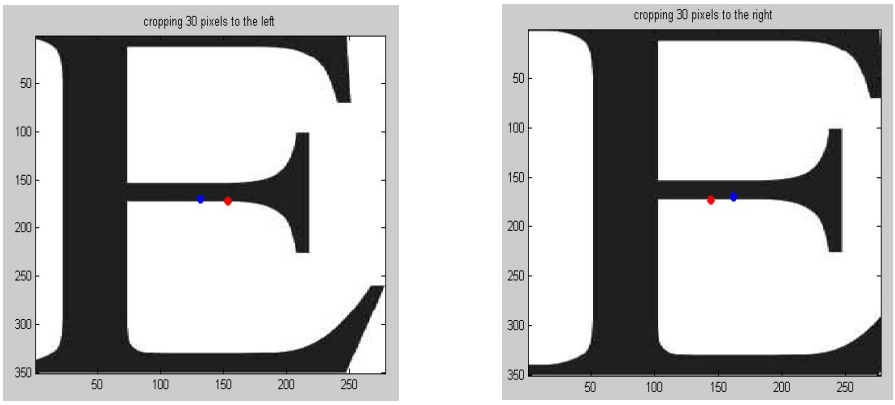


Figure 27. Cropping the Sides Causes Centroid shifting. (a) Cropping Left Side (b) Cropping Right Side (Blue: Original Centroid, Red: Centroid after Cropping)

As shown in Figure 27 above, for the two different types of cropping, the image centroids have shown significant shifts. The choice of only cropping one side of the image in the above example is intentional, since this unbalanced operation has more impact on the shifting of the image centroid. The same type of shifting can be observed when the image is padded on the sides.

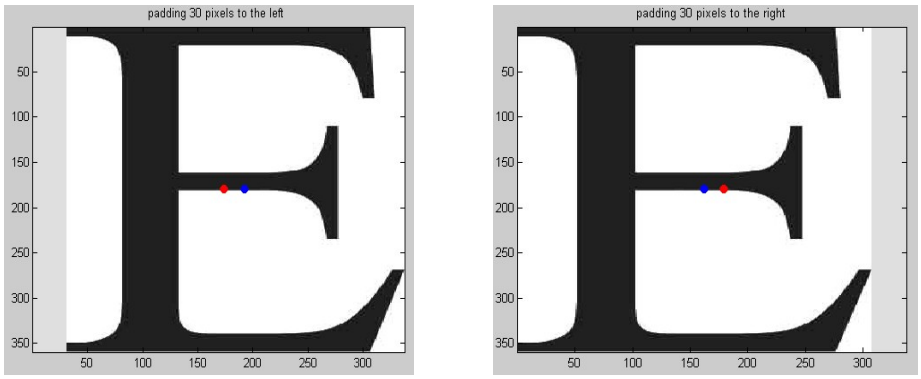


Figure 28. Padding the Sides Cause Centroid Shifting (Blue: Original Centroid, Red: Centroid after Padding) (a) Left, (b) Right

As expected, unbalanced padding to the side of the image has the same impact on the image centroid as unbalanced cropping, as shown in Figure 28 above. The reason behind the centroid shifting is that the whole image is used in the computation of the geometric moments. Therefore, when the image content changes by cropping or padding, then the geometric moment values will change too. As a result, we end up with a different image centroid and a different origin to compute Zernike Moments or Radial Tchebichef Moments. As such, the attack now successfully destroys the synchronization between the watermark embedding at the source and watermark detection in the receiver.

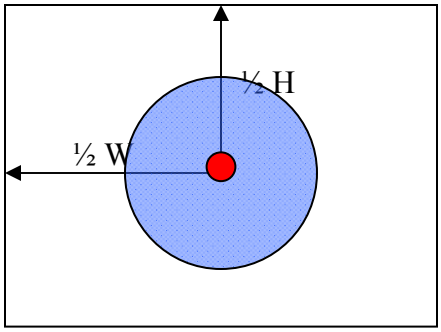


Figure 29. Using Image Spatial Center and a Sub-region for Embedding.

One potential approach to fend off image cropping or padding attack is to localize the computation of Zernike Moments or Radial Tchebichef Moments to within a pre-defined sub-region of the image using the spatial center of the image, as shown in Figure 29 above. This way, if the attacks are cropping/padding to the right or at bottom of the image, then the region with embedded watermark can still be recovered. However this approach will not help the general cropping/padding attacks. Especially if attacks are a combination of cropping/ padding and rotation and scaling, then using spatial center as origin will not work either.

In summary, using either the image centroid or image spatial center as origin intrinsically opens up the existing image moments-based watermarking algorithm to geometric attacks that involve cropping/padding. In fact, all the attacks highlighted in Table 2 will present challenges for the existing moments-based watermarking techniques. In a sense, image cropping/padding is the Achilles' heel for these techniques.

However, such limitation does not imply that image moments are not a good medium to embed watermark. On the contrary, such an approach offers a very good remedy to the handling of image moments that presents problems for DCT or wavelet-domain

watermark algorithms. What is needed is a framework that can provide a structure that is robust enough to survive not only image rotation but also any attacks that involves cropping, padding or any combinations of such operations.

Even though using a region around the spatial center may not handle general cropping/padding type attacks, it does open the gate to develop a new robust watermarking framework that can resist such attacks. However, before we can introduce the proposed new framework, we need to establish another theoretical foundation – scale space based affine invariant interest detection.

5.2 Overview of Scale Space Analysis

An inherent property of real-world objects is that they only exist as meaningful entities over certain ranges of scale. For example, examining tree branches is meaningful from a distance of a few centimeters to a few meters; however, from 10 kilometer away, it makes more sense to discuss the forest in which the tree grows. This fact shows that the notion of *scale* is of utmost importance for someone aims at describing them. Scale is a critical factor for image processing also to extract desired information from image data. While in certain situations, appropriate scales for analysis may be known *a priori*. In many situations it may not be obvious what the proper scales to us are. One such example is a vision system with the task of analyzing unknown scenes, As Figure 30 below shows,, bricks at a fine scale, becomes texture when examined at a coarse scale further away.

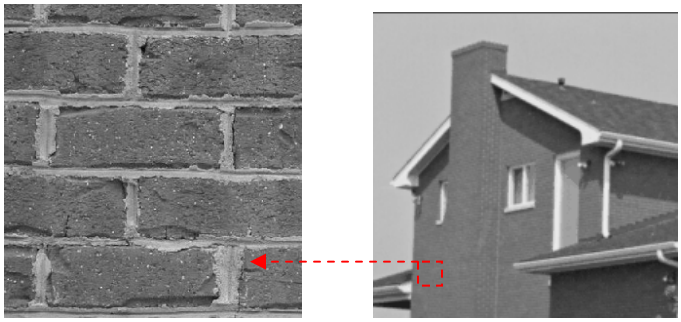


Figure 30. Image Analysis at Different Scales.

Scale-space theory is a framework for early visual operations to handle the multi-scale nature of image data([23], [24], [34], [35]) ., the idea being that if no prior information is available to determine the appropriate scales to use for a given data set, we will need to represent the input data at multiple scales. This is accomplished by embedding the original signal into a one-parameter family of derived signals, in which fine-scale structures are successively suppressed (see Figure 31) [34], a crucial requirement being that structures at coarse scales should be simplifications of the corresponding structures at finer scales.

We define f or any N-dimensional signal $f: R^N \to R$, its scale-space representation $L: R^N \to R$ as:

$$L(\mathbf{x};\ t) = f(\mathbf{x}) \otimes g(\mathbf{x};t) = \int_{R^N} f(\mathbf{x} - \xi) g(\xi;t) d\xi \qquad (5.76)$$

where $g: R^N \times R_+ \to R$ denotes the scale-space kernel, $\otimes$ is the convolution operation in $\boldsymbol{x}$ and t is the *scale parameter*.

For an image *f(x, y)*, its scale space representation can be simplified as

$$L(x,y;\ t) = f(x,y) \otimes g(x,y;t) \tag{5.77}$$

It has been shown by Lindeberg [34] that under a variety of reasonable assumptions the only possible scale-space kernel is the Gaussian function:

$$g(\mathbf{x};\ t) = \frac{1}{(2\pi\sigma^2)^{D/2}} e^{-(x_1^2+x_2^2+\ldots+x_D^2)/2t} \tag{5.78}$$

where the variance t of this kernel is the *scale parameter*.

Two properties of the Gaussian kernel are evident from its definition:

- Positivity: $g(x,y;t) > 0$
- Normalization: $\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} g(x,y;t)dxdy = 1$

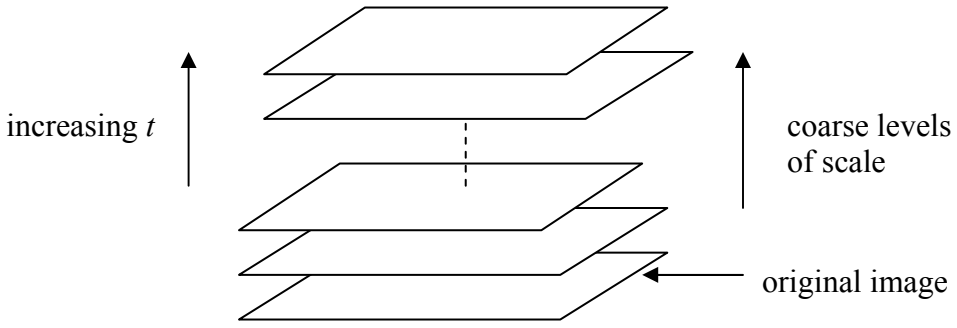


Figure 31. Scale-space Representation as a Multi-scale View of the Original Image

The following Figure 32 demonstrates the result of Gaussian smoothing of a one-dimensional signal. As expected, with larger scale parameter, less detail can be observed.

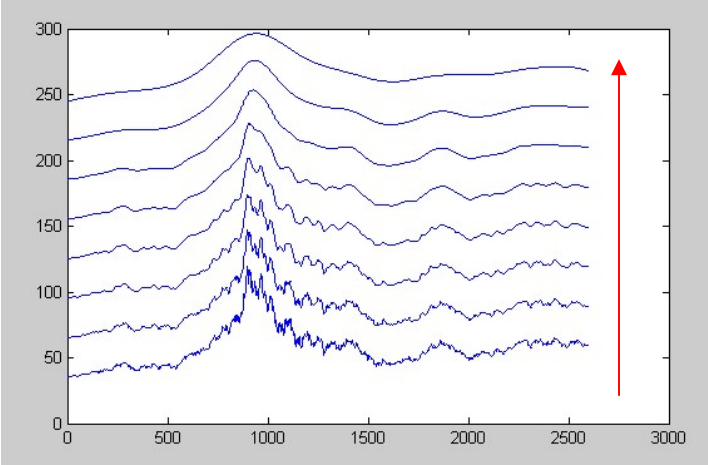


Figure 32. 1-D signal Successively Smoothed by Gaussian Kernels of Increasing Width (TI Stock from 1996-2007)

The same effect can be observed in 2-D image, as shown in Figure 33 below. The image detail gets blurrier with larger scale parameter. In addition, it can be seen that while some extrema changes disappear with the change of scale, the extrema that describe the characteristics of the image are invariant to the scale change. This is the general idea behind the affine invariant interest point detection to be discussed later.

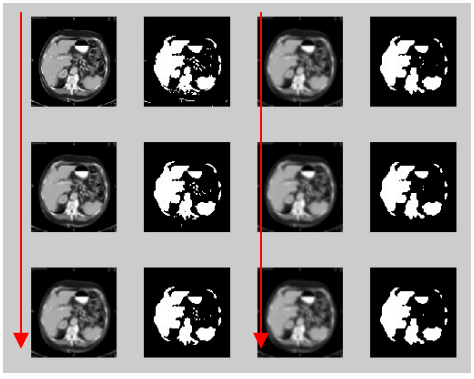


Figure 33. Successive Smoothing of CAT-scan Image and Associated Scale-space Extrema

5.3 Relation between Scale Space and Wavelet representations

Before we discuss the applications of scale space representation, it is interesting to compare scale-space representation and the wavelet representation.

Scale-space is a multi-scale representation; wavelet on the other hand uses a multi-resolution pyramid representation. The difference is that scale-space representation preserves the same spatial sampling at all scales, while in the wavelet representation, the number of grid points is reduced from one layer to the next by pre-filtering followed by sub-sampling. Therefore, in a scale-space representation, the representations at all levels of scale are immediately accessible without any need for further computations. The task of operating on the data will be successively simplified, since a feature existing at a coarse scale will in general correspond to a larger number of grid points than a feature at a fine scale. On the other hand, the wavelet representation is efficient and compact, while a scale space representation successively becomes more redundant as the scale parameter increases. An orthogonal wavelet representation is in fact non-redundant while a scale-space representation can be said to be maximally redundant. The decreasing image size in pyramids reduces the computational work and memory storage both in the actual computation of the representation and in the subsequent processing. There are commercially available hardware accelerated wavelet transformation implementations.

The wavelet representations imply a fixed sampling step in scale, while the scale-space concept possesses a continuous scale parameter. Therefore, one can expect the task of following or tracking features across scales to be easier in a multi-scale than in a multi-resolution representation, since refinements of the scale sampling can be performed

whenever required. Furthermore, the wavelet representations may under-sample the signals along the scale direction.

Another important property with the scale-space representation is that the behavior of structure across scales can be analytically described with a simple formula. By definition, it is given as the solution to the diffusion equation, which means that features at different scales can be related to each other in a precise manner.

Another difference between scale-space and wavelet representations is the definition of the kernel. In a wavelet representation, the kernels can be chosen from a vast kernel family for different environments, whereas in a scale-space representation, the Gaussian kernel has been proven to be the only choice.

Finally, the scale-space representation is invariant to translations. While the discrete wavelet transform (DWT) coefficients is not translation invariant for any translation that is not proportional to the discrete scale factor 2^k at level k [17].

5.4. Scale Selection Principle

The problem of choosing an appropriate scale has been handled by formulating algorithms which rely on the information present in the data at a small set of manually chosen scales. An alternative approach is to employ a coarse-to-fine algorithm – start at a given coarse scale and propagate down to a given finer scale. This, however, leads to the introduction of free parameters that requires tuning. Notably, image descriptors can be highly unstable if computed at inappropriately chosen scales, whereas a proper tuning of the scale parameter can improve the quality of an image descriptor substantially. Therefore it is essential to complement traditional multi-scale processing by techniques for automatic scale selection.

Lindeberg has studied this problem in [34] and presented a scale selection principle for image descriptors which can be expressed in terms of Gaussian derivative filters. The general idea for scale selection is to study *local extrema over scales* of the normalized differential descriptors. Such extrema are likely to correspond to interesting image structures. This technique provides a unified framework for scale selection for detecting image features such as blobs, corners and edges.

In the scale-space representation, the amplitudes of spatial derivatives generally decrease with scale, i.e., if a signal is subject to scale-space smoothing, then the numerical values of spatial derivatives computed from the smoothed data can be expected to decrease. In practice, it means that the amplitude of the variations in a signal will always decrease with scale. Normalized derivatives can provide a way to compensate for such decrease.

By using the normalized derivative, a one-to-one correspondence between the matching response of the Gaussian derivative kernels and the frequency content of the signal is established. Selecting the scale at which the maximum occurs corresponds to selecting the pattern (or the scale) for which the operator response is strongest. This property applies to a large class of image descriptors which can be formulated as scale-space representation.

Given two signals f and f' related by $f(\mathbf{x}) = f'(s\mathbf{x})$ and their corresponding scale-space representations:

$$\begin{aligned} L(\cdot;t) &= g(\cdot;t) \otimes f \\ L'(\cdot;t) &= g(\cdot;t') \otimes f' \end{aligned} \tag{5.79}$$

where spatial variables $\mathbf{x}$ and the scale parameters t satisfy:

$$\boldsymbol{X}' = s\boldsymbol{X};\ t' = s^2 t \tag{5.80}$$

we introduce a γ-normalized derivative operator:

$$\partial_{\xi} = t^{\gamma/2}\partial_x \tag{5.81}$$

It can then be derived that the *m*th order normalized derivative of *L* and *L'* satisfies the following equation [23]:

$$\partial_{\xi^m} L(\boldsymbol{X};t) = s^{m(1-\gamma)}\partial_{\xi'^m} L(\boldsymbol{X}';t') \tag{5.82}$$

From the above equation it can be observed that provided that the scale parameters and the spatial scales are matched according to Eq. (5.4), when γ=1, the normalized derivatives of the original and scaled signals are equal, which leads to scale invariance. The above observations are summarized as the **Principle for scale selection:**

In the absence of other evidence, assume that a scale level, at which some (possibly non-linear) combination of normalized derivatives assumes a local maximum over scales, can be treated as reflecting a characteristic length of a corresponding structure in the data.

The automatic scale selection principle has contributed to many feature detection applications that detect features at different scales more accurately. This principle is one of the important foundations in the affine invariant point detection [22] that will be discussed in the next section.

5.5 Affine Invariant Interest Point Detection

In addition to global characteristics such as image moments, local characteristics such as interest points have also been widely used in pattern recognition and matching. One difficulty has been to achieve rotation, translation, and scale invariance of these

characteristics under image transformations. Mikolajczyk and Schmid proposed an affine invariant interest point detector [22], [23] (which we denote as M-S detector from here on) that provides an approach to overcome these difficulties. This M-S detector is based on two foundations: 1) the principle for scale selection – the local extrema over scale of normalized derivatives indicate the presence of characteristic local structures; 2) an interest point detector with strong invariance to rotation, scale, and image noise. For the latter, the Harris detector is chosen in their implementation,.

5.5.1. Harris Detector

Before we discuss the M-S affine interest point detector, we provide an overview of the Harris detector [6].

The Harris detector is formulated by considering a local window in the image and determining the average error of image intensity E that resulted from shifting the window by a small amount in various directions:

$$E(x,y) = \sum_{W} w(x_i, y_i)\left[f(x_i, y_i) - f(x_i + \Delta x, y_i + \Delta y)\right]^2 \tag{5.83}$$

where $w(x_i,y_i)$ is a window function centered at (x, y). We choose a smooth circular Gaussian function: $w(x,y) = g(x,y;\sigma) = \frac{1}{2\pi\sigma} e^{-(x^2+y^2)/2\sigma^2}$.

The shifted image can be approximated by a Taylor expansion truncated to the first order terms:

$$f(x_i + \Delta x, y_i + \Delta y) = f(x_i, y_i) + \begin{bmatrix} \frac{\partial}{\partial x} f(x_i, y_i) & \frac{\partial}{\partial y} f(x_i, y_i) \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \tag{5.84}$$

Substituting Eq. (5.9) into Eq. (5.8) yields:

$$E(x,y) = \sum_W g(x,y;\sigma)\left[f(x_i,y_i) - f(x_i+\Delta x, y_i+\Delta y)\right]^2$$
$$= \sum_W g(x,y;\sigma)\left[f(x_i,y_i) - f(x_i,y_i) + \begin{bmatrix} f_x(x_i,y_i) & f_y(x_i,y_i)\end{bmatrix}\begin{bmatrix}\Delta x \\ \Delta y\end{bmatrix}\right]^2 \quad (5.85)$$

where $f_x(x,y) = \frac{\partial}{\partial x} f(x,y)$ and $f_y(x,y) = \frac{\partial}{\partial y} f(x,y)$ are the first order derivatives of *f(x, y)* with respect to *x* and *y*. With further consolidation, *E(x, y)* can be expressed as:

$$E(x,y) = \begin{bmatrix}\Delta x & \Delta y\end{bmatrix}\mu(x,y)\begin{bmatrix}\Delta x & \Delta y\end{bmatrix}^T \quad (5.86)$$

where $\mu(x,y)$ is the second moment matrix:

$$\mu(x,y) = g(x,y;\sigma) \otimes \begin{bmatrix} f_x(x,y)^2 & f_x(x,y)f_y(x,y) \\ f_x(x,y)f_y(x,y) & f_y(x,y)^2 \end{bmatrix} \quad (5.87)$$

Let λ_1 and λ_2 be the eigenvalues of *μ(x, y)*. The eigenvalues form a rotationally invariant description. As shown in Figure 34, there are 3 different scenarios that can be considered:

1. Both λ_1 and λ_2 are small, so that there is little change in $\mu(x,y)$ in any direction; the windowed image region is flat.
2. If one eigenvalue is high than the other, then local shifts in one direction cause little change in *μ(x,y)* and significant change in the orthogonal direction; therefore the windowed image region contains an edge.
3. If both eigenvalues are high, then shifts in any direction will result in a significant increase; this indicates the windowed region contains a corner.

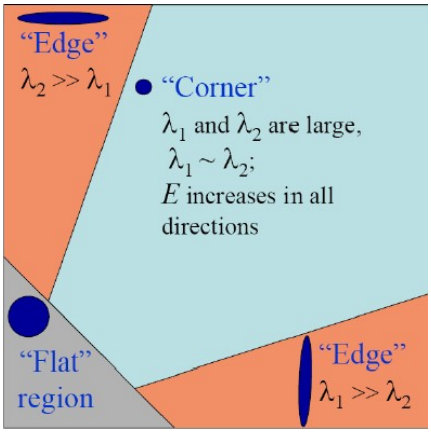


Figure 34. Partitioning of Harris Measure

The measure of corner response can be expressed as following:

$$R = det(\mu) - \alpha(trace(\mu))^2 \tag{5.88}$$

where $det(\mu) = \lambda_1\lambda_2$ and $trace(\mu) = \lambda_1 + \lambda_1$, α is an empirical constant, usually chosen to lie between 0.04 and 0.06.. R is referred to as the Harris Measure. The Harris corner detector finds the point with large Harris Measure: $R > threshold$; then the local maxima among these points will be the corner point of the local region.

One important attribute of the Harris corner detector is its rotation invariance property. Since the two eigenvectors associated with λ_1 λ_2 can be viewed as major and minor axes of an ellipse, an image rotation will cause the ellipse to rotate but the shape of the ellipse will remain the same, and the eigenvalues remain the same, therefore the Harris Measure will not change.

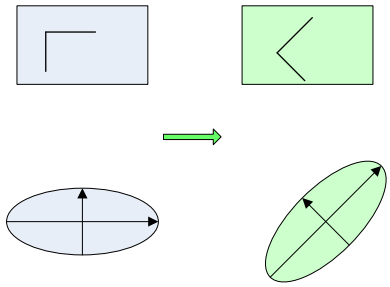


Figure 35. Rotational Invariance of the Harris Measure

Figure 36 shows the results of the Harris corner detector against CT Scan image. The following parameters were used in the detector: α=0.04, σ=3 and Threshold = 2000.

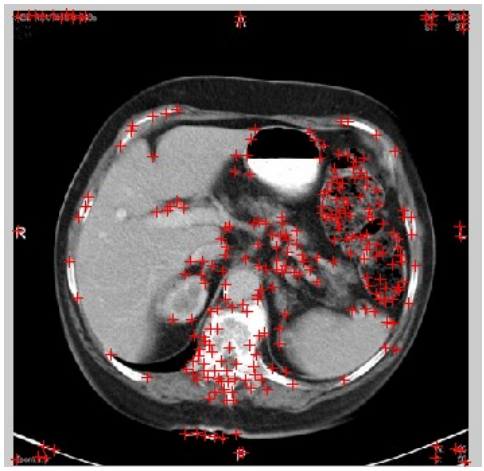


Figure 36. Harris Corner Detector result

The Harris detector can be extended to a multi-scale Harris Detector. This can be achieved by replacing the original image *f(x)* with its scale-space representation $L(\boldsymbol{x};\sigma_D)$; then we will have the multi-scale version of the Harris Measure:

$$\mu(\boldsymbol{X};\sigma_I,\sigma_D)=\sigma_D^2 \quad g(\boldsymbol{X};\sigma_I)\otimes\begin{bmatrix} L_x^2(\boldsymbol{X};\sigma_D) & L_xL_y(\boldsymbol{X};\sigma_D) \\ L_xL_y(\boldsymbol{X};\sigma_D) & L_y^2(\boldsymbol{X};\sigma_D) \end{bmatrix} \tag{5.89}$$

$$\det(\mu)-\alpha \;\; trace^{\,2}(\mu) \;\; > \;\; threshold$$

where σ_I is the *integration scale*, associated with the Gaussian kernel $g(\boldsymbol{X};\sigma_I)$ used to smooth the local window over which the Harris Measure is evaluated; σ_D is the *derivative scale*, which is associated with the Gaussian kernel in the scale-space representation and L is the scale-space representation of the image $f(\boldsymbol{X})$.

5.5.2. Affine Invariant Interest Point Detector

We now discuss the implementation of the affine invariant interest point detector. The automatic scale selection principle outlined in section 3.x can be employed to achieve scale invariance. Points are extracted at several scales and the characteristic scale is automatically determined based on the maximum of the normalized Laplacian $\left|\acute{o}_D^2(L_{xx}(\boldsymbol{X};\acute{o}_D)+L_{yy}(\boldsymbol{X};\acute{o}_D))\right|$, where derivatives are computed with a uniform Gaussian kernel. However one of the unique characteristics of the multi-scale Harris detector is that the local Harris maxima will have different spatial locations when extracted at different detection scales. Thus, the Harris measure will need to be evaluated at each different scale. This will introduce some complexity in the environment that involves affine transformation.

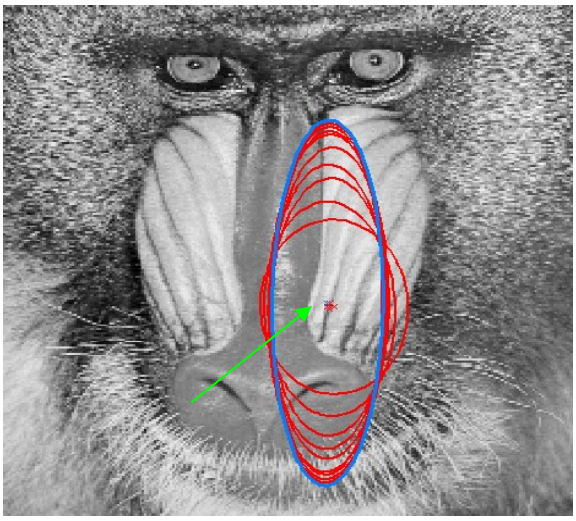


Figure 37. Harris Measure Maxima at Different Scales

To deal with affine transformations requires the detection scales in the x and y directions to vary independently so that both scales can be adapted to the local image structure. This in turn introduces the problem of computing the multi-scale Harris Measure in the affine Gaussian scale space instead of the linear Gaussian scale space, where a circular window is replaced by an ellipse. An affine scale-space representation can be generated by convolution of the image with non-uniform Gaussian kernels:

$$g(\boldsymbol{X};\Sigma) = \frac{1}{2\pi\sqrt{\det(\Sigma)}} e^{-\frac{\boldsymbol{X}^T\Sigma^{-1}\boldsymbol{X}}{2}} \tag{5.90}$$

where $\boldsymbol{X} \in R^2$. With the affine Gaussian kernel, the scale-space representation corresponds to convolution with a rotated elliptical Gaussian kernel:

$$L(\boldsymbol{X};\Sigma) = g(\boldsymbol{X};\Sigma) \otimes f(\boldsymbol{X}) \tag{5.91}$$

If the matrix Σ is an identity matrix, then the kernel is the linear Gaussian kernel, which can determined by one scale parameter σ. For the case where Σ is a symmetric positive

definite 2x 2 matrix, the kernel has three degrees of freedom. In the affine scale space the second moment matrix can expressed as:

$$\mu(\boldsymbol{x};\Sigma_I,\Sigma_D) = g(\boldsymbol{x};\Sigma_I) \otimes (\nabla L(\boldsymbol{x};\Sigma_D)(\nabla L(\boldsymbol{x};\Sigma_D))^T) \tag{5.92}$$

where Σ_I,Σ_D are the covariance matrices that determine the Gaussian kernels for local window as well as the scale space representation. To reduce the search space, first a constraint $\Sigma_I = k\Sigma_D$ is imposed, where k is a scalar constant. Furthermore, the following conditions are set which the second moment matrix and the scale matrices are required to satisfy:

$$\mu(\boldsymbol{x};\Sigma_I,\Sigma_D) = M_L \tag{5.93}$$

where Σ_I and Σ_D are scalar multiples of M_L^{-1} : $\Sigma_I = tM_L^{-1}$ and $\Sigma_D = sM_L^{-1}$. Assume Σ_I' and Σ_D' defines another affine transformation for $\boldsymbol{x}'$: $\mu'(\boldsymbol{x}';\Sigma_I',\Sigma_D') = M_L'$, $\Sigma_I' = tM'^{-1}_L$ and $\Sigma_D' = sM'^{-1}_L$. Then the matrices M_L and M_L' are related by:

$$M_L = A^T M_L' A \quad A = M_L'^{-1/2} R M_L^{-1/2} \quad \Sigma_L' = A\Sigma_L A^T \tag{5.94}$$

where R is an arbitrary rotation matrix. It was also shown in [44] by Baumberg that if $\boldsymbol{x}$ and $\boldsymbol{x}'$ are normalized by transformations:,

$$\boldsymbol{x}_N = M_L^{-1/2}\boldsymbol{x}; \quad \boldsymbol{x}'_N = M_L'^{-1/2}\boldsymbol{x}' \tag{5.95}$$

then in the normalized domain, $\boldsymbol{x}'_N$ and $\boldsymbol{x}_N$ are related by a pure rotation: $\boldsymbol{x}_N = R\boldsymbol{x}'_N$,that is, M_L and M_L' are both rotational matrices. In other words, in the normalized domain, the local window will be converted from an ellipse back to a circular window.

With the above normalization process, the framework for affine invariant interest point detector can now be established. The overall iteration procedure is outlined as follows. In

the following, a point in the original image domain is denoted as $\boldsymbol{x}$ and the point in the transformed image domain is denoted as $\widetilde{\boldsymbol{x}}$.

For each extracted point using the Harris measure, choose a circular window in the transformed image domain with that point as origin. The radius of this window is proportional to the integration scale σ_I.

1. *Initialize* $U^{(0)}$ *to the identity matrix;*
2. *Normalize window* $W(U^{(k-1)}\widetilde{\boldsymbol{X}}) = I(\boldsymbol{X})$ *centered in* $U^{(k-1)}\widetilde{\boldsymbol{X}} = \boldsymbol{X}$ *;see Eq. (5.20)*
3. *Select the integration scale* σ_I *in the* $\widetilde{\boldsymbol{x}}$ *;*
4. *Set* $\mu = \mu(\widetilde{\boldsymbol{X}}^{(k-1)}, \sigma_I, \sigma_D)$ *and select the derivation scale* $\sigma_D = s * \sigma_I$*, which maximize* $\frac{\lambda_{\min}(\mu)}{\lambda_{\max}(\mu)}$ *with the constant* $s \in [0.5, \ldots, 0.75]$ *;*
5. *Detect spatial localization* $\widetilde{\boldsymbol{X}}^{(k)}$ *of the maximum of Harris measure (Eq. 5.13) nearest to* $\widetilde{\boldsymbol{X}}^{(k-1)}$ *and compute the location of interest point* $\boldsymbol{X}^{(k)}$ *;*
6. *Compute* $\mu_i^{(k)} = \sqrt{\mu(\widetilde{x}^{(k-1)}, \sigma_I, \sigma_D)}$ *;*
7. *Concatenate transformation* $U^{(k)} = \mu_i^{(k)} U^{(k-1)}$ *and normalize* $U^{(k)}$ *such that* $\lambda_{\max}(U^{(k)}) = 1$*;*
8. *Loop back to step 2 if* $\frac{\lambda_{min}(\mu_i^{(k)})}{\lambda_{max}(\mu_i^{(k)})} > \varepsilon_C$

We now describe this procedure in some detail. The key in the implementation is the computation of the shape adaptation matrix. An iterative approached is adopted where the shape adaptation works in the transformed image domain. Instead of applying an affine Gaussian kernel, the image is transformed as described above Eq. (5.16)—(5.19) [22]. Then a recursive implementation of the linear Gaussian kernel can then be used for computing L_x and L_y. The second moment matrix μ is computed according to Eq. (5.17). A local window is transformed by U-transformation:

$$U^{(k-1)} = (\mu^{-\frac{1}{2}})^{(k-1)} \cdots (\mu^{-\frac{1}{2}})^{(1)} U^{(0)} \tag{5.96}$$

in step 7 of the iteration procedures outlined above. A new second moment matrix μ is computed at each iteration step and that the *U* matrix is the concatenation of square roots of the second moment matrices; *U* is then normalized so that $\lambda_{max}(U) = 1$.

Of the two scale parameters in the multi-scale Harris detectors, the integration scale σ_I is selected using the automatic scale selection principle to preserve scale invariance. Therefore σ_{I^*} is the scale that attains a local maximum for the normalized Laplacian $\left|\sigma_{I^*}^2 (L_{xx}(\sigma_{I^*}) + L_{yy}(\sigma_{I^*})\right|$. The local derivation scale σ_D is less critical and is set proportional to the integration scale $\sigma_D = s*\sigma_I$. Factor *s* is commonly chosen in the range [0.5, 0.75]. To determine the optimum σ_D with a given integration scale σ_I, the ratio of eigenvalues of the second moment matrix μ at iteration step (*k*):, namely, $(\lambda_{min}(\mu)/\lambda_{max}(\mu))$ is evaluated for different values of *s* in the range $s \in [0.5, \ldots, 0.75]$. The optimum value for *s* is the one for which the ratio assumes a maximum. The factor *s* also influences the convergence of the second moment matrix. The iterative procedure converges toward a matrix with equal eigenvalues. Therefore the smaller the difference between the eigenvalues $(\lambda_{max}(\mu), \lambda_{min}(\mu))$ of the initial matrix, the faster the procedure converges. Thus, the selection of the factor *s* according to the criterion above allows for a reasonable eigenvalue ratio and improves the chance for the second moment matrix to converge.

As mentioned above the local maxima of the Harris measure change their spatial location if the detection scales change. This can also be observed if the scale change is different in each direction. The affine normalization of a point neighborhood slightly

changes the local spatial maxima of the Harris measure. Consequently, the local maximum Harris measure is re-detected in the affine normalized local window W.

A vector of displacement to the nearest maximum is then obtained in the U-normalized image domain. The location of the initial point is corrected with the displacement vector back-transformed to the original image domain $\boldsymbol{x}^{(k)} = \boldsymbol{x}^{(k-1)} + U^{(k-1)} \cdot (\boldsymbol{x}^{(k)}_w - \boldsymbol{x}^{(k-1)}_w)$, where $\boldsymbol{x}_w$ are the coordinates of the detected interest point in the transformed image domain during the current iteration. Each point inside the window will also be back-transformed to the original image domain using Eq. (5.20).

In the iteration procedure, the convergence measure is based on the μ matrix computed in each iteration step. The termination requirement is that this matrix is sufficiently close to a pure rotation matrix. This implies that $\lambda_{max}(\mu)=\lambda_{min}(\mu)$. In practice the following condition is used: $|\lambda_{min}(\mu)/\lambda_{max}(\mu)| > \varepsilon_c$. To prevent the procedure from diverging, any second moment matrix that is associated with an unstable elongated ellipse is rejected, that is, if $\lambda_{max}(\mu)/\lambda_{min}(\mu) >$ some threshold ε_l.

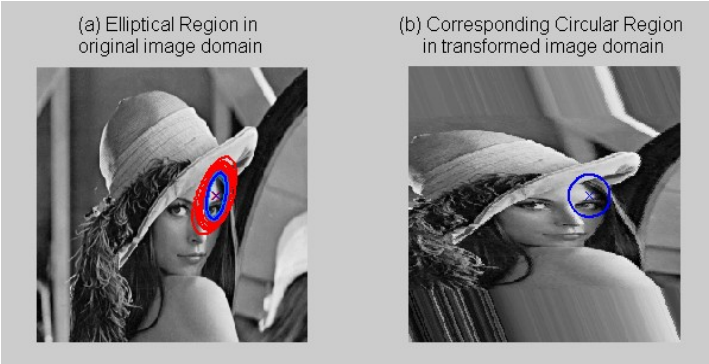


Figure 38. Affine Invariant Interest Point Detection

This procedure will output the spatial location of the interest point, the associated integration scale factor and the second moment matrix describing the orientation of the elliptical region associated with the interest point. Figure 38 above shows the adaptation process of obtaining the final interest point and the elliptical patch in the original image domain, as well as the corresponding circular patch in the transform image domain.

After the treatment above of the affine interest point detector, we now are ready to introduce the proposed image watermarking framework in Chapter 6.

Chapter 6

A UNIFIED WATERMARKING FRAMEWORK

6.1 Watermarking based on Affine Invariant Point and image moments

As discussed in Section 5.1, the main issue for the existing Zernike Moments based watermarking scheme is that Zernike moments are not translation invariant and under "cocktail" type geometric attacks involving cropping, rotation or scaling, it is impossible to re-establish the relation between the altered moments and the original ones. A straightforward approach to overcome the problem discussed in Section 5.1 is to use some image registration technique to determine the original image centroid. However, as discussed in Chapter 4, under the condition of blind watermarking, such an approach is subject to the same drawbacks as for general spatial domain watermarking schemes – susceptible to noise; not robust against dithering or other LSB randomization type attacks.

We therefore focus on alternate approaches in which the goal is to find a region that is close to the center of the image, occupies a regular shaped sub-region in the image, and is such that the detection of the region can be repeated at different scales.

The first attempt was to utilize image segmentation methods, such as various watershed transformation techniques. The intent was to identify a main object in the image and embed the watermark in this object. However, the resulting object boundaries tend to vary at different scaling levels and the resulting shape turns out to be more complicated

than needed – the perfect region for the subsequent watermark embedding would be a circular region that complies with the definition of Zernike Moments or Radial Tchebichef Moments. Also such a region will need to be invariant to scale change. These requirements lead us to the study of the technique proposed by T. Lindeberg [30] since the affine invariant interest point detection algorithm based on scale space representation described in Chapter 5 will be a good fit for our purpose. Finally it is determined that the M-S affine invariant interest detector which extends Lindeberg's technique is the optimum choice for this research.

As discussed in Chapter 5, a set of interest points as well as an oval shaped region that reflects the characteristic scale associated with each of these interest points will be detected by the M-S affine invariant interest point detector. Furthermore, for each interest point, a second moment matrix that transforms this oval region into a circular region and the characteristic scale of this region is generated by the detector.

To simplify the description of the algorithm without sacrificing generality, we assume that the watermark in embedded in only one region in the image. In this framework, we replace computing the image centroid with the detection of an affine invariant point. The first step would be to detect an initial set of interest points using the Harris detector with a pre-defined default initial scale, as discussed in Chapter 5. However, since our intention is to extract one interest point to use as origin, and potentially there could be large number of interest points detected in this process as shown in Figure 39 below, further processing is undertaken to select only one interest point and associated patch for watermark processing.

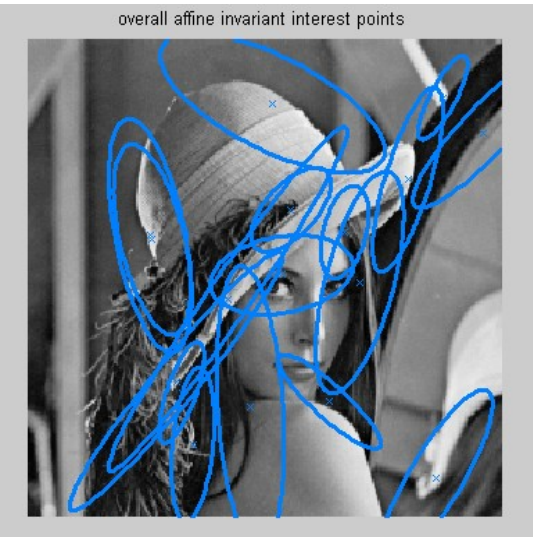


Figure 39. Affine Invariant Interest Points before Filtering

As a first step, a circular region with radius R and with the image centroid as center is established. Only affine invariant interest points that fall within the circle will be retained for further processing, as shown in Figure 40. That is, only interest points satisfying the following condition will be retained:

$$\sqrt{(x_I - x_C)^2 + (y_I - y_C)^2} < R \tag{6.97}$$

where (x_I, y_I) and (x_C, y_C) are the coordinates of the interest point and image centroid respectively.

It should be noted that even though the image centroid is used in the proposed algorithm, its function is fundamentally different from that in the original Zernike Moments based schemes. Here the image centroid is used to establish a filter region. Therefore, the proposed algorithm is insensitive to the shifting of the image centroid that can be caused by image cropping/padding type attacks. The choice of the filtering radius is also fairly flexible in general. The only requirement is that the source and receiver

reach agreement on the definition of the radius R_{filt}. In practice, R_{filt} can be set proportional to the image size, i.e., ¼ of the maximum resolution of the image; or it can be set to an absolute distance in number of pixels, for example, R_{filt} = 60 (pixels). Choosing R_{filt} proportional to image size may be more appropriate since this may further reduce impact due to the effect of image rescaling.

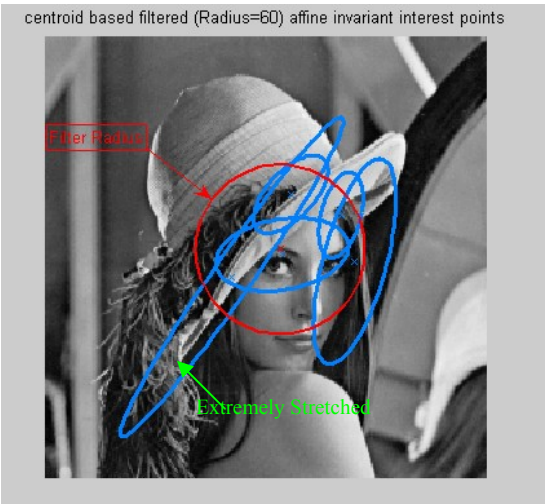


Figure 40. Affine Invariant Interest Points after Centroid Filtering

Following the centroid based filtering; the remaining interest points will undergo the iterative detection process outlined in Chapter 5. Points that converge within a pre-determined number of iteration steps will be retained for further processing.

The next step is to reject any interest points with an extremely "stretched" oval region, such as the one in Figure 40 pointed to by the green arrow. Such a region tends to follow edges on the image. However, the rejection of such points will be done after the iterative adaptation, since even though the initial oval shape is stretched, the adaptation process may drastically adjust the shape of the oval region, and at the end produce a "normal"

oval region. The criterion for such rejection is based on the ratio of the two eigenvalues of the second moment matrix μ: if $\lambda_{max}/\lambda_{\min} > T_L$, where T_L a pre-defined threshold, then the point will be rejected.

Any points still retained will be ordered by their Harris measure $det(\mu) - \alpha\ trace^2(\mu)$, where μ is defined in Eq. (5.12). As noted previously, a higher Harris measure indicates sharper corner response in contrast to the background, therefore would be more robust against any subsequent image processing or compression operations. In a sense, such points can be regarded as "more interesting" points since they bear more information about the local features of the image. Hence the interest point with the highest Harris measure will be chosen as the origin of the region in which the watermark is to be embedded.

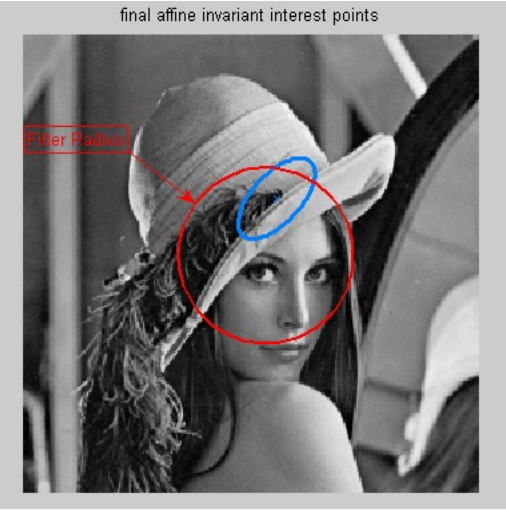


Figure 41. Selected Affine Invariant Interest Point

Now that we have successfully selected the affine invariant interest point as the origin, the next issue to be addressed is determining the region to embed watermark. As shown

in Figure 41, the region associated with this origin in the original image domain is in general an oval region instead of the circular one that is required for the determination of both Zernike Moments and Radial Tchebichef Moments. As pointed out earlier , we use the second moment matrix associated with the chosen interest point to transform the image from affine Gaussian space (oval region in original image) into a linear Gaussian space (circular region in the transformed domain).. As shown in Figure 42, this circular region, with radius R_{mark} proportional to the characteristic scale associated with this interest point, will be the one over which the image moments are computed.

Choosing R_{mark} proportional to the characteristic scale: $R_{mark}=u*\sigma_I$, instead of exactly the same as the characteristic scale, serves two purposes:

- It allows the sub-region to cover a desirable amount of image content, which is purely customer driven;
- Since factor u does not necessarily need to be an integer, it can serve as part of a secret key that the source can pass along to the receiver.

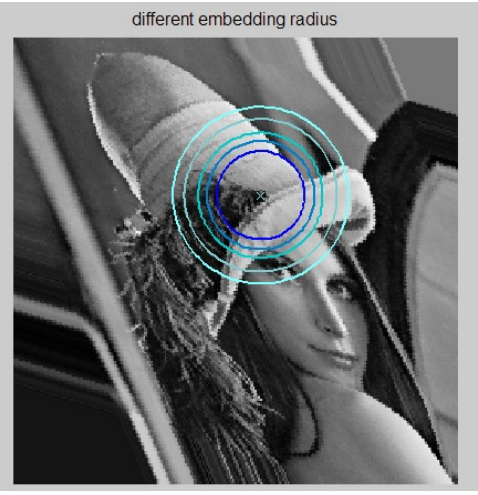


Figure 42. Different (non-integer) Radius Factor u as Security Measure

We now describe the approach to embed the watermarks in the image moments. Following the reasoning in all the existing image moments-based watermarking implementations, the watermarks will be embedded in the amplitude of the image moments up to a certain order, to take advantage of their rotation invariant property. The 0th and 1st order moments are excluded because these moments contain the image DC level and any variations would show up as visible degradation of the image. Similarly, we exclude moments with extremely small amplitude as these are more easily impacted by noise or image processing operations, thereby impairing watermark detection.

The amplitudes of the image moments A_i are adjusted using a geometric sequence-based modulation, as described below:

Define a geometric sequence of real numbers [42],

$$s_{i+1} = \frac{1+\alpha}{1-\alpha} s_i, \qquad s_0 = 1, \tag{6.98}$$

parameterized by $\alpha \in (0, 1)$. This sequence posses this unique property:

For a real number $s > 1$, there will exist two consecutive elements s_i and s_{i+1} in the sequence such that: $s \in [s_i, s_{i+1}]$.

Now, define an index function: $index(s) = (-1)^i$, where i is the subscript or index of s_i. It is easy to see that any real number $s > 1$ can be modified by adding or subtracting at most $\alpha * s$ to change to a different lower bound s_i, and hence to a possible different $index(s)$ value. The index function for $\alpha = 0.1$ is depicted in Figure 43 below.

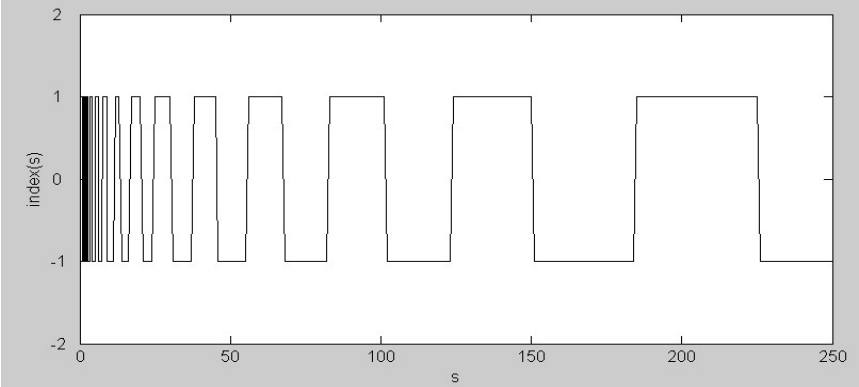


Figure 43. Geometric Sequence

The reverse of the index function is a look up table (*LUT*) that takes *s* and target binary bit *w* as input to give

$$LUT(s,w) = \begin{cases} (s_i + s_{i+1})/2 & if\ index\ (s) = w \\ (s_{i+1} + s_{i+2})/2 & if\ index(s\ ) \neq w \end{cases} \tag{6.99}$$

where $s_i < s_{i+1} < s_{i+2}$ are 3 consecutive values in the geometric sequence and $s \in [s_i, s_{i+i}]$.

Prior to embedding, the amplitudes of the image moments are normalized:

$$|\hat{Z}_{pq}| = C|Z_{pq}/Z_{00}| \tag{6.100}$$

where the constant $C>1$. Z_{00}, Z_{11}, Z_{n0} and $|\hat{Z}_{pq}| < 1$ are excluded from potential candidates. To encode a pseudo random sequence $\{w_i\}$, $w_i \in \{-1,1\}$, of length N: the first N amplitudes $\{|\hat{Z}_{pq}|\}$ that fit the above criteria are selected. The ith term in the sequence is noted as $|A_i| = |\hat{Z}_{pq}|$

The watermark is embedded using *LUT* function:

$$| A_i^w | = LUT(| A_i |, w_i) \tag{6.101}$$

The corresponding watermarked image moments are related to $|A_i^w|$ as follows:

$$Z_{pq}^w = (Z_{00}/C) | A_i^w | exp(-j\ phase(Z_{pq})) \tag{6.102}$$

Because of the properties of the index function, each moment will be modified by at most $100\alpha\%$. The values of α and C will need to be adjusted empirically so that the watermark is perceptually invisible.

Watermark detection is performed by converting the target image in the same manner as for embedding the watermark. N image moments are selected based on the same criteria. These image moments are then normalized by Eq. (6.4). Since all N moments were modified in the same manner by at most $100\alpha\%$, a simple correlation between $index(|A_i'|)$ and w_i would produce a non-robust technique because the contribution from the small amplitude moments will be the same as from the larger, more significant moments. A weighted scheme therefore in which the correlation is computed by weighting by the absolute values of the amplitude of the moments will be more effective:

$$corr = \frac{\sum_{i=1}^{N} | A_i' |^{\beta}\ index\ (| A_i' |) w_i}{\sum_{i=1}^{N} | A_i' |^{\beta}}. \tag{6.103}$$

The weight puts more emphasis on large amplitudes while suppressing the insignificant moments that are more likely to be affected by image processing operations.

As in [24], we now compute the difference between the moments of the original and watermarked images $\Delta Z_{pq} = (Z_{pq}^w - Z_{pq})$ to construct a difference image. This difference image is then superimposed on the original image to complete the watermarking process.

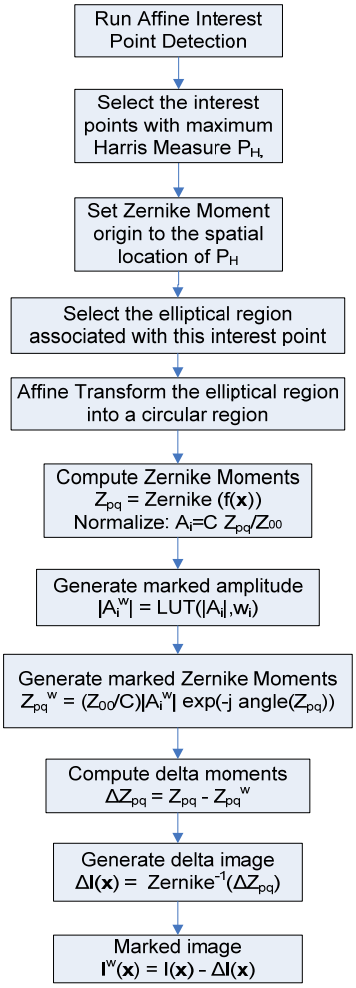


Figure 44. Watermark Embedding Process

On the detector side, the first step again is to run the affine interest point detection algorithm to find the origin and associated circular region of the watermark as in the embedding process. Because the interest point detection is invariant to image transformation, the synchronization of the origin and watermarked region is maintained between the transmitted and received images. The image moments $Z_{pq}^{'}$ over this region are normalized by Eq. (6.4) using the same constant *C*. The first *N* moments that fit the criteria above are chosen to form the sequence $\{A_i^{'}\}$. The associated binary sequence is

obtained by running $A_i^{'}$ through the index function $index(|A_i^{'}|)$ using the same values of α. The correlation of the original watermark sequence w_i and $index(|A_i|^{'})$ using Eq. (6.7) is compared with a threshold to determine if the current image contains the watermark.

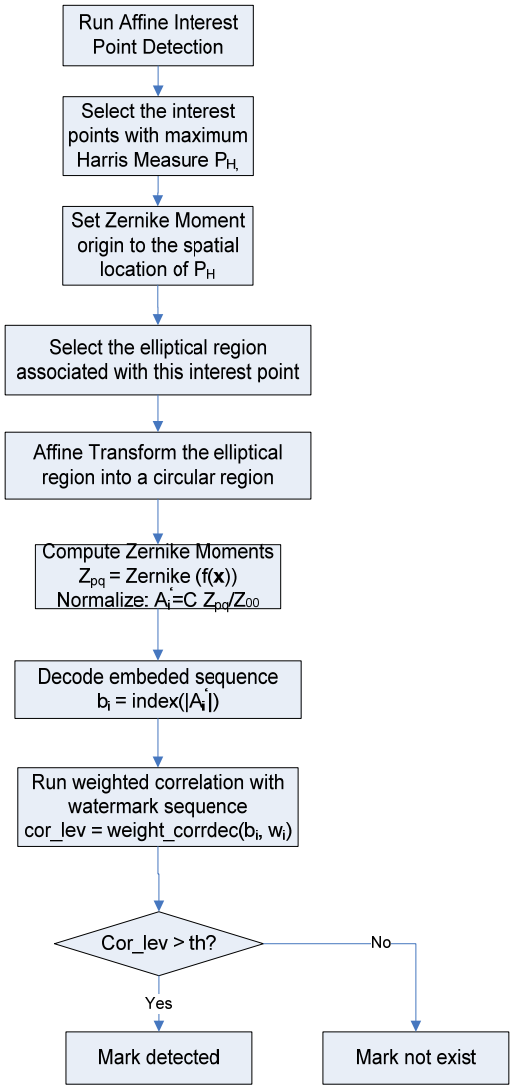


Figure 45. Watermark Detection Process

One of the potential drawbacks of both ZM- and RTM-based image watermarking is the computational load and time required, especially for higher order moments. As shown in Figure 46, the computation load grows exponentially with *N*.

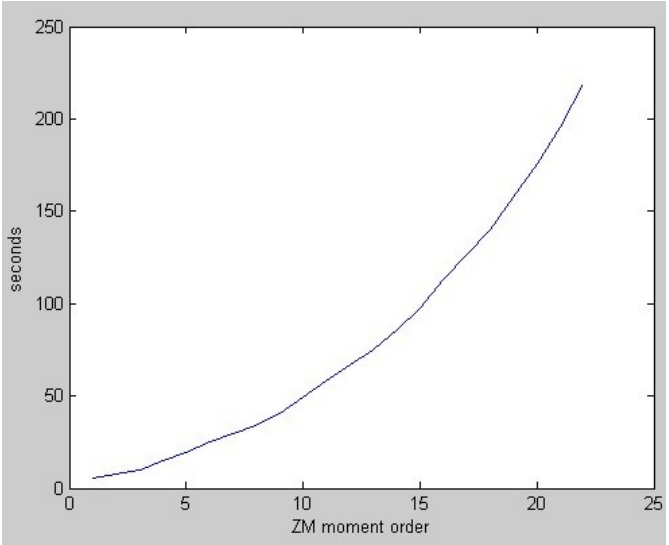


Figure 46. Zernike Moments Computation Load

While high computation load is not suitable for real-time type watermarking applications, the robustness of the new framework will be beneficial in many cases where time or computational load is not a huge concern. One such area is film fingerprinting and piracy forensic analysis. In such application, watermarks can be embedded as part of film production. When a film is sent for analysis, the watermark authentication can be done off-line. The utmost priority here is the robustness of the watermark with speed as a secondary consideration.

6.2 Video Watermarking

Video watermarking has some unique challenges. One is that video tends to undergo a series of transcoding operations; therefore the watermark will need to survive video

transcoding. Furthermore, in the video watermarking application, in order for the scheme to be robust, it has to be able to survive the varying bit rate reduction/compression during the video transmission. For example even if the watermark is to be embedded in the raw video, one may need to look ahead to see if the watermark is required to survive another video format such as H.264 or MPEG2 with potentially much lower bit rate.

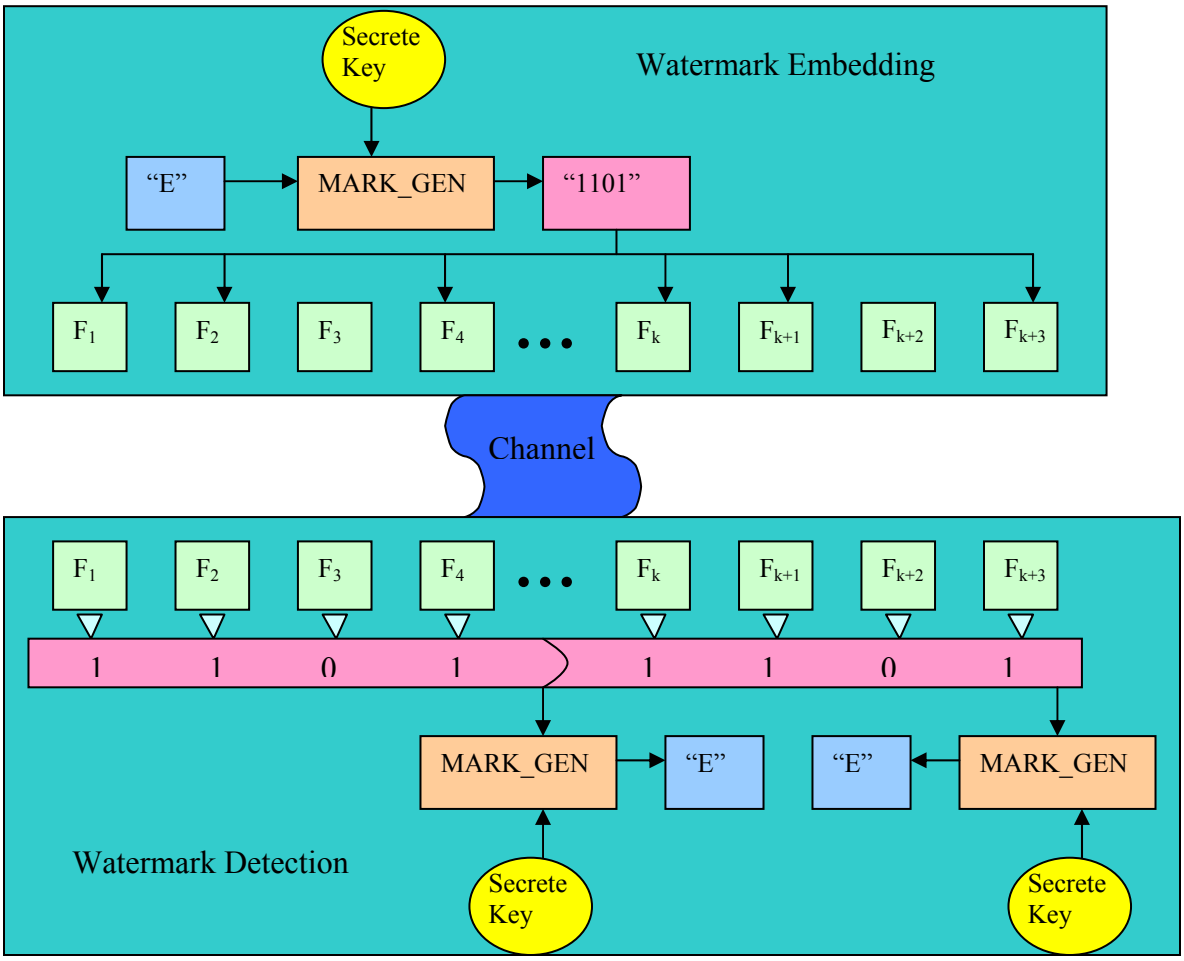


Figure 47. Temporal spreading of Watermark Bits

In this section, we will expand the previously discussed framework into the realm of video watermarking. Here we intend to introduce two implementations that use two different image moments functions – RCM and Hu's Moment Invariants (MI). These two implementations share some common areas. As in image watermarking, the embedding is done in the original video frame, instead of in the DCT/DWT transform domain. The watermark is converted to a bit sequence. We can spread the multiple bits of the watermark into different frames by embedding one bit of watermark in one frame. The detector would be a two-stage implementation. First each frame will undergo detection and the result will be fed into a secondary buffer. Then the bit stream in the buffer will then be fed into watermark information extractor to obtain the actual embedded information, as in Figure 47 above.

Embedding one bit in one frame improves the robustness of the watermarking against downstream compression as described in Figure 48. Furthermore, since the algorithm is implemented on the original video independent of the video coding format, there is less impact by video transcoding.

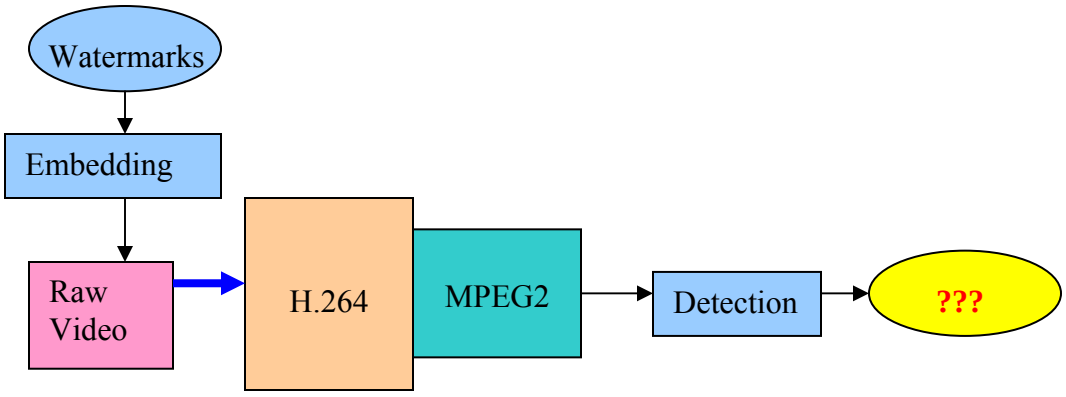


Figure 48. Robustness of Video Watermarking –Surviving the Smallest "Pipe"

In the first approach to video watermarking using the proposed framework, we adopt the RCM based image watermarking approach described in Section 6.1. On the source side, we first choose the frame in which the watermark is to be embedded and then identify an embedding region using the affine interest point detection. . The watermark is then embedded using the process described previously by modulating a subset of the RCM amplitudes. . At the receiver, we again identify the watermark region using the same affine interest point detection process, and evaluating the amplitudes of the RCM subset to detect the existence of the watermark. To reduce the impact of the inter-frame motion compensation, watermark is embedded at regular frame intervals.

For the second implementation, we will attempt to demonstrate the flexibility of our proposed framework by integrating a MI based watermarking technique into our framework. As described in Chapter 4, another promising watermarking algorithm proposed by Alghoniemy and Tewfik [4] utilizes the properties of MI to establish the watermarking embedding and detection algorithm. Their technique was shown to be robust against many geometric attacks as well image processing attacks and is simple to implement and with low computation complexity.

However, it suffers some deficiencies. The first, as with many existing moments based algorithms is its dependency on the image centroid. As described in Chapter 2, Hu's formula is derived based on the central moments:

$$\mu_{pq} = \sum_{\Gamma} (x - x_c)^p (y - y_c)^q f(x, y) \tag{6.104}$$

where x_c and y_c are the image centroid. As such, their watermarking technique will fail under the same image cropping/padding type attacks. However, our proposed framework

comes to the rescue – first we define the region that is based on the scale and location of an affine invariant interest point, and then convert the region to the transform domain using the second moment matrix. Then the same watermark embedding and detection procedures outlined by Alghoniemy and Tewfik can be applied in this circular region in the transform domain. This will help to overcome the dependency on the image centroid of the original approach while retaining its simplicity and robustness.

Another deficiency of the original moment invariant based approach is that it inherently can only embed **one** bit of information within the watermark. For image watermarking, such low capacity can be problematic; however, it fits perfectly with our general approach in video watermarking since here our intention is to embed one bit of the watermark in one video frame and relying on the temporal domain to increase the watermark capacity.

The third deficiency in the original algorithm is that it is really not a blind watermark technique – the watermark level will need to be supplied externally to the receiver in order for the detection to work properly. In our proposed video watermark framework, this obstacle is overcome by carefully arranging the embedding frames. We utilize the temporal redundancy in the video stream. Since there are, in general, a group of frames that are highly correlated within the video stream, the variations of the moments within these frames should be small. We will use a terminology in Film to Video Conversion – *cadence* to describe a set of frames over which the moments are essentially constant [21]. Within the cadence, the first frame will be used as a *reference frame* to compute the expected MI level; the last frame will be the *mark frame*, where the watermark embedding/detection are conducted. As shown in Figure 49, only cadences that have

"flat" MI distribution will be selected for watermark embedding, otherwise, the current cadence will be skipped and next one will be examined. This essentially converts a non-blind watermarking technique into a blind watermarking algorithm.

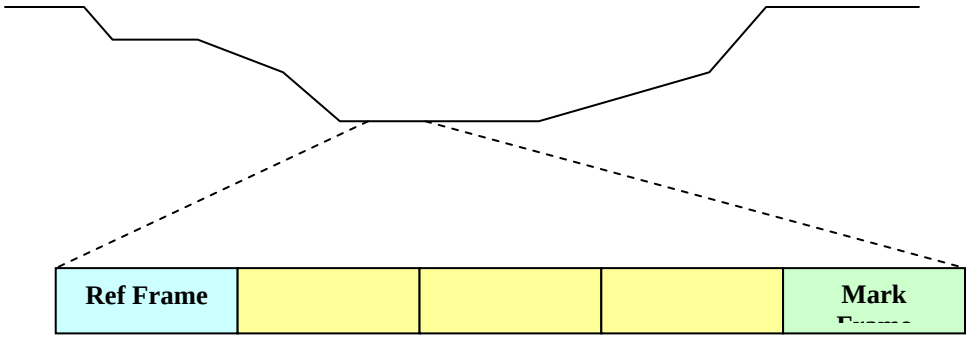


Figure 49. Use Flat Cadence for MI based Video Watermarking

Another topic that is worth discussing is frequency of applying affine invariant interest point detection. This issue is common to both video watermarking implementations proposed above. We list several options here with increasing order of computational complexity but improved robustness:

1. *Apply affine invariant interest point detection on the first frame and re-use the same interest point throughout the video stream;*
2. *Apply affine invariant interest point detection on each new cadence;*
3. *Apply affine invariant interest point detection for each Intra frame and re-use until next Intra frame;*
4. *Apply affine invariant interest point on each frame;*
5. *Apply affine invariant interest point detection on the first frame of each scene and re-use the same interest point throughout the scene;*
6. *Apply affine invariant interest point detection on each new cadence in each scene;*

Evidently, the issue for option 1 is that if first frame is lost and the content of its neighboring frames ware drastically different, then there is no way to detect the watermark successfully, therefore, when choosing this option it is desirable to select a frame from a range of frames that contain very slow motions. Option 4 is obviously the most robust one, since we can guarantee the accuracy of the watermark origin and region for every frame, but it is at the expense of unnecessarily high computation load. Option 2 may be the optimum choice in terms of computational load though Option 6 can indicate if any particular scene within the video stream has been manipulated or altered. However, ultimately the choice will depend on the actual situation – if the goal is to prevent movie piracy, then option 4 may be a favorable choice, since robustness outweighs any other consideration in that environment; but for application such as Broadcast Monitoring, then option 1 would be sufficient.

6.3 Summary

In the above three sections, we have described a proposed image watermarking framework and extended it to video watermarking. In Chapter 7, we will discuss some experimental results and demonstrate the effectiveness of both the image watermarking and video watermarking frameworks proposed here.

Chapter 7

EXPERIMENT RESULTS

In this chapter, we present some simulation results illustrating the applicability of the watermarking technique described in this research. In all our experiments, the Stirmark package is heavily used to generate various types of attacks. Therefore, we start the discussion of our experimental results with a brief overview of the StirMark Benchmark package.

7.1 StirMark overview

StirMark is a generic benchmark tool that is widely used in the evaluation of new digital watermarking techniques. It provides a framework that allows the developers to easily generate a variety of attacks that can be used to evaluate the performance of their algorithm. The C++ source codes are freely available under the GNU General Public License from: *http://www.petitcolas.net/fabien/watermarking/stirmark/*. The package includes some built-in attacks that the developers can immediately apply to their work. These tests covers most common types attacks such as geometric attacks, image processing attacks, compression attacks. It also provided hooks to allow new customer attacks/tests to be incorporated into the package.

The directory structure of the StirMark is as following:

/	Bin/		Benchmark/		*Executable*
	Profiles/				*Configuration files describing the tests to run*
	Media/	Input/	Images/	**Folder1**/	*Input Images*
				…	
		Output/	Images/	Folder1/	*Test/attacked results for the input image.*
				…	

The source code is mainly oriented toward Windows platform. User can use the MS Visual Studio to generate executables that runs on Windows. Once compiled, user will need to tailor the configuration file in the Profiles directory to include the tests intend to run. The default configuration file is SMBsettings.ini. The configuration file consists of the following sections:

[ImageFolders] *Folder1=Images/Set1* *Folder2=Images/Set2* *Folder3=Images/Set3* *…*	*Input Image Folders*
[Tests] *Test1= Test_Cropping* *Test2=Test_EmbedTime* *Test3=Test_AddNoise* *…*	*Lists of test applied*
[Test_Cropping] *; cropping ratio in %* *list=70 75 80 85 90 95* *…*	*Parameters for each test*

The configuration file is a plain text file can be easily altered using any text file editor. The content can be altered before each run or the user can store different configuration in separate configuration files and specify the file at the command line when run the test.

The user will first run the watermarking embedding process, put the watermarked image in one of the input directories specified in the configuration file above and run the test to generate the attacked results. Then the results can be fed into the watermark

detector and the detected results evaluated. In general, the development process can be described in the following flow chart:

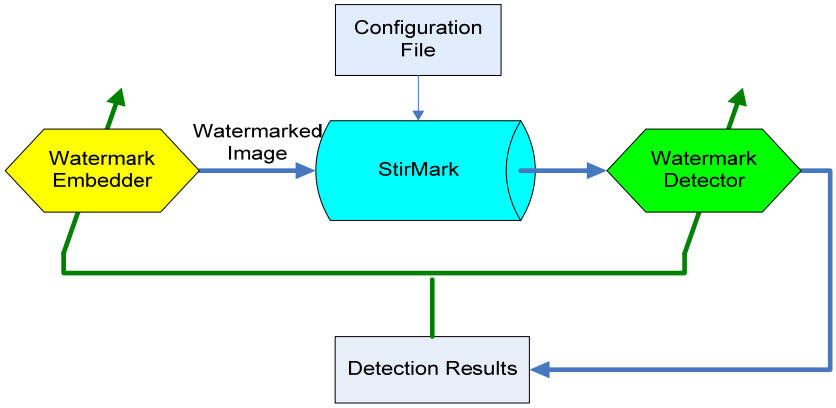


Figure 50. Development Flow Using StirMark

As mentioned in Chapter 6, there are some new attacks that are very effective in exposing the vulnerability of the existing moments based watermarking schemes. For this purpose, two new tests were added to StirMark tests:

- One-sided image cropping;
- One-sided image padding;

In StirMark package, adding new tests is fairly straightforward. To create a new test OneSideCropping, an object CSMBTest_OneSideCropping derived from CSMBTest. Most of the default method in the parent class is sufficient for the new object. The main operation DoCropping will be the only method that needs to be created for this new test. With the built-in Cropping test as reference, creating this method was also pretty straightforward. The OneSidePadding was created in similar fashion.

The StirMark executable is essentially a MS-DOS command. It takes two parameters as input. One specifies the profile to be used; the other specifies the name of the log file:

StirMarkBench [myINIFile.ini] [myLOGfile.log]

If no parameters are specified, then SMBsettings.ini will be used as the default profile, and SMBReport.log will be the log file.

We now use StirMark to demonstrate the vulnerability the existing moments based watermarking schemes by demonstrating the impact of image cropping and padding on the location of the image centroid.

7.2 Centroid Shifting due to Image Padding/Cropping

In this experiment, we use StirMark to generate One Sided cropping and One Sided Padding of images and then compute the distance by which the image centroid between the original image and the attacked image has shifted. The centroid in the original image is computed. Its location and the intensity of a 5x5 neighborhood of the point are stored. Then for each attacked image, its image centroid is computed, a 50x50 window around the centroid is searched by computing the sum of absolute difference (SAD) of the 5x5 window with the image centroid of the original image; the point with lowest SAD values is the location of the original image centroid. The shifting distance can then be computed between the new and original image centroids.

Since the primary cause for centroid shifting is the enlargement or reduction of the image area, we did not implement any attacks that combine cropping/padding with other geometric transformations such as scaling or rotation. The following are some results of image centroid shifting due cropping and padding. The original centroid is shown in green, and the centroid of the attacked image is shown in red. The cropping and padding

percentage ranged from 2%, all the way up to 20%. Left column of images are after padding and right are after cropping.

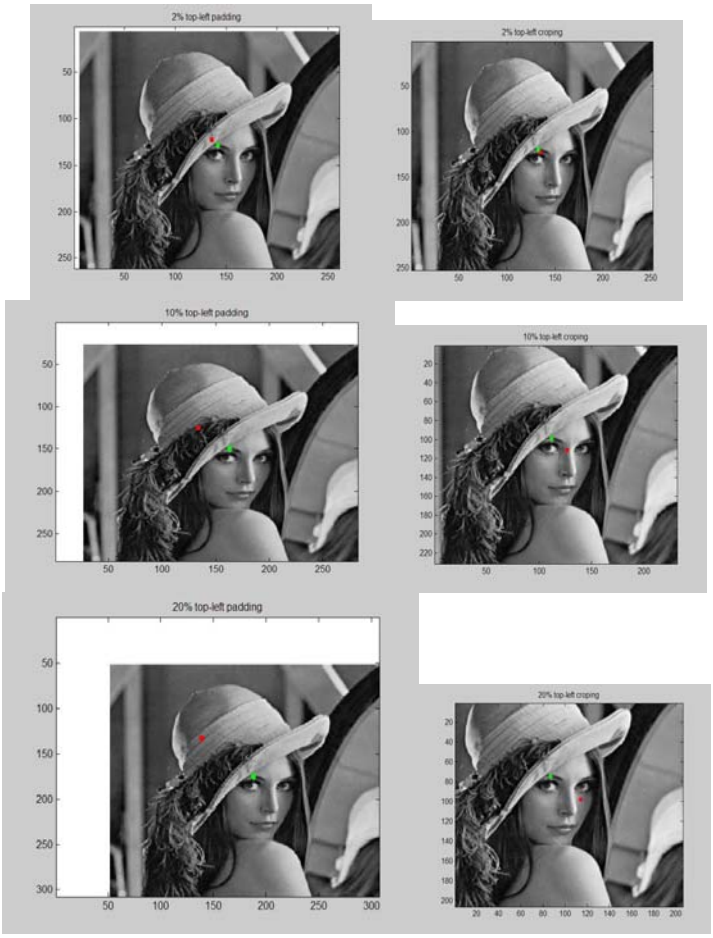


Figure 51. Image Centroid Shifting due to Padding and Cropping (2%, 10% and 20%)

The shift in the image centroid causes the amplitudes of the moments of the cropped images to differ from those of the original image. Table 3 shows the summary of the centroid shift and variation of the Zernike moments amplitudes for the 256x256 Lena image.

The shift in the centroid would obviously cause the receiver to lose synchronization with the source about the location of the watermark, thereby preventing successful watermark detection. Between these two different types of attacks, image padding poses a greater threat, since in this case even though the information in the original image is not lost, however, it causes watermark synchronization to be destroyed successfully.

Table 3. Moments Variation due to Cropping and Padding

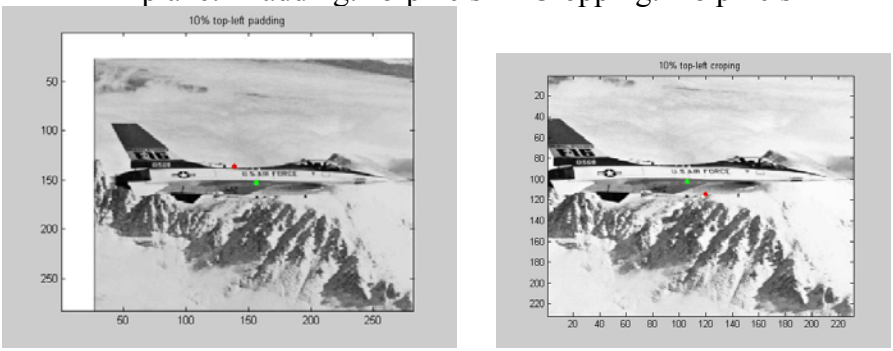
Percentage	Centroid Shift Distance (pixel)		Average Moments Variation (%)	
	Cropping	Padding	Cropping	Padding
2%	4	8	39	114
4%	6	17	169	222
6%	9	23	317	311
10%	19	38	356	430
14%	27	49	363	427
16%	30	55	396	437
20%	36	65	390	453

In this experiment, only simple max intensity white band is used for padding, which obviously can be easily recognized and mitigated. However, there are almost infinite numbers of choices for the padding pattern that can make it very difficult to separate the original image and padding.

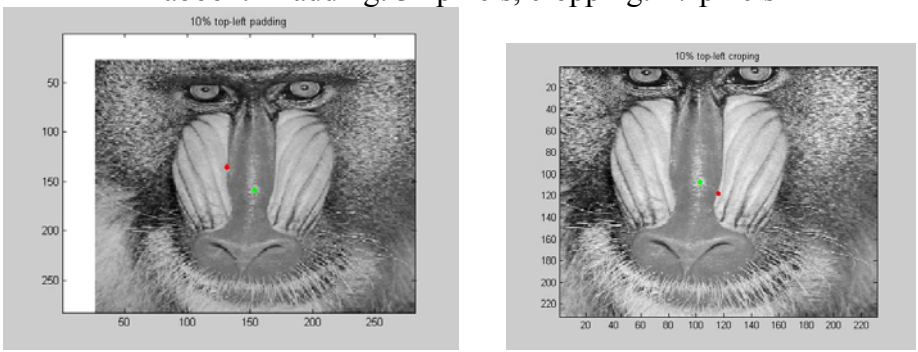
The examples below show the result of 10% cropping and padding for three different types of image scenes. They all confirm the similar level of centroid shifting due to the cropping and padding attacks. Again, image padding induced more centroid shifting than image cropping. What is also interesting is that the resulting shift seems independent of the type of scene the original image. All four different types of scenes show almost same shifting distance with the same amount of cropping/padding. From these results, one can

conclude that these types of attacks used either by themselves or in conjunction with other types of attacks will have a similar impact on moment-based watermarking schemes.

Airplane: Padding: 25 pixels Cropping: 18 pixels



Baboon: Padding: 32 pixels, cropping: 17 pixels



Water: Padding: 34, Cropping: 18

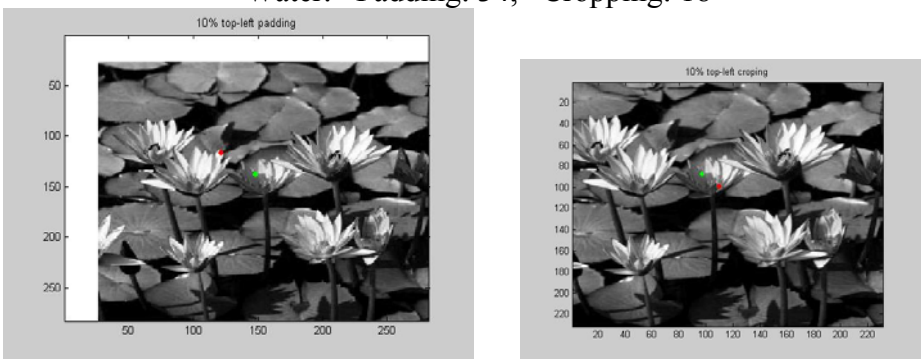


Figure 52. Image Centroid Shifting due to Padding and Cropping

After having demonstrated the deficiencies of existing algorithms, now we will demonstrate the results of the proposed watermarking framework in next section.

7.3 Experimental Results from the Proposed Watermarking Framework

The first step of the proposed algorithm is the detection of the affine invariant point to be used as the origin. The process outlined by Mikolajczyk and Schmid [35] is in general adopted. However, some special consideration has to be given to address the particular issues in this application. In order to better adapt to the variety of image processing type attacks, and improve the speed, the image is pre-"soften". While the iterative affine point detection process can eventually converge to the affine interest point that reflects the characteristic scale of the image, the softening of the image can allow the location of the initial point to be closer to that of the final location and reduce the required iterations. In this research, a 5x5 Gaussian low pass filter is used for this purpose:

$$LPF = \begin{bmatrix} 0.0369 & 0.0392 & 0.0400 & 0.0392 & 0.0369 \\ 0.0392 & 0.0416 & 0.0424 & 0.0416 & 0.0392 \\ 0.0400 & 0.0424 & 0.0433 & 0.0424 & 0.0400 \\ 0.0392 & 0.0416 & 0.0424 & 0.0416 & 0.0392 \\ 0.0369 & 0.0392 & 0.0400 & 0.0392 & 0.0369 \end{bmatrix} \tag{7.1}$$

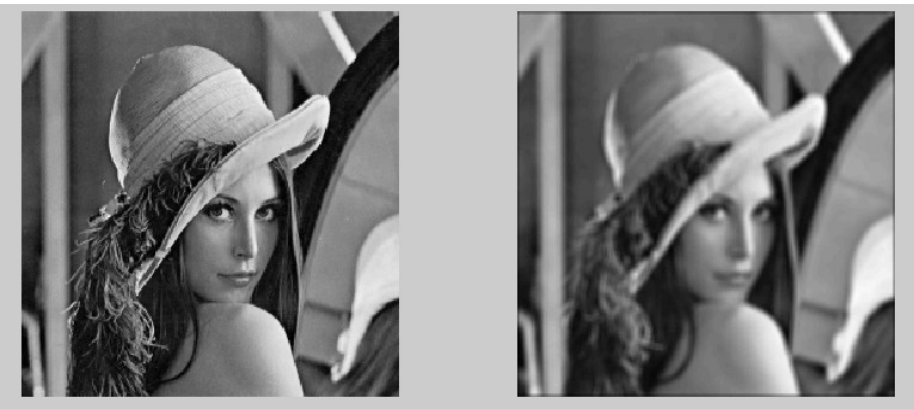


Figure 53. Image Blur before Affine Point Detection

Another issue that needs to be resolved is that in the general affine interest point detection, the intention is to detect all affine interest point of the image. However, for our research, the goal is to locate one such point to be used as the origin of the region which is to be used for watermarking. The centroid based filtering serves this purpose. This filter has one parameter filter radius R_F that needs to be tuned according to the environment of the application. This filter has one parameter, the filter radius R_F that needs to be tuned according to the anticipated intensity of the attacks that cause the centroid to shift. The radius R_F will need to be large enough to accommodate the potential threats.

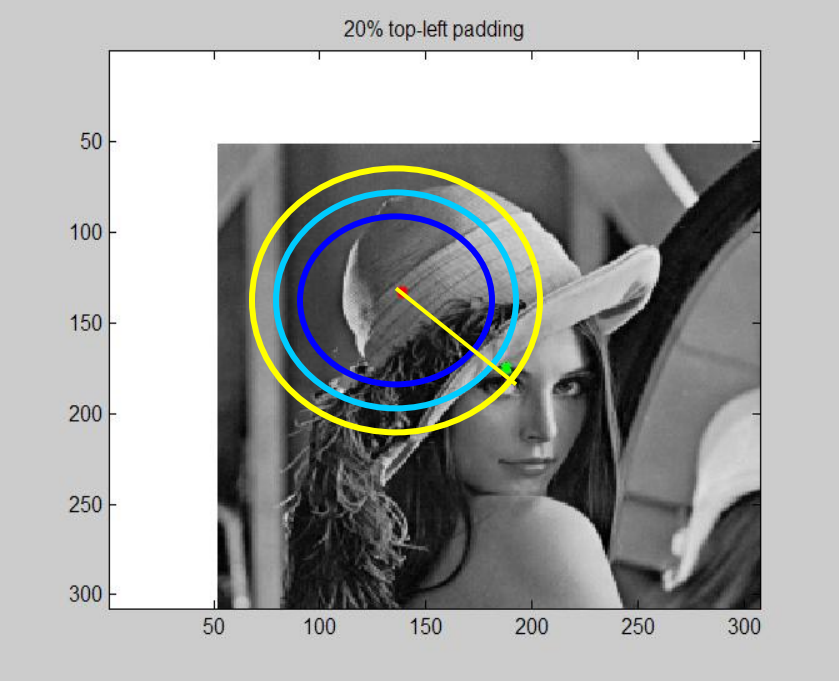


Figure 54. Filter Radius Needs to Account for Largest Potential Centroid Shifting

However, increasing the filter radius will resulted in more initial detected points fall into the filtering circle, as a result, will increase the computation load.

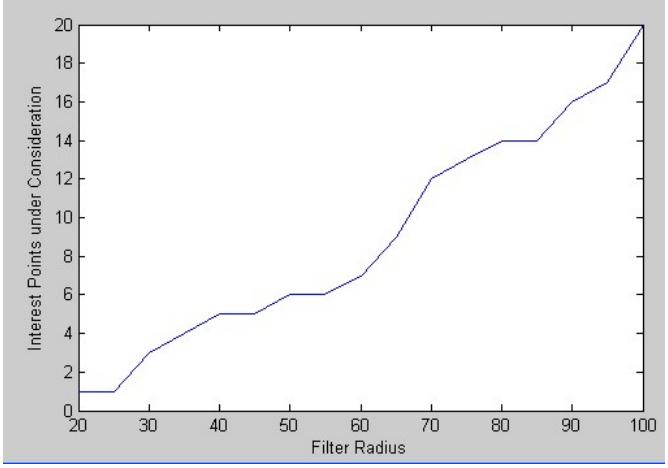


Figure 55. Computation Load Increase due to Larger Filter Radius

As shown in Figure 55 above, the interest points to be considered will increase from 1-2 to above 20 if the radius is opened up from 20 pixels to 100 pixels. Since the iterative process will need to be repeated fro each of the identified points, therefore, the additional computation load due to increased filter radius can be substantial. In our experiments, we have chosen the radius R_F =60, this being generally equal to the amount of centroid shift when the image undergoes 20% padding. This will result in about 5-6 points to be considered for the iterative affine point detection. Again, in practice, this parameter will need to be tuned by the users to balance the severity of the potential threat and the tolerable computational load increase.

The iterative affine point detection portion is applied against each of the points that fall inside the filter circle. The other parameter to be chosen is the maximum number of

allowable iterations. This parameter was chosen empirically. In our experiments, the affine interest point on converged, on the average, within 10 iterations and in the worst case took 20 iterations. For this reason, we set the maximum number of iterations to 25.

As mentioned in Chapter 6, another type of elimination is required to remove points with over stretched support region, since such points turn to fall on the edges rather than the corners that the Harris detector is intend to find. For this, we will remove any points with ratio of the eigenvalues of the second moment matrix over 10: $\lambda_{\max} / \lambda_{\min} > 10$. Since λ_{max} and λ_{min} are proportional to the long and short axis of the elliptical support region, therefore $\lambda_{\max} / \lambda_{\min} > 10$ corresponds to an extremely flat elliptical region that follows an edge on the image.

The reason that this step is done after the iterative affine point detection is that the shape of the support region may alter during this process, as show in Figure 56 below. The initial support region actually is not stretched, but the final region is.



Figure 56. Final Support Region Shape Different from Initial Region

After all the necessary filtering, the origin is chosen by first ranking the interest points that are left, and then the point with highest Harris measure is chosen as the origin. The integration scale σ_I and the second moment matrix that are associated with this point are used to define the region to be used for embedding. While it is possible to use $k*\sigma_I$ as the scale used to define the watermark embedding region, where k is a constant factor that is agreed upon between source and receiver, in our experiments, we have used k=1.. Figure 57 below shows the chosen origin and associated region for some examples.

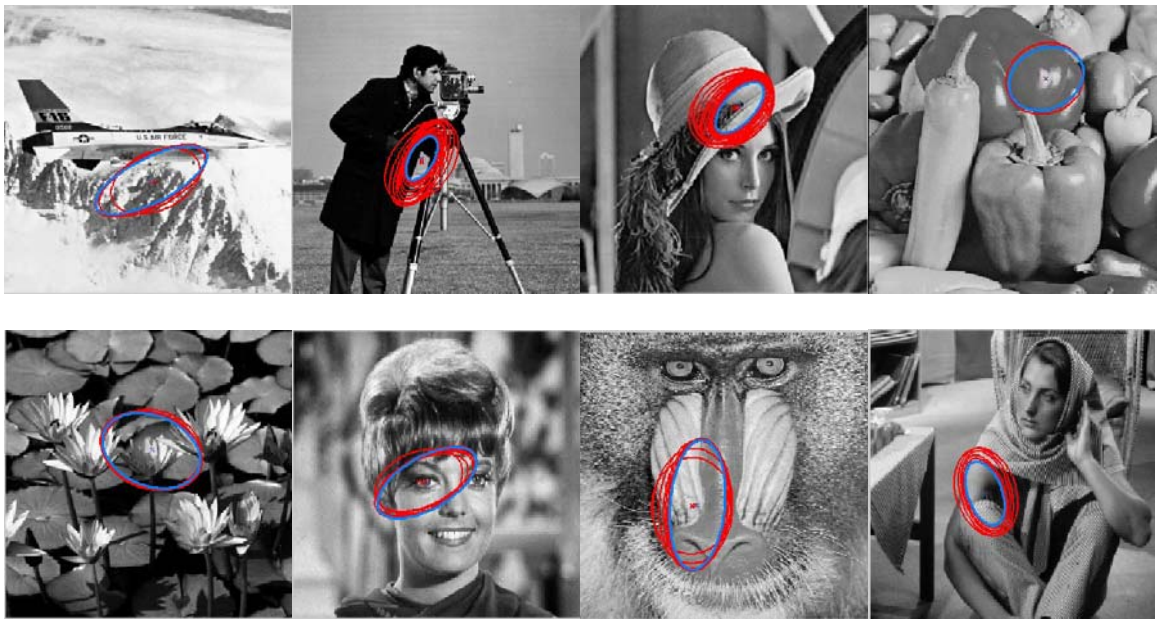


Figure 57. Examples of Selected Origin and Embed Region.

With the region and the origin defined in the image domain, the next step is to transform this region into the affine domain using the second moment matrix obtained during the iterative affine point detection. The resulting circular region is perfect for such computing the Zernike or Radial Tchebichef moments, see Figure 58 below.



Figure 58. Embedding in the Circular Region in Transform Domain

At this point, it is necessary to specify the set of individual moments to be used for watermark embedding/detection. The 0th and 1st order moments are excluded for reasons noted earlier. We also avoid low energy moments by rejecting moments which amplitudes below $2e^{-3}$ and its vicinity two points, as shown in Figure 59.

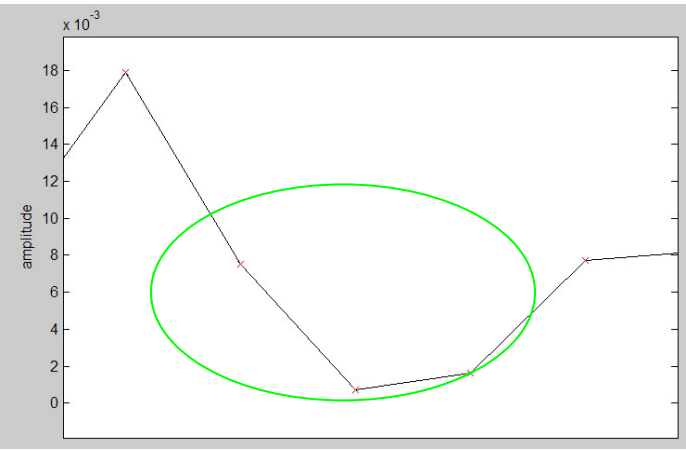


Figure 59. Moments Rejected for Watermark Embedding

Since the moments are normalized using DC term, therefore $|A_i|<1$. In order to embed the watermark by modulating the amplitude of each of the qualified moments using the geometric integer sequence based look up table (LUT) described in Chapter 6, the amplitude of the moments will need to be multiplied by a constant C and then rounded into an integer that can be fed into the LUT. Since the minimum amplitude is $2e^{-3}$, therefore C is set to 3000; this will allow enough dynamic range even for the smallest moments. The factor α for the geometric sequence is chosen to be 0.1. In the experiments, a 32 bit pseudorandom sequence is used to modulate the amplitude of first 32 moments that meet the above conditions. In the watermark detection, the weighting factor β for the cross-correlation is set to 0.9. The watermark detection threshold is chosen to be 0.6.

To evaluate the proposed algorithm, a subset of 32 images from USC-SIPI Image Database Misc Volume is chosen as shown in Figure 60 below:





Figure 60. Images Used in the Experiment

During the experiment, the following 10 different attacks were performed on all the images selected for evaluation:

- One Sided Padding: 5%, 10%, 15% and 20%;
- One Sided Cropping: 5%, 10%, 15% and 20%;
- JPEG: 0.80 and 0.85, 0.9 and 0.95;

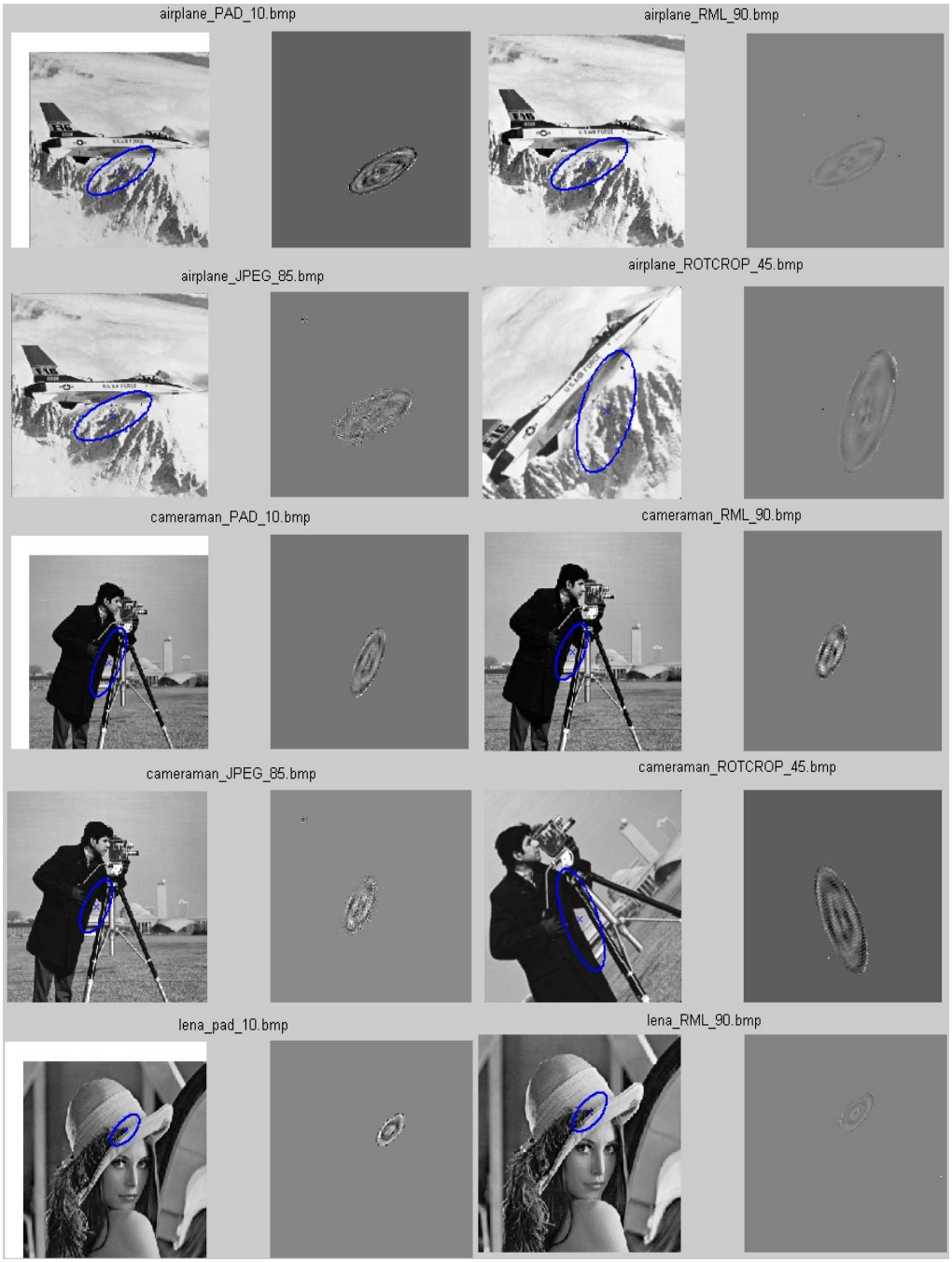
- Remove Lines: 1/10, 1/20;
- Median Filter: 3x3 filter size;
- Add Noise: 1 ~-45dB, ~-20 dB;
- Rescale: 80%, 90%, 110% and 120%
- Rotation: 10°, 45° and 90°
- Rotation-Crop: 10°, 45° and 90° and cropping 20%;
- Rotation-Scale: 10°, 45° and 90° and scale 110%;

The two "One Sided" attacks were new attacks added during this research, the rest are built-in attacks in Stirmark package. 30 different attacks were done against each of the image shown above. This generated a sample size of 960 attacked images to analyze the algorithm performance.

The actual implementation utilized a modular approach to take advantage of the flexibility of the proposed framework. The embedding moment selection and moment computation are grouped into one plug-in module that can be easily swapped in/out of the implemented framework. As a testament to the flexibility of the proposed framework, two types of image moments: Zernike Moments and Radial Tchebichef Moments. Another set of image moments – Orthogonal Fourier-Mellin moments were evaluated initially but was discarded due to their inability to reconstruct grey scale images properly due to the reasons described in Chapter 5.

The images in the following figure demonstrate the ability of the proposed framework in correctly detecting watermark under some typical types of attacks. Each case is shown as an image pair; the left side is the watermarked image and with the watermarked region detected by the receiver superimposed. The right side shows the embedded watermark after undergoing attacks. . It is generated by applying the same attack to both the

watermarked and original images through and then computing the frame differences of the two images.



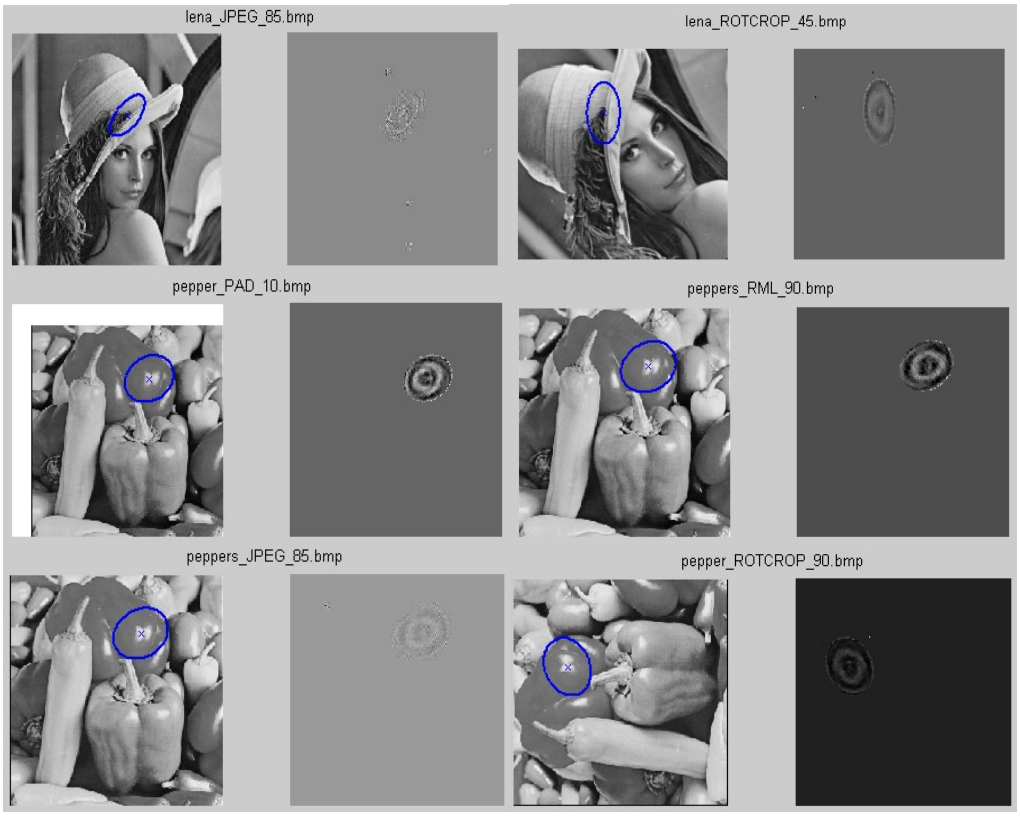


Figure 61. Affine Invariant Interest Point Detection after Attacks

In all, four different cases of attacks are shown for each image: Padding by 10%, Line Removal of 10% (remove 1 out of 10 lines), JPEG compression with 0.85 quality factor and a combined geometric attack – Rotate then Crop (rotate 45° for first 3 image and 90° for last one, followed by 10% cropping).

As shown in the summary of the detection rate against each type of attacks in Table 4, the proposed algorithm performed well against all the attacks, especially (one sided) image cropping or padding attacks. Furthermore, in general, the rate of successful detection is with RTM is somewhat higher compared to those achieved using ZM. This is to be expected for reasons discussed earlier in Chapter 5.

Table 4. Comparison of RTM and ZM detection performance

OPERATIONS	SUCCESS RATE	
	RTM	ZM
CROP	100%	95%
PADDING	95%	92%
JEPG	88%	86%
RML	92%	91%
MEDIAN	100%	78%
NOISE	81%	75%
RESC	88%	84%
ROT	89%	83%
ROTCROP	83%	88%
ROTSCALE	84%	77%
AVERAGE	90%	86%

One of the main factors for the failure is due to wrong affine invariant interest point picked by the receiver, which results in lost of synchronization between source and receiver. We would like to discuss it further here. As can be seen in the table below, in the vast majority cases, the correct point was detected. However, since the point that used to embed the watermark is chosen by first sorting detected points based on their Harris measure and then picking the one point with highest value when the detected points have Harris measures which are very close, it is possible that attacks may cause the ranking to be shifted, which results in a different point ending up at the top of the rank. However, as can be seen from the data shown in Table 5, if the receiver checks the two highest points instead of just one, such failures can be reduced by more than 80%, albeit at the expense of receiver computational load.

Table 5. Rank of Correct Origin

2	3	>3
81%	14%	5%

Another issue that can help to reduce failures due to wrong origin being selected is that the application will need to take into consideration, the anticipated attack intensity. For example, if there is a potential of 20% cropping, then this factor will need to be taken into consideration when choose the origin for watermark embedding. Otherwise, if cropping will cut into the watermarked region, then it will be impossible for the receiver to recover the watermark. The following is an example of such failure We do not investigate this further in this research, but just point out that it may be necessary to take such a case into account in implementing this algorithm.

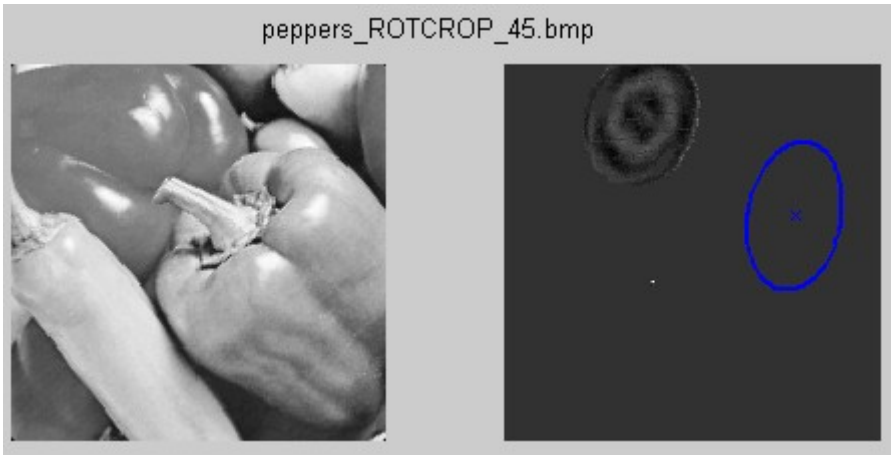


Figure 62. Origin Synchronization Failures due to Poor Watermark Location

In the next section we discuss the extension of the proposed framework for image watermarking to video sequencers.

7.4 Extending Proposed Watermarking Framework to Video Watermarking

In experiments for both video watermarking implementations (RTM based and MI based), we used a 12x10 "TI" binary image shown in Figure 63 below as the tested watermark. As described in Chapter 6, the image is flattened to a 120 bit sequence and embedded in to the video stream at one bit per frame. Then at the receiver end, the detected 120 bits are rearranged to a 12x10 binary image. The recovered image is then compared with the original marked image to evaluate the algorithm performance.

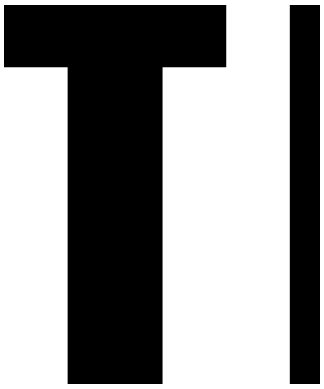


Figure 63. Mark Image

Since we already thoroughly studied the performance against geometric attacks in last chapter, we would like to study the impact of video transcoding and bit rate reduction on the watermark performance. During the experiment, "coastguard" clip is used to test both implementations. The bit rate of the raw video is 3.6Mbps. The watermark is embedded in the raw video using these two methods.. Then the marked video is encoded into H.264 with bit rate ranging from 100kbps to 3Mbps. Finally, the transcoded video clips are fed into the detector to extract the embedded watermarks. In this experiment, we intend to set up a watermarking interval of 10, therefore for RTM based approach, we embed the

watermark every 10 frames, , and for MI approach, the cadence is set to 10 frames. The original "coastguard" clip only has 300 frames. In order to have long enough duration for watermark embedding, the clips were repeated 10 times to create a 3000 frame clip for the experiment. As mentioned in Chapter 6, in both implementations, the first step is affine interest point detection to define the watermarking region. For simplicity and without lost of generality, we adopted option 1 described in Chapter 6 – conduct affine interest point on the first frame and use the detected region to embed watermark throughout the stream.

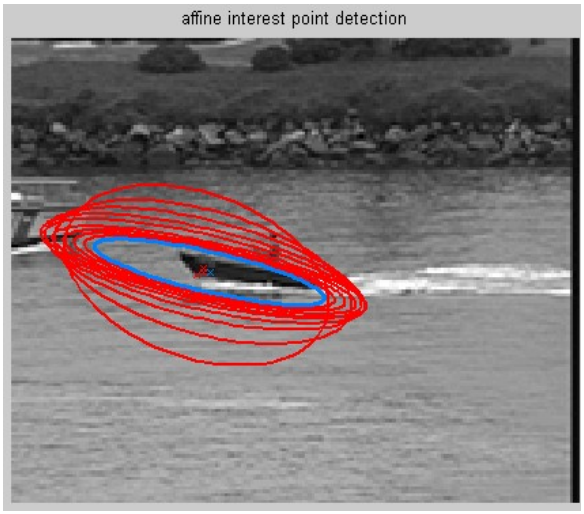


Figure 64. Affine Interest Point Detection

For RTM implementation, the same parameters used in image watermark detection are adopted. The detected results for several different rates are shown in Figure 65 below.

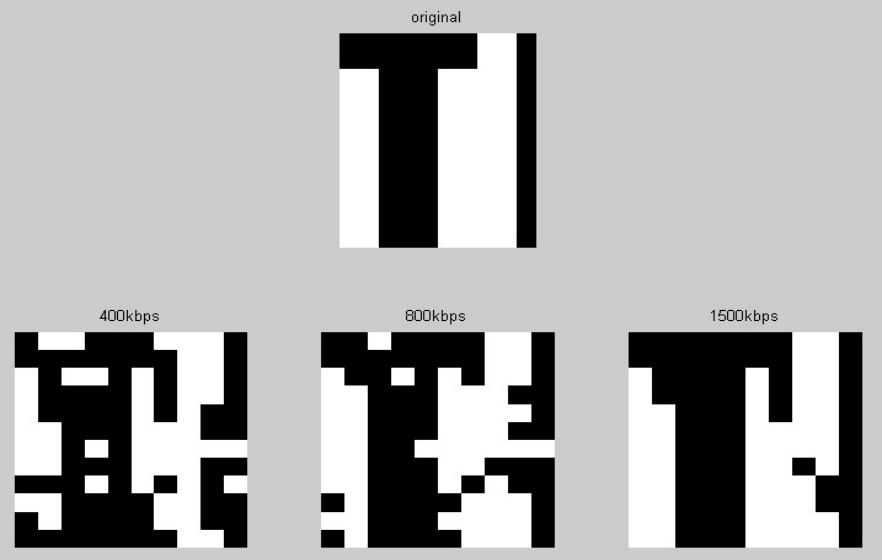


Figure 65. RTM Detection Results under Different Bit Rate.

As shown in Figure 65, the mark is recognizable when bit rate is at 800kbps. When the bit rate is at 400kbps, the high bit error will make the mark barely recognizable.

We now examine the moment invariant based video watermarking approach. The original algorithm described in [4] is a non-blind image watermarking scheme since it requires an external reference level to be used in watermark detection. As discussed in Chapter 6, we will use the fact that there exist clusters of frames over which the variation in moment invariants is small; therefore, the moment invariant functional value of an unmarked neighboring frame can be used as the reference level. Here we use the average of the seven Hu's moment invariants:

$$f(\Phi) = (\phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6 + \phi_7)/7 \tag{7.105}$$

where ϕ_1, $\phi_2 \ldots \phi_7$ are defined in Eq. (3.13)

We target watermark strength of 0.5% * $f(\Phi)$. Therefore, the expectation is that a cadence within the video stream can be found with variations of $f(\Phi)$ below half the level of the intended watermark, that is 0.25%* $f(\Phi)$ so that this cadence can be used for watermark embedding.

To validate our assertion that there will exist cadences in video streams with sufficiently small variations of $f(\Phi)$ to allow selection of reference and mark frame from within these cadences, we will first conduct an experiment to look at the frame to frame variations of $f(\Phi)$ in the following 8 video clips of difference scenes:

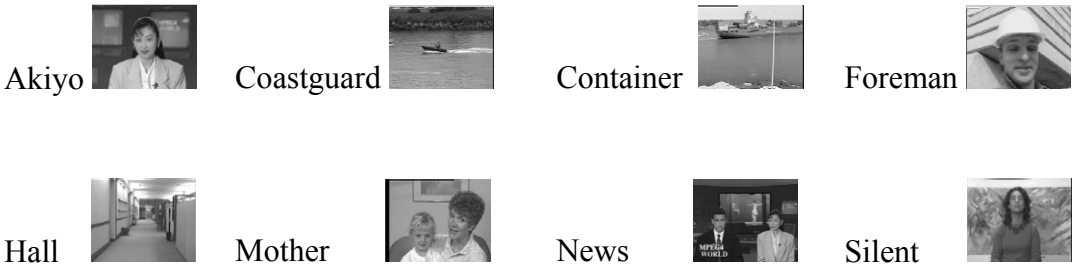


Figure 66. Clips used to Examine Moment Invariant Distributions

As with our previous applications, the first step in our algorithm is the affine invariant interest point detection to determine the watermark embed region. The function $f(\phi)$ mentioned above would be computed over the region determined in this step. For simplicity, during this experiment, we will only conduct the affine interest point detection on the first frame of each clip and reuse the same region for all subsequent frames.

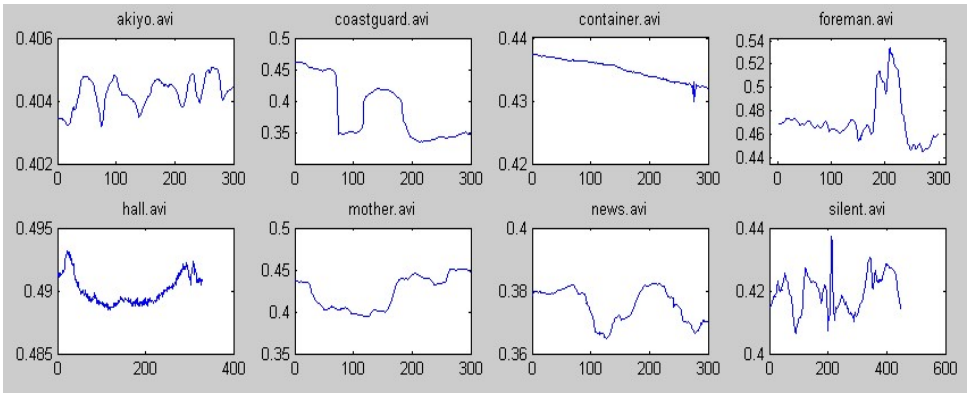


Figure 67. Average Moment Invariant for Each Frame in the 8 Video Clips

From Figure 67, it can be seen that except for "silent" clip, the rest of the clips all contains enough frames in which the variation is less than 0.25%. It is clear that some of the clips can support embedding more bits than the others due to the different types of scene content. When the video does not consist of a lot of fast motion, especially near the image center region, where the embedding region is most likely to be chosen from, then the previous stated stability requirement can be satisfied. As shown above, the proposed algorithm may have some difficulty with the "silent" clip, since there is a lot of motion (sign language gestures) near the image center region. However, even for this clip, there are some regions that qualify for the above requirement, as shown in Figure 68 below.

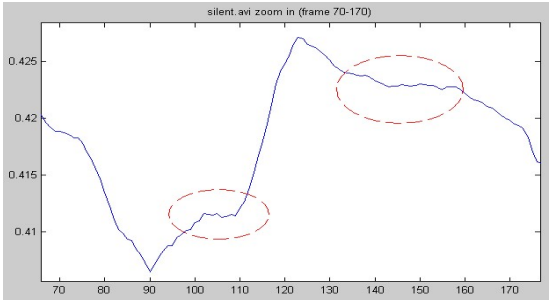


Figure 68. Existence of Flag Region in Video with Fast Motion

Having validated the possibility of selecting the reference frame from a group of frames, we now use one of the clips, “coastguard.avi”, to demonstrate the implementation process.

We initialize the frame counter to 0, and then use a 10 frame sliding window to select the next cadence. We examine the moment invariant function value of all 10 frames selected; if the variation is less than 0.25% of the maximum moment invariant function value, then these 10 frames are declared a cadence, otherwise we will skip these frames and slide to next 10 frames. This assures that the scene change boundaries can be avoided in the selection process.

The following is example of such the resulted watermark embedding using frame #280 of the coastguard clip as reference frame, and watermark embedded in frame #284.

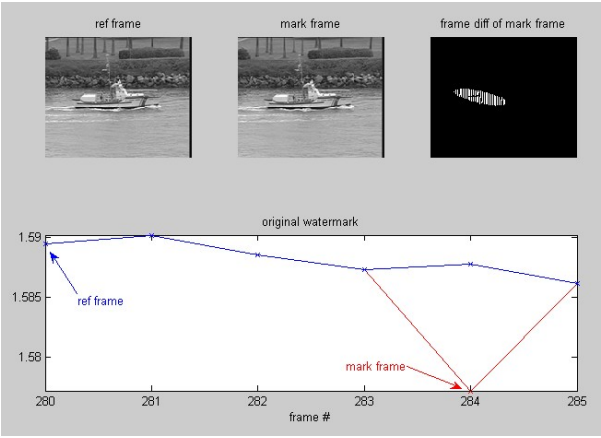


Figure 69. Watermarking Embedding using Reference Frame Within a Cadence

During watermark detection, the same frame cadence agreement needs to be established between the receiver and source (i.e., both uses same number of frames in the defining one watermark bit). The first frame in the cadence is used as the reference frame and the affine invariant interest point detection process is applied to this frame to identify the

watermarked embedding region. The corresponding moment invariant function value is computed to use compute the watermark reference level. The same region is extracted from frame 5 in the cadence and tested against the reference level to determine the presence of the watermark.

One goal to extend the proposed framework to video watermark domain is to provide an alternative method that is robust against Video Transcoding. In this experiment, we first stored both the original and marked "coastguard" clip in AVI format with a bit rate of 3500kb/s, and then both clips are transcoded to H.264 with same bit rate and transcoded back to AVI. In this experiment, several different bit rates of H.264 encoding were tested.

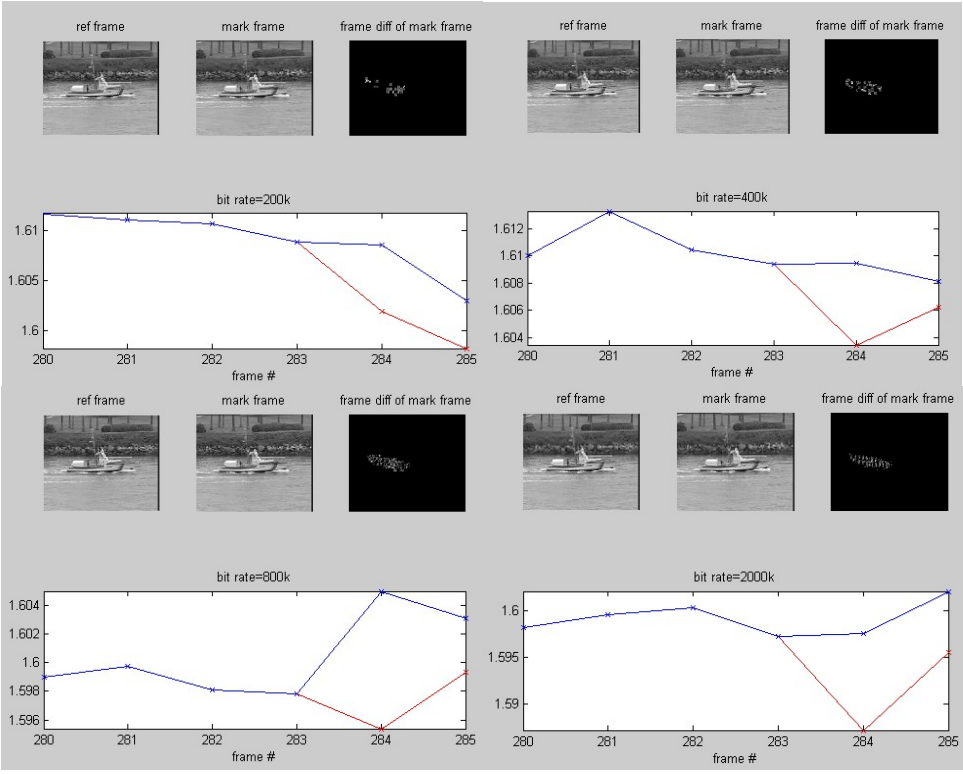


Figure 70. Watermark Detection after Transcoding with Different Bit Rate.

The above results shows that if the bit rate is sufficiently high, the watermark embedded using the proposed algorithm can survive the video transcoding. When the bit rate is below 200kb/s, the watermark may become too weak to be detected. The following detected results in Figure 71 also confirm the above observations.

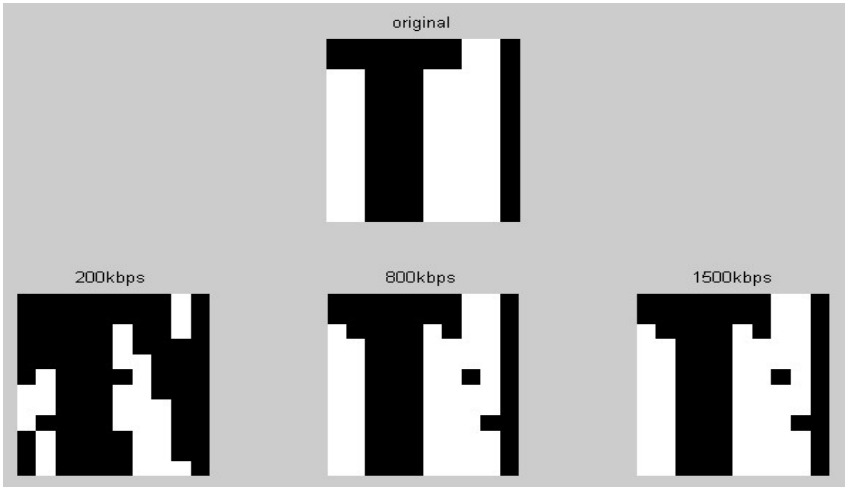


Figure 71. Detected Patterns with Different Bit Rates.

Another interesting observation is the watermark can "leak" into its adjacent frame due to the motion compensation of the underlying video coding scheme. Therefore, there is a need for padding frames at the start/end of the cadence to avoid watermark from previous cadence leaking into the next to improve the watermark robustness.

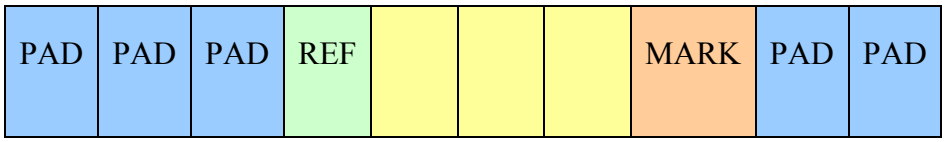


Figure 72. Cadence Padding to Improve Watermarking Robustness.

In comparing the performances of both algorithms, it is clear that MI approach is more tolerant of low bit rate video transcoding and can be used at rates as low as about 200kbps. On the other hand, RTM in general will be hard to be recognized below 800kbps. However, RTM approach requires total of 1200 frame to encode to 120 bits. MI approach on the other hand required total of 2650 frames to complete the task. Furthermore, the frames required for MI is scene dependent. In RTM approach, the required frame are fixed. Speed wise, MI approach is about 5 times faster than RTM approach, since it requires computing moments up to 3 only.

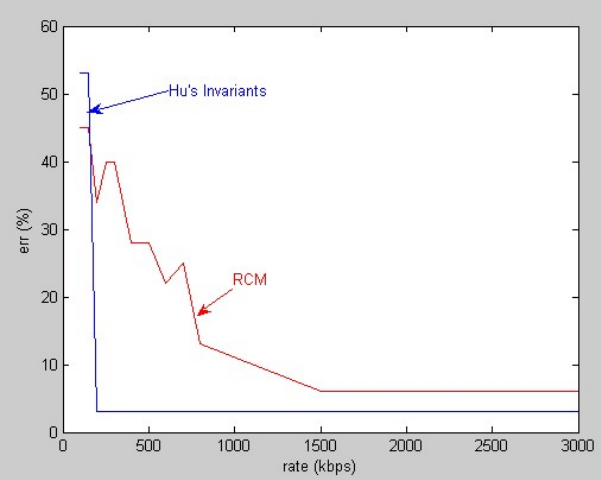


Figure 73. Detection Bit Error vs. Bit Rate

7.5 Summary

The experiments in this chapter attempted to show some of the implementation considerations involved in using the proposed algorithm for either image or video watermarking. The results demonstrated the effectiveness of the proposed framework and offered an improvement over existing moments-based watermarking algorithms.

Chapter 8

SUMMARY AND CONCLUSIONS

This work studies the watermarking schemes based on image moments. One of the interesting properties of some of the image moments such as Zernike Moments is the rotational invariance. This unique property offers Zernike Moments based watermarking algorithms an intrinsic edge over watermark embedding in spatial, DCT/DFT or Wavelet domain, since lacking rotation invariance is the main weakness in these approaches. However, one issue remains for the existing Zernike Moments based watermarking schemes is that the paradigm utilized in pattern recognition applications – choosing image centroid as the origin of the moments to synchronize between source and receiver. However, geometric attacks such as image cropping/padding can effectively destroy such synchronization by shifting the image centroid. Therefore, the choice of moment origin becomes an Achilles' heel for the existing Zernike moments (and indeed any other type of moments) based watermarking algorithms.

A watermarking framework for blind watermarking that integrates global image descriptor such as image moments and local image descriptor such as affine invariant interest point is developed. Instead of using image centroid as the moment origin, an affine interest point is detected and used as the origin. By applying same affine interest

point detection at both source and receiver, the synchronization of the moment origin can be achieved in blind watermarking applications. Another new development in this work is that the watermark embedding and detection is done in a circular region in the affine transform domain that fits perfectly with the requirements for Zernike Moments computation, instead of the elliptical region in the original image domain.

Experiment results have demonstrated that under this new framework, both source and receiver can synchronize to the same origin in the majority of the tests conducted. There were some cases that receiver has chosen wrong origin point due to the variation in the ranking of the Harris measure in selecting such point. However, further analysis shows that if receiver can expand the potential origin list from one to three, such issue can essentially be eliminated. Improperly choosing the size and location of the watermark region can contribute to the lost of synchronization. For example, if the region is too close to the image boundary, then image cropping may cut into the watermark region. However, that is more an implementation detail that can be addressed in the actual application.

Besides Zernike Moments, there are other image moment descriptors such as Radial Tchebichef Moments or Hu's Moment Invariants are also suitable for image/video watermarking applications. Both of the two moments were tested to demonstrate the flexibility of the proposed framework in accommodate different types of image moments. In head to head comparison of the performance between Zernike Moments and Radial Tchebichef Moments in image watermarking application, both performed well, with Radial Tchebichef Moments offer slight improvement in the success rate. This edge can

be attribute to that Radial Tchebichef Moments can better describe small region than Zernike Moments.

Hu's Moment Invariants is used in an attempt to further expand the proposed framework in video watermarking domain. In this implementation, the fact that video in general consists of sequences of frames that are highly correlated is used to construct cadences of frames. This helps to expand an original moment invariant based watermarking that requires prior knowledge of the moment invariant level (i.e., non-blind watermarking) into a true blind watermarking scheme. Moment Invariant level from the reference frame within the cadence is used to derive the expected watermarking level in the subsequent mark frame. Padding frames is included before the reference frame and after the mark frame to avoid the watermark from previous cadence leaking into the next one. Similar to the image watermarking counterpart, this scheme will be very effective against image geometric attacks such as cropping/padding or scaling. In addition, the experiment results show that the unique arrangement of reference frame – mark frame is effective against video transcoding with sufficiently high bit rate. Only one watermark bit per cadence is embedding, but the temporal natural of the video provides a richer medium for higher watermark information capacity.

8.1 Directions for Future Research

It is hoped that this body of work forms the basis for future research. Potential candidate topics for this are the following:

- Integration of affine interest point detection with DCT/DFT or Wavelet based watermarking. In this work, the focus was on making image moments based

watermarking more robust. However, the information associated with affine interest point detection: orientation, region, scale etc. can be used to pre-process the image at both the source and receiver to allow Wavelet based watermarking is conducted in the same region with same orientation.

- A spatial watermarking algorithm that uses the affine interest points. Such scheme may lead to another effective video watermarking scheme.
- A more efficient method to compute Zernike Moments and Radial Tchebichef Moments. The moment computation is the most time consuming portion of the algorithm.
- Hardware implementation of proposed algorithm, especially the video watermarking implementation. (Affine interest point based spatial watermarking may be an interesting one to study).

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